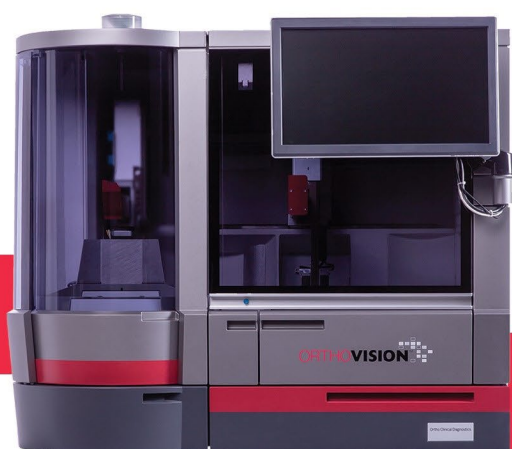


ORTHO BioVue® Cassettes

Laboratory Information System (LIS) Guide

Analyzer
ORTHO VISION®



Analyzer
ORTHO VISION®
Max



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1. Introduction

1.1 Purpose

This Guide defines the communications interface between the ORTHO VISION® System and a customer's Laboratory Information System (LIS). This Guide also explains the supporting requirements for the configuration of these interfaces.

Note: This Guide contains the Laboratory Information System (LIS) interface for ORTHO VISION® System and ORTHO VISION® Max System. Throughout the document where ORTHO VISION® System is mentioned, it refers to both ORTHO VISION® System and ORTHO VISION® Max System. System specific information will be notated e.g. (ORTHO VISION® Max System only).

1.2 Audience

This document serves as a reference for Information Technology personnel who are responsible for creating and maintaining the communication between the ORTHO VISION® System and the Laboratory Information System (LIS).

1.3 Revision History

Version	Section Number	Section Title	Revision Details
2025-08-01	3.4.4	Vision ASTM O Order Record	Added "The order creation time is provided from the ORTHO VISION® System."
	3.4.9.17	Quality Control Examples	- Replaced the examples for AlbaQ-Chek J with ORTHO DAILY QC. - Removed the example for Quality Control Result (Managed by VISION®). - Removed the example for Quality Control Result (Managed by LIS).
	3.6	Compatibility with ORTHO AutoVue® and ORTHO AutoVue® Ultra Systems	Added the note statement "Note: The order creation time is provided from the ORTHO VISION® System".
	4.1	BioVue Tests	Added tests 60065, 70003, 70025, 70029, 70033, 70037, 70041, 70045, 70049, 70053, and 70059.
	4.2	Cassettes	- Added DVI Newborn Cassette - Replaced ABO-DD(DVI) Donor Grp with DVI Grouping
	4.3	Reagents	Added AntiToxin reagents.

Version	Section Number	Section Title	Revision Details
	4.5	Sample Types	• Changed from 0.8CELLS to 0.8CELLS_R • Changed from 3CELLS to 3CELLS_R • Removed “AlbaQ-Chek J” and its liquid family.
2023-11-01	3.3	LIS Messages	• Before the “Profiles” section, added the following statement “Operators manually entering text that is included in LIS messages (patient names, comments, and other fields), must be supported by the configured LIS interface character encoding. Unsupported characters will not be converted correctly. Recommend using UTF-8 if issues occur”.
	3.4.8	Vision ASTM C Comment Record	• Added new section
	3.4.9.9	Result with 2 Samples	• All messages are replaced.
	4.1	BioVue Tests	• Removed Rh from Analyses column and Ctrl from Rh column for Test ID 10023
	4.5	Sample Types	• Added the following sample types “ORTHO DAILY QC, ORTHO DAILY QC JAPAN, and ORTHO DAILY QC SERUM JAPAN” with description as “QCFluid”.
2022-09-01	2.2.2	Data Elements	• Updated “Automatic Partial” description to match current software
	3.4.4	Vision ASTM O Order Record	• Modified description for “Number of Donor Samples”
	3.4.5	Vision ASTM R Result Record	• Added Abnormal Result Flag - Negative Reaction Threshold "N" flag
	3.4.8.1	ASTM Keep-Alive Message	• Added example for new Keep-Alive Message
	3.5.4	ASTM O Order Record	• Modified description for “Number of Donor Samples”
	3.5.8.1	ASTM Keep-Alive Message	• Added example for new Keep-Alive Message
	4.1	BioVue Tests	• Added tests 10289, 10290, 10291,

Version	Section Number	Section Title	Revision Details
			10292, 10462, 10463, 10464, 60002, 60007, 60008, 60025, 60029, 60033, 60037, 60041, 60045, 60049, 60053, and 60059
	4.2	Cassettes	Added ABD(DVI) Donor Conf and ABO-DD(DVI) Donor Grp
	4.3	Reagents	Added liquids Undefined and Undefined Decontamination.
	4.4	Unit Result Names	Added Anti-D(DVI) and Undefined
	N/A	Back Cover Page	Added Importer Symbol
2021-02-01	1.4	Definitions	- Added definitions for “Message” and “Partial Results / Partial Profile Results”. - Revised “A Profile is a name assigned to one or more tests.” to “A Profile is a name defining a group of one or more tests. A Profile is the smallest orderable unit.” for Profile definition. - Revised “A determination of a single analyte or a combination of values from other determinations of observations which constitute a measure of a single system attribute.” to “The smallest reportable unit within a Profile. Tests are selected from the test menu.” for Test definition.
	3.3	LIS Messages	- Revised “VISION® supports the following message formats (structure and content):” to “VISION® supports the following ORTHO ASTM message formats (structure and content):” - Revised “Enhanced ASTM -- The ORTHO VISION® System is capable of sending and receiving ASTM messages in the predecessor format.” to “Enhanced ASTM -- The ORTHO VISION® System can send and receive ASTM messages in the predecessor format.” - Revised “Each ABO R-record would

Version	Section Number	Section Title	Revision Details
			be followed by two M-records; one for Anti-A and one for Anti-B.” to “Each ABO R-record would be followed by two M-records: one for Anti-A and one for Anti-B.”
	3.4.8	Sample Vision ASTM Messages	All messages are replaced, and some new messages are added in its subsections.
	3.4.8.4	Example Result	Revised “The Order record report type (O.26) is set to ‘F’ for final results and the Result record Result Status field (R.6) is set to ‘F’ for final result.” to “The Order record report type (O.26) is set to ‘F’ for final results and the Result records Result Status fields (R.6) are set to ‘F’ for final result.” and “The Order record report type (O.26) is set to ‘R’ and the Result record Result Status field (R.6) is also set to ‘R’ for repeat results.” to “The Order record report type (O.26) is set to ‘R’ and the Result records Result Status fields (R.6) are also set to ‘R’ for repeat results.”
	3.4.8.8	Result with 2 Samples	- Revised in bullet 1 “40^M^0^soladmin” to “20^M^-10^soladmin”. - Revised in bullet 2 “The interpreted result for the Rh result was manually set; this is indicated by the Result record ‘Result Abnormal Flags field (R.7) containing the Manual flag (M) along with all the Rh Manufactures record’s ‘Manual Correction Flag’ set to ‘A’ (automatic) in field (M.6.2).” to “The interpreted result for the Rh result was manually set; this is indicated by the Result record ‘Result Abnormal Flags field (R.7) containing the Manual flag (M) along with the Rh Manufactures record’s ‘Manual Correction Flag’ set to ‘A’ (automatic) in field M.6.2.”

Version	Section Number	Section Title	Revision Details
			- Revised “All the results were manually accepted by ‘soladmin’; this is indicated by the Result records ‘Operator Identification’ field (R.11) set to the operator ID of ‘soladmin’ and not set to ‘Automatic.’” to “The ABO results were manually accepted by ‘soladmin’; this is indicated by the Result records ‘Operator Identification’ field (R.11) set to the operator ID of ‘soladmin’ and not set to ‘Automatic.’” - Added “The ABScr results were automatically accepted; this is indicated by the Result record ‘Operator Identification’ field (R.11) set to ‘Automatic.’”
	3.4.8.9	Order with Multiple Profiles	Revised “ORTHO VISION® will create six orders from the message above; one for each profile.” to “ORTHO VISION® will create six orders from the message above: one for each profile.”
	3.4.8.10	Profile with multiple tests	- Revised “Type & Screen” to “T&S”, “Uplode test name” to “Upload test name”, “When the same Ctrl well is listed under more than one analysis result, the M-records will be identical with the possible exception of the M-record sequence number.” to “When the same Ctrl well is listed under more than one analysis result, the M-records will be identical, with the possible exception of the M-record sequence number.” - Added “When sending a message that uses delimiters in a text field, the sender must escape the delimiter. The profile “T&S” is written as “T&E&S” in the order to escape the ‘&’ in the profile name (ASTM standard escape sequence). The ASTM standard specifies “T&E&S” or “T&X26&S” for escape sequences.”

Version	Section Number	Section Title	Revision Details
	3.4.8.11	Donor Confirmation Test	Revised “Therefore the well number fields (M.4.2) will be either 1-2-3 (above) or 4-5-6 (below).” to “Therefore, the well number fields (M.4.2) will be either 1-2-3 (above) or 4-5-6 (below).”
	3.4.8.13	Pheno Test	Added new section
	3.4.8.16	Quality Control Examples	- In 3.4.8.16.1, removed “BRC tests cannot be ordered by the LIS, however their results are uploaded to the LIS.” and added “Quality Control orders have the Order Action Code (O.12) set to Q.” and revised “See Cards and Reagents tables in section 4.” to “See Cassettes and Reagents tables in section 4.”. - In 3.4.8.16.2, revised “An LIS must include expected results in field O.14 when ordering Non-Ortho QC Sample IDs (result not known from barcode).” to “The LIS must include expected results in field O.14 when ordering non-Ortho QC Sample IDs (result not known from barcode). Expected results must be included even if expected results are configured in Vision.” - In 3.4.8.16.3, revised “A quality control order from the LIS is the same as any other order. Since barcode interpretation is turned off, the LIS order needs to contain a sample fluid type that aligns with the QC fluid; use ‘CENTBLOOD’ instead of ‘AlbaQ-Chek J’.

Version	Section Number	Section Title	Revision Details
			aligns with the QC fluid; use 'CENTBLOOD' instead of 'AlbaQ-Chek J'. Because the LIS is managing QCs, the action code in the order should not be set to 'Q' (treat specimen as Q/C test)." In 3.4.8.16.4, revised "BRC tests cannot be ordered by the LIS; however their results will be uploaded to the LIS." To "BRC tests cannot be ordered by the LIS, however their results are uploaded to the LIS."
	3.4.8.18	Include Operator is set to 'Yes'	- Revised "Below, user1 loaded the samples and reagents, the ABO results were automatically accepted, and user2 manually accepted the ABScr results." to "Below, user1 loaded the samples and reagents. The ABScr results were automatically accepted, and user2 manually accepted the ABO results". - Added "In some cases, SYSTEM will be returned as the operator."
	3.4.8.19	Partial Results	Added new section.
	3.5.8	Sample ASTM and Enhanced ASTM Messages	- All example messages are replaced. - Revised messages in Example Order, Example ASTM Result Message, Example Result, Crossmatch Order, and Crossmatch Result. - In 3.5.8.7 Crossmatch Result, revised "echos" to "echoes"
	4.1	BioVue Tests	- Removed the rows for Test ID's 10062, 10063, 10064, 10065, 10176 and removed "C3, IgG, Poly" column for Unit Results in the table. - Added new table to show unit results for Test ID's 10062,10063,10064,10065, 10176, 10262, 10263, and 10264.
	4.6	Analyses	Rearranged the rows of the table in alphabetical order.

Version	Section Number	Section Title	Revision Details
2018-10-01	N/A	N/A	Throughout the document, changed the following: • Consolidated / Combined “ORTHO VISION® Analyzer” and “ORTHO VISION® Max Analyzer”. • Changed “ORTHO VISION® Analyzer” and “ORTHO VISION® Max Analyzer” to “ORTHO VISION® System”. • Changed Header “ORTHO VISION® Analyzer”, “ORTHO VISION® Max Analyzer” and “ORTHO BioVue® System Cassettes” to “ORTHO VISION® (BioVue) System”.
	1.1	Purpose	• Added Note.
	2.2.2	Data Elements	• Added “or in multiple message files” in Host Query Mode.
	3.2.1	ASTM 1381	• Added a paragraph “The LIS has the option to reject the receiver interrupt and keep sending frames” in Receiver Interrupts.
	3.4.5	Vision ASTM R Result Record	• Updated in table #11 • Added #11.2 and #11.3
	3.4.8	Sample Vision ASTM Messages	• Added Note “All results are simulated”. • All messages are replaced. Some new messages are added. • All example messages are replaced. • Revised “W13131200096700” to “W13131200096100” in Host Query. • Replaced 3.4.7.15.2 • Added 3.4.8.15.3, 3.4.8.17 and 3.4.8.18
	3.4.8.9	Order with Multiple Profiles	• Changed from “VISION® will create two orders from the message above” to “VISION® will create six orders from the message above. All Profiles have the sample ID from their Order records and all Profiles inherit their patient demographics from their preceding Patient records.” • Revised Illustration.

Version	Section Number	Section Title	Revision Details
	3.4.8.10	Profile with Multiple Tests	- Deleted “If there were two or more tests with identical analysis types, there would be one R-record for each test. For example, if a profile was defined with two ABO tests (test IDs 10015 and 10019), the result message would contain” - Revised “(well 4)” to “(well 4 in M.4.2)”.
	3.4.8.11	Donor Confirmation	Revised “The “ABD Conf” profile is defined to use the “ABD Conf-10” test which consumes” to “The “Group+Rh” profile is defined to use the test 10007 “ABD Conf-10” which consumes”
	3.4.8.14	Cancelled Orders	Added “The LIS cannot cancel and order once is has started processing.”
	3.4.8.15.1	When VISION® is managing quality controls:	- Revised “Non-Quality Control Order with Cassette and reagent lots specified:” to “Non-Quality Control Order with Cassette and reagent lots specified:” - Revised “An LIS can send orders to VISION® and specify required Cassette and reagent lots” to “When VISION® is managing Quality Controls, an LIS can send Quality Control orders to VISION® and specify required Cassette and reagent lots” - Added “See Cards and Reagents tables in section 4.” - Revised “^00061” to “^00016”
	3.5.5	ASTM R Result Record	- Revised in table #11 - Added #11.2 and #11.3
	3.5.8	Sample ASTM and Enhanced ASTM Messages	- Revised messages in Example Order, ASTM Result Message, Example Result, crossmatch Result and BRC Results. - Added Crossmatch Order. - Added “Quality Control Order (Managed by VISION®): and Quality Control Result (Managed by LIS):”

Version	Section Number	Section Title	Revision Details
			with message in 3.5.5.16. • Changed “3.5.8.16.3” to “3.5.8.16.1”
	3.5.8.5	Example Result	• Replaced “Result with “Strip trailing delimiters” set to “No”.” by “Enhanced ASTM Result with “Strip trailing delimiters” set to “No”.”
	3.5.8.6	Crossmatch Order	• Added “Enhanced ASTM and ASTM crossmatch orders are the same as Vision ASTM.”
	4.1	BioVue Tests	• Updated ABO in Test ID 30258.
	4.3	Reagents	• Added “Vision does not guarantee that the list of reagents in an M-record will be reported in the same sequence everytime. The reagent name and not its position in the uploaded result message should always be used to access reagent lots and expiration dates. See 3.4.8.18 Results with multiple reagents for an example message.”

1.4 Definitions

AD	Application Data. Media that contains test or protocol rules that controls the ORTHO VISION® System when tests are processed.
Assay	Part of a Test that uses one column of the Cassette to analyze a Sample.
Cassette	Plastic container that contains six columns where reactions occur; Cassette is the informal name for the ORTHO BioVue® System Cassette.
Column	One of 6 microtubes of a Cassette where the reaction occurs.
Device	A specialized part of the ORTHO VISION® System that performs a function, such as centrifuge, incubator, or pipettor.
Download	Data transfer from the Laboratory Information System to the ORTHO VISION® System.
Error	1. A general type of error related to a handling strategy. 2. A specific error event related to a handling history.
Frame	The basic unit of communication at the Data Link Layer for ASTM protocol.
LIS	Laboratory Information System. In relation to the ORTHO VISION® System, the LIS is a computer system in the laboratory responsible for tracking sample orders and results.
Messages	ASTM 1381, section 3.3.2, defines a message as one ASTM 1394 Record. ASTM 1394, section 3.1.5, defines a message as one Header Record through a message terminator (L) Record.
OCD	Ortho Clinical Diagnostics.
Order	A request to perform a profile (group of tests) for samples of a single patient (and several donors).
ORTHO VISION® System	An instrument designed to automate in vitro Immunohematology testing of human blood.
Partial Results / Partial Profile Results	A result message containing tests results for some but not all of the tests in a profile. Partial results may be uploaded when the upload mode is configured for Automatic Partial or when an operator manually sends a partial result to the LIS. Partial results can only occur for Profiles defined with two or more tests. A result message will always contain complete set of test results. Test 1007 (10: ABD Conf-10) has two analyses: ABO and Rh. Result messages containing a test result for test 1007 will always contain the ABO and Rh results.
Profile	A Profile is a name defining a group of one or more tests. A Profile is the smallest orderable unit.
Sample	A tube of patient or donor blood.
Test	The smallest reportable unit within a Profile. Tests are selected from the test menu.
Upload	Data transfer from the ORTHO VISION® System to the Laboratory Information System.
User	The operator of the ORTHO VISION® System.

1.5 References

ASTM E 1381-02	ASTM E 1381-02 – Standard Specification for Low Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems
ASTM E 1394-97	ASTM E 1394-97 – Standard Specification for Transferring Information Between Clinical Laboratory Instruments and Computer Systems
LIS1-A	ANSI/CLSI LIS1-A– Standard Specification for Low Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems
LIS2-A	ANSI/CLSI LIS2-A – Specification for Transferring Information Between Clinical Laboratory Instruments and Information Systems

2. External Interface Design

2.1 Overview

The Laboratory Information System (LIS) interface is used to communicate between the ORTHO VISION® System and a LIS.

2.2 The LIS Interface

The ORTHO VISION® System implements an interface to a remote LIS connected through TCP/IP, RS232, or shared folders.

2.2.1 Purpose

The LIS Interface simplifies the management of multiple ORTHO VISION® Systems. Analysis results can be managed centrally.

2.2.2 Data Elements

The LIS Interface supports four working modes: Upload, Download, Host Query and Broadcast.

Upload Mode – Also called unidirectional mode. Tests are manually ordered on the system. Results are transferred to a connected LIS. The user cannot release unaccepted results to the LIS; the user can manually resend accepted individual results to the LIS.

With this mode, the user can select one of the following LIS result transmission options:

- Disabled – The result transmission is disabled.
- Manual – Accepted results are only transmitted upon user request. The user can release accepted results to the LIS.
- Automatic – The results of an order are transmitted when all the results of the order are available, and all the results have been automatically accepted. (Manually accepted results are only transmitted upon user request.)
- Manual with Auto Upload – The results of an order are transmitted when they are manually accepted.
- Automatic Partial – Automatically accepted test results are transmitted upon availability. Manually accepted test results are transmitted on acceptance of the test result.

Note: A result is only automatically accepted if all results of an order using the same test analysis are consistent, for example, are available.

Download Mode – Also called bidirectional mode. This functions the same as Upload Mode, additionally, the LIS can transmit orders to the ORTHO VISION® System. As soon as all required resources such as samples and reagents are available, the order is processed.

Host Query Mode – Functions the same as Download Mode, additionally, during an inventory check, if the instrument detects a sample without an order, the instrument sends a host query to the LIS for that sample. If the LIS does not respond, the instrument resends the query based on configuration settings (see details below). The response becomes a normal order ([See Download Mode](#)). The system immediately queues one ASTM 1394 message per sample without an order (H-Q-L). These messages are sent to the LIS in one or more ASTM 1381 sessions, or in multiple message files. The sequence of messages will be repeated for all samples without an order up to the configured retry value.

When enabled, VISION® will send Host Query messages to the LIS:

- Query messages are only sent when an order is not associated with a sample on VISION®. If an order comes back from the LIS, VISION® will stop sending repeated query messages for that sample.
- VISION® sends one query message per sample every 30 seconds for a maximum of 11 messages per sample. VISION® can be configured to resend query messages anywhere from 30-999 seconds and resend 1-11 query messages per Sample ID.
- If the LIS receives more than one query message for a sample and responds to each one, sending VISION® duplicate orders; VISION® will treat each duplicate as an additional order and will process them all. The resend interval should be set in VISION® to allow the LIS enough time to respond.

Broadcast mode (for multiple instrument configurations) - The LIS sends orders to all instruments. When a sample is loaded and processed on one instrument, the instrument uploads the results. When the LIS receives the results from one instrument, the LIS may cancel the duplicate orders on all other instruments. The system can also be configured to automatically cancel orders that have not run from 1-99 hours. This mode is based on Download Mode.

The functionality of Download (Bidirectional) Mode supports all other modes. When the VISION® LIS interface is enabled, VISION® will be in Download Mode. Host Query mode can be enabled or disabled. All other modes are based on how the LIS interacts with VISION®.

3. External Interface Communication Protocols

The ASTM layers involved in transferring data from the ORTHO VISION® System to the LIS and from the LIS to the ORTHO VISION® System are divided into these components:

Physical Layer – The physical layer is comprised of the actual hardware and software configuration used to communicate between the two systems.

- For serial communications, this corresponds to Section 5 in ASTM E 1381.
- For Ethernet communications, this corresponds to Sections 7 and 8.2.1.1 in ASTM E 1381.
- For file sharing, this corresponds to network file sharing protocols that are supported by the ORTHO VISION® System.
- For specifications, refer to the appropriate Interface (Ethernet, Serial, or File Sharing).

Datalink Layer – The Datalink layer is responsible for the logical data frames comprised of the raw data exchanged with the physical layer. It ensures that packets can be transferred accurately between computers.

- For serial communications this corresponds to the Datalink layer specified in Section 6 in ASTM E 1381.
- For Ethernet communications, this corresponds to Section 8 in ASTM E 1381.
- For file sharing this corresponds to custom protocols implemented internally by the ORTHO VISION® System.

3.1 Physical Layer

The ORTHO VISION® System can be configured to communicate to the remote LIS by either TCP/IP, Serial RS232 or File sharing interface.

3.1.1 Physical TCP/IP

The ASTM protocol supports the use of TCP/IP communication for the physical transport layer. The ORTHO VISION® System establishes a connection to the LIS at a configured IP address and port. This interface is bidirectional but is only initiated by the ORTHO VISION® System. Under the client/server model, the LIS acts as the server and the system acts as the client (LIS1-A, section 8.2.1.1).

3.1.2 Physical Serial Interface

The ASTM protocol described in ASTM E 1381 supports the use of serial communication (RS232) for the physical transport layer.

The System uses 1 start bit.

The System supports 1 or 2 stop bits.

The System supports EVEN, ODD, NONE, SPACE and MARK parity.

The System supports the following baud rates: 300, 600, 1200, 2400, 4800, 9600, 19200, 38400, 57600, and 115200.

The System uses 8 data bits.

Access to the serial connection is provided by a DB9 male connector (plug). A cable with a DB9 female connector (socket) will be required to connect to the serial port. Some sites may require a null-modem adapter or cable.

Serial (RS232) Connector			
DB9 Connector	Pin	Signal	Name
	1	DCD	Data Carrier Detect
	2	RXD	Receive Data
	3	TXD	Transmit Data
	4	DTR	Data Terminal Ready
	5	GND	Signal Ground
	6	DSR	Data Set Ready
	7	RTS	Request to Send
	8	CTS	Clear to Send
	9	RI	Ring Indicator

3.1.3 Physical File Sharing Interface

While the LIS1-A/ASTM E 1381 standard specifies the low-level serial and TCP/IP protocols for the exchange of ASTM messages, the LIS2-A/ASTM E 1394 standard applies only to the structure of ASTM messages and does not specify what communication protocols can be used. Systems are free to exchange ASTM messages using any defined communication protocol.

With the file sharing interface, ASTM messages are exchanged using message files and shared folders. An application sends an ASTM message by writing the message to a file in a shared folder. An application receives an ASTM message by first reading a file from a shared folder followed by deleting the file once it has been read. File naming conventions and the use of separate upload and download folders are methods used to avoid conflicts when exchanging messages.

The ORTHO VISION® System is configured to use folders that are shared on the network for the exchange of messages, sharing of Cassette images, and a message trace folder for troubleshooting.

The LIS sends an ASTM message to the ORTHO VISION® System by writing the message to a file in the download folder. The ORTHO VISION® System sends an ASTM message to the LIS by writing the message to a file in the upload folder.

The ORTHO VISION® System deletes all files in the download and upload folders that are older than 7 days.

The LIS can read Cassette images associated with results from the ORTHO VISION® System's shared image folder. When the LIS connection is started, all images that are older than 7 days are deleted. After startup, images that are older than 7 days are deleted periodically.

The ORTHO VISION® System deletes trace files that are older than 7 days and that are stored in the shared folder when the LIS connection is started. After startup, trace files that are older than 7 days are deleted periodically.

The upload and download folders can be combined into the same folder. When using a combined folder, care must be taken to ensure that the file patterns for upload and download do not overlap (see details below).

The upload, download, and trace shared folders can reside on the ORTHO VISION® System or on the LIS or another system. The images shared folder resides on the system. Remote shared folders are accessed by ORTHO VISION® System through a UNC path, such as:

`\\ServerComputerName\ShareName`

Where ServerComputerName is the WINS name, DNS name or IP address of the server computer, and ShareName is the shared folder name, or its path. The LIS can access the ORTHO VISION® System shared folders using UNC paths, as shown in the example below. The actual folder names are configurable when the ORTHO VISION® System is set up.

Example:

```
\\ SystemName\download
\\ SystemName\upload
\\ SystemName\traces
\\ SystemName\images
```

Where SystemName is the WINS name, DNS name or IP address of the server computer, and download, upload, traces, and images are the shared folder name. If the ORTHO VISION® System is configured to share a download folder on the system located at C:\Lan\upl, the UNC path will be:

\\ SystemName \C_Lan_upl

Errors can occur if one computer is writing to a message file while the other computer tries to read it. To avoid such conflicts, the LIS and the ORTHO VISION® System first write an ASTM message file to a temporary file in the designated shared folder. The ORTHO VISION® System creates temporary files named tmp-<timestamp>.tmp (for example, tmp-20131018143212.tmp). Once the complete ASTM message has been written to the file and the file has been closed, the temporary file is renamed to its final name. Configured file names and patterns should not match/overlap the temporary file pattern used by the LIS or ORTHO VISION® System.

Because of limitations with the Windows WIN32 API, LIS temporary file names/patterns should not contain a substring that matches the final file name. The file extension .tmp is reserved for use by the ORTHO VISION® System in the upload or combined folder.

File naming and the use of separate upload and download folders are methods used to avoid conflicts when exchanging messages. There are 4 types of file names the LIS and VISION® will create:

- VISION® creates an intermediate upload (result/query) file (for example tmp-20131018143212.tmp).
- VISION® renames the intermediate upload (result/query) file to the next upload file name using the upload file pattern.
- LIS creates an intermediate download (order) file.
- LIS renames the intermediate download (order) file to the next download file name based on the download file pattern.

A message file must be completely written before the other system reads it. When separate upload and download folders are used:

- Upload file patterns should never match/overlap the VISION® intermediate upload file pattern.
- Download file patterns should never match/overlap the LIS intermediate download file pattern.

When using a combined folder:

- Upload and download file patterns should never match/overlap the VISION® intermediate upload file pattern.
- Upload and download file patterns should never match/overlap the LIS intermediate download file pattern.
- The LIS intermediate download file pattern should never match/overlap the VISION® intermediate upload file pattern.

When using a combined folder, care must be taken to ensure that the file patterns do not overlap. A mixture of two messages can occur if one system is writing to a message file while the other computer tries to read the message.

3.2 Datalink Layer

The VISION® instrument supports the following communication protocols:

- ASTM E 1381 over Serial RS232
- ASTM E 1381 over TCP/IP
- Network shared folders

3.2.1 ASTM 1381

The ASTM 1381 and LIS1-A standards defines the low-level protocol used for both serial and TCP/IP data exchange between laboratory instruments and information systems.

Maximum frame length -- For backward compatibility, the ORTHO VISION® System supports configuring the maximum length of an ASTM frame. ASTM records (plus overhead) bigger than maximum configured record length are sliced into intermediate frames. Some LIS contain hard coded maximum frame length limits when reading upload messages.

The ORTHO VISION® System supports a configurable maximum frames size (8...64000) that supersedes values specified in ASTM E 1381.

Multiple messages per session – VISION® can send one or more ASTM 1394 messages in one ASTM 1381 session.

The ASTM 1381 standard defines a message as one ASTM 1394 record, whereas the ASTM 1394 standard defines a message as one header (H) record through a message terminator (L) record.

The Message Terminator Record (L) is the last record in the message; “A header record may be transmitted after this record signifying the start of a second message” (ASTM 1394, section 13.1).

An ASTM 1381 session is a total unit of communication activity, used to indicate the events starting with the establishment phase (ENQ) and ending with the termination phase (EOT). The ASTM 1381 standard does not restrict sending only one ASTM 1394 message per ASTM 1381 session.

The system will send one message when it first becomes available. However, when the system has more than one message to send, the system will send multiple ASTM 1394 messages in one ASTM 1381 session.

Example showing multiple messages in one session:

```
VISION: <ENQ>
LIS   : <ACK>
VISION: <STX>1H|\^&|||OCD^VISION^ ... <CR><ETX>71<CR><LF>
LIS   : <ACK>
VISION: <STX>2P|1| ... <CR><ETX>71<CR><LF>
LIS   : <ACK>
VISION: <STX>30|1| ... <CR><ETX>71<CR><LF>
LIS   : <ACK>
VISION: <STX>4R|1| ... <CR><ETX>71<CR><LF>
LIS   : <ACK>
VISION: <STX>5M|1| ... <CR><ETX>71<CR><LF>
LIS   : <ACK>
VISION: <STX>6L|1|N<CR><ETX>07<CR><LF>
```

```

LIS : <ACK>
VISION: <STX>7H|\^&|||OCD^VISION^ ... <CR><ETX>71<CR><LF>
LIS : <ACK>
VISION: <STX>0P|1| ... <CR><ETX>71<CR><LF>
LIS : <ACK>
VISION: <STX>10|1| ... <CR><ETX>71<CR><LF>
LIS : <ACK>
VISION: <STX>2R|1| ... <CR><ETX>71<CR><LF>
LIS : <ACK>
VISION: <STX>3M|1| ... <CR><ETX>71<CR><LF>
LIS : <ACK>
VISION: <STX>4L|1|N<CR><ETX>07<CR><LF>
LIS : <ACK>
VISION: <EOT>

```

Contention -- When both the LIS and VISION® simultaneously send an <ENQ>, the data link is in contention (LIS1-A, sections 6.2.7.1).

Contention is resolved as follows (LIS1-A, sections 6.2.7.1 and 8.2.7.1):

1. The LIS stops trying to send and prepares to receive.
2. The LIS waits at least 20 seconds to allow VISION® to send an <ENQ>.
3. VISION® waits 1 second before sending an <ENQ> to allow the LIS to prepare to receive.
4. After 1 second, VISION® sends <ENQ>.

Example showing contention resolution:

```

VISION: <ENQ>
LIS : <ENQ>
VISION: <ENQ> - After 1 second delay
LIS : <ACK>
VISION: <STX>1H|\^&|||OCD^VISION^ ... <CR><ETX>71<CR><LF>
LIS : <ACK>
VISION: <STX>2P|1| ... <CR><ETX>71<CR><LF>
LIS : <ACK>

```

Receiver Interrupts -- VISION® will generate Receiver Interrupts when the LIS is sending messages and VISION® has a message to send. After a frame is sent to VISION®, VISION® will reply with an <EOT> in place of an <ACK>, signifying that the frame was received successfully, and that VISION® is ready to receive another frame; but VISION® is also requesting to interrupt the LIS transfer to send its own messages. The LIS can either accept the interrupt or simply treat the <EOT> as an <ACK> to the frame and continue sending (LIS1-A, sections 6.3.5.2 and 8.3.5.2).

The LIS has the option to reject the receiver interrupt and keep sending frames. If the LIS decides to accept the receiver interrupt, then when the LIS returns to the idle state, it should request to send and resend the message that it failed to completely send when it was interrupted. The LIS should not accept the receiver interrupt if it cannot resume by resending the message that was interrupted.

3.2.2 ASTM Datalink Serial Communication

The ORTHO VISION® System supports the Establishment Phase (Link Connection) specified in Section 6.2 of ASTM E 1381. This includes establishment and contention.

The ORTHO VISION® System supports the Transfer Phase specified in Section 6.3 of ASTM E 1381. This includes:

- Frame Format in Section 6.3.1 of ASTM E 1381
- Frame Numbering in Section 6.3.2 of ASTM E 1381
- Frame Checksums in Section 6.3.3 of ASTM E 1381
- Frame Acknowledgements in Section 6.3.4 of ASTM E 1381
- Frame Receiver Interrupts in Section 6.3.5 of ASTM E 1381

The ORTHO VISION® System supports the Termination Phase (Link Release) as specified in Section 6.4 of ASTM E 1381.

The ORTHO VISION® System supports Error Recovery specified in Section 6.5 of ASTM E 1381. This includes:

- Detecting and handling defective frames in Section 6.5.1 of ASTM E 1381
- Timeouts in Section 6.5.2 of ASTM E 1381

The ORTHO VISION® System supports the Restricted Message Characters requirement specified in Section 6.6 of ASTM E 1381.

3.2.3 **ASTM Datalink Ethernet Communications**

The ORTHO VISION® System establishes a connection to the LIS at a configured IP address and port as specified in Section 8.2 of LIS1-A. This interface is bidirectional and is only initiated by the ORTHO VISION® System.

The System supports the Establishment Phase (Link Connection) specified in Section 6.2 of ASTM E 1381. This includes establishment and contention.

The ORTHO VISION® System supports the Transfer Phase specified in Section 6.3 of ASTM E 1381. This includes:

- Frame Format in Section 6.3.1 of ASTM E 1381
- Frame Numbering in Section 6.3.2 of ASTM E 1381
- Frame Checksums in Section 6.3.3 of ASTM E 1381
- Frame Acknowledgements Section 6.3.4 of ASTM E 1381
- Frame Receiver Interrupts in Section 6.3.5 of ASTM E 1381

The ORTHO VISION® System supports the Termination Phase (Link Release) specified in Section 6.4 of ASTM E 1381.

The ORTHO VISION® System supports Error Recovery specified in Section 6.5 of ASTM E 1381. This includes:

- Detecting and handling defective frames in 6.5.1 of ASTM E 1381
- Timeouts in Section 6.5.2 of ASTM E 1381

The ORTHO VISION® System supports the Restricted Message Characters requirement specified in Section 6.6 of ASTM E 1381.

3.2.4 ASTM Datalink Folder Communications

At startup if shared folder communications are enabled, the ORTHO VISION® System validates the format of configured ASTM message file names and verifies that the ORTHO VISION® System download and upload folders exist and can be written to. The upload format must not end with “.tmp” or “*”. The ORTHO VISION® System communicates any error conditions to the operator.

A **download file pattern** is defined for the ORTHO VISION® System during system configuration and is used to monitor the download folder for message files. A download file pattern can be a file name or a file name pattern. Valid file names are file names that are valid for both the LIS and the ORTHO VISION® System. A download file name can contain any number of alphanumeric characters, underscores, and the dot character (‘.’), up to the maximum file name length.

A download file name pattern can contain any number of alphanumeric characters, underscores, the dot character (‘.’), or wildcard characters (‘*’, ‘?’), up to the maximum file name length. The file pattern ??? .dnl is valid.

The maximum length of an upload or a download file name is 30 characters.

The following are acceptable wildcard characters:

A question mark (?) matches one character.

An asterisk (*) matches zero or more characters.

Note: An asterisk * at the directory level matches all files in the directory and, therefore, should never be used by itself. It will not only match the download files, but also match all the files in the directory.

Note: The ORTHO VISION® System download file name/pattern is case sensitive; a file named LIS.DNL is considered a different file from lis.dnl.

Examples:

If the LIS writes a fixed file name, use that file name for the download file pattern: LIS writes LIS.dnl. Use LIS.dnl for the file pattern.

If the LIS writes a fixed length file name with some varying characters, use the question mark (?) for the characters that vary. LIS writes LIS01.dnl, LIS02.dnl, LIS03.dnl and so on. Use LIS?? .dnl as the pattern, which mean the word LIS followed by two characters followed by the .dnl extension.

If the LIS writes a varying length file name, use a question mark (?) and an asterisk (*) for the characters that vary. LIS writes LIS1.dnl ..., LIS9.dnl, on up to LIS9999.dnl, use LIS?* .dnl, which means the word LIS followed by one or more characters with a .dnl file extension.

When a matching file is present in the download folder, ORTHO VISION® System does the following:

1. Copies the download file to a trace file in the folder "\\< system computer name>\TRACES", (DNL<YYYYMMDDHHMMSS>.TRA). The TRACES folder is written to only by the ORTHO VISION® System.

2. Reads the ASTM message file and import orders.
3. Deletes the ASTM message file.

Upload and query file patterns are defined for the ORTHO VISION® System with choices on the Setup software screen during system configuration; the file patterns are used to create upload message files that the LIS can recognize. Upload and query file patterns can be regular file names or file name patterns. Valid file names must be valid for both the LIS and the ORTHO VISION® System. Upload and query file names can contain any number of alphanumeric characters, underscores, and the dot character ('.'), up to the maximum file name length. Upload and query filename patterns can contain any number of alphanumeric characters, underscores, the dot character ('.'), question marks ('?'), or asterisk (*), up to the maximum file name length. One or more consecutive question mark characters ('?') means a sequence of filenames where each file contains a sequence number that is incremented for each file. For example, if "LIS???.upl" is the file pattern, then files are named LIS001.upl through LIS999.upl. All question mark characters must be together in the file pattern (the pattern "LIS??ABC???.upl" is invalid). The file pattern???.upl is valid.

An asterisk (*) is used for a timestamp (YYYYMMDDhhmmss) with a resolution of one second. An upload pattern of LIS???.*.upl generates an upload file such as LIS004-2013103170016.upl.

Because the ORTHO VISION® System can generate more than one upload file per second, a timestamp without a sequence number could cause messages to be lost. With the exception of the last upload message (within the same second), all messages would be overwritten. The ORTHO VISION® System prevents messages from being overwritten by not overwriting existing messages and generating an error "Apsw29: Unable to communicate with the LIS".

To prevent errors when an upload file pattern contains a timestamp without a sequence number, the ORTHO VISION® System appends a sequence to the end of the pattern. An upload pattern of *.upl is equivalent to *???.upl. An upload pattern of LIS*.upl generates an upload file such as LIS2013103170016004.upl, where the last 3 digits is a sequence number.

Error "Apsw29: Unable to communicate with the LIS", will occur if the Host Query File Pattern contains a fixed filename (such as: Query.upl) and there is more than one sample without orders in the sample racks.

The ORTHO VISION® System completes these steps to write an upload file:

1. Check if a file with the same file name is present, if yes, then the ORTHO VISION® System communicates an error condition to the operator. This condition might indicate that the LIS is not reading message files.
2. If the file is not present, write the message to a temporary filename so that the filename extension does not contain a substring that matches the upload file pattern extension.
3. Close the file.
4. Copy the file to a trace file in the folder "\\<system computer name>\TRACES", (UPL<YYYYMMDDHHMMSS>.TRA or QU<??-YYYYMMDDHHMMSS>.TRA).
5. Rename the file to its final name, so that the LIS will recognize the file only after it has been fully written and closed.

The LIS deletes the file after reading it.

3.3 LIS Messages

This section describes the structure and content of messages (orders, results, and queries) exchanged between an LIS and VISION®. The ASTM E 1394-97 and LIS2-A standards describe the general message structure and content, whereas this guide describes VISION® implementation of these standards.

VISION® supports the following ORTHO ASTM message formats (structure and content):

- Vision ASTM
- Enhanced ASTM
- ASTM

Vision ASTM -- Vision ASTM is the most recent ASTM version. Vision ASTM is a super set of the ASTM and Enhanced ASTM message formats.

Enhanced ASTM -- The ORTHO VISION® System can send and receive ASTM messages in the predecessor format. The ASTM mode for the system must be explicitly set to Enhanced ASTM to enable this backward compatibility mode. Some sources also call this format “AV2G” (AutoVue 2nd Generation).

ASTM -- This format, another predecessor ASTM format, is identical to the Enhanced ASTM format with one exception. The ASTM format does not support manufacturer records. If ASTM mode is enabled, no manufacturer records are sent by the ORTHO VISION® System (contains well grading results).

All of these message formats are specific to Ortho Clinical Diagnostics (Ortho) instruments and are based on the ASTM E 1394-97 and LIS2-A standards.

Comment, Result, Request Information, Scientific and Manufacturer Information record types are accepted and ignored on download. In the record field descriptions, all values are text, and meet the ASTM Common Field Types specification Section 6.6 of ASTM E 1394. If a field value has multiple components, they are differentiated by the component delimiter ('^') character Section 6.4.5 of ASTM E 1394.

The ORTHO VISION® System supports the hierarchical message structure as defined in Sections 5.1.8 through 5.1.11 of ASTM E 1394.

The ORTHO VISION® System does not support the Logical Information Storage and Logical Transmission Error Recovery Requirements defined in Sections 5.2.1 and 5.2.2 of ASTM E 1394.

The ORTHO VISION® System uses the following record specifications:

- For all records, fields not supported are ignored on a download.
- The ORTHO VISION® System uploads null values for any field listed as *Not supported* for all records.

The ORTHO VISION® System can transmit and receive messages using

- Unicode/UTF-8
- ISO8859/1 (ISO Latin-1)
- Windows-31J (Microsoft Shift-JIS CP932)
- Windows 1252 Code Page

The character encoding is configurable.

Only one character encoding is supported at a given time.

Operators manually entering text that is included in LIS messages (patient names, comments, and other fields), must be supported by the configured LIS interface character encoding. Unsupported characters will not be converted correctly. Recommend using UTF-8 if issues occur.

Profiles – A Profile is a name assigned to one or more tests. VISION® does not enforce a standard set of Profile names and does not have defined default Profiles. Profiles are defined based on the unique requirements of each site; therefore, the name given to a profile can be instrument, site or LIS specific.

- A profile can only contain unique test names; one cannot put a test in a profile twice. A profile cannot be defined to run the “00: ABO(FWD)/Rh-00” test twice.
- A profile named “Type & Screen” can be defined with one or more tests such as “00: ABO(FWD)/Rh-00” and “22: 08 AbScr Sel Poly”.

Trim trailing null fields

The ORTHO VISION® System is configurable to include or trim delimiters for all trailing null fields (see Section 6.4.8 of ASTM E 1394). This configuration setting is ignored for the L record in Enhanced ASTM and ASTM message formats. The L record is always sent without trailing fields in Enhanced ASTM and ASTM. Some LIS systems will reject upload messages if a specific number of “pipes” (field delimiters) is not contained in each record. However, the system is capable to receive records where trailing null fields were omitted.

The ORTHO VISION® System uploads records with the following number of delimiters. Note that all field delimiters are the same as the number of fields specified in ASTM E 1394.

H records	13 field delimiters
P records	34 field delimiters
O records	30 field delimiters
R records	13 field delimiters
Q records	12 field delimiters
M record	5 field delimiters

Escape Sequences – Escape sequences are used to represent certain special characters within text fields of ASTM messages. Some characters are not allowed in ASTM text fields and escape sequences are used to convert them into a valid sequence of characters. The sender of an ASTM message converts restricted characters to an escape sequence and the receiver of the message converts escape sequences back to the original text.

If the ORTHO VISION® System is configured for the ASTM escape sequences, then the ORTHO VISION® System accepts and ignores the following escape sequences:

&H&	Start highlighting text
&N&	Normal text (end highlighting)
&Zcccc&	Local (manufacturer) defined escape sequence

If the ORTHO VISION® System is configured for ASTM escape sequences, then the ORTHO VISION® System accepts and supports the following escape sequences:

&F&	Embedded field delimiter character
&S&	Embedded component field delimiter character
&R&	Embedded repeat field delimiter character
&E&	Embedded escape delimiter character
&Xhhhh&	hexadecimal data (e.g., &XA& is a linefeed character)

When configured for ASTM escape sequences:

- 'Type & Screen' should be written to an ASTM message as 'Type &E& Screen' or 'Type &X26& Screen'
- 'Type &E& Screen' should be read from an ASTM message as 'Type & Screen'
- 'Type &X26& Screen' should be read from an ASTM message as 'Type & Screen'

If the ORTHO VISION® System is not configured for ASTM escape sequences, then the ORTHO VISION® System accepts and supports the following escape sequences:

&	Embedded field delimiter character
&^	Embedded component field delimiter character
&\	Embedded repeat field delimiter character
&&	Embedded escape delimiter character

When not configured for ASTM escape sequences:

- 'Type & Screen' should be written to an ASTM message as 'Type && Screen'
- 'Type && Screen' should be read from an ASTM message as 'Type & Screen'

Date fields -- All date fields are formatted as specified by Dates and Times defined in Section 6.6.2 of ASTM E 1394. The ORTHO VISION® System rejects Date and Time values downloaded with an appended Time Zone value as defined in Section 6.6.2.1 of ASTM E 1394.

Note: Valid dates range from 1/1/1753 and beyond.

Sequence of records in result messages -- The sequence of records (O-R-M) in a result message is dependent on the definition of the profile. The sequence of records in a result message will always be the same for each result message for a profile. If the profile definition is changed, their order may also change.

Results with multiple reagents -- Vision result messages include a list of reagents which include the name, lot, and expiration dates of the reagents used by an assay. LIS software should always use the name of the reagent when collecting lot and expiration information. Doing so will protect the LIS from any variations and from any future changes to Vision software.

Vision does not guarantee the list of reagents used by an assay will be reported in a specific sequence.

The reagent name and not its position in the uploaded result message should always be used to access reagent lots and expiration dates.

Analysis interpretations -- A Profile can be defined with more than one test, but analysis interpretations are performed on one test at a time and only with the wells from that test. An analysis interpretation may use one or more well results. All wells that are used in the interpretation of an analysis result are listed in M records following the analysis interpretation result (R) record. A single Ctrl well result may participate in more than one analysis interpretation and will be listed for each one. This means that the same Ctrl well may be listed more than once in a result message.

When the same Ctrl well is listed under more than one analysis result, the M-records will be identical with the possible exception of the M-record sequence number. The M-record sequence number is specific to its position under each R-record.

Well identification -- The well number and Cassette can be identified by the information in the M-record. The M-Record contains the Cassette name, Well Number (1-6), Cassette ID, Cassette Lot Number, and the Cassette Expiration Date. All of these components uniquely identify the Cassette and well result. One can determine if two M-records describe the same well by comparing the Cassette information and well numbers.

Multi-test Profiles -- Profiles that are defined with more than one test may produce result messages with multiple, identically named analysis and well results. These analyses and well results will be specific to each test. For example, a Profile defined with two tests, each with an ABO analysis and each ABO analysis result containing Anti-A and Anti-B unit results. This would produce a result message with two R-records, one for each ABO analysis. Each ABO R-record would be followed by two M-records: one for Anti-A and one for Anti-B. There is nothing in the R-record or the M-record that identifies the test that they are associated with. Tests names can be included in the Result and Manufacture records when configured for Vision ASTM format and "Upload test name" is enabled.

ISBT donor barcodes -- When VISION® scans an ISBT donor barcode, VISION® stores the complete ISBT barcode, including the "=" sign and Flags.

When VISION® compares an ISBT donor barcode to a donor ID from an order (manually entered or downloaded from an LIS), it compares it to the complete ISBT barcode (= W13131200096800). An ISBT donor ID downloaded to VISION® in an order from an LIS must also contain the complete ISBT barcode. If an operator has to manually enter a donor barcode and cannot use the handheld barcode scanner, then the operator will have to enter the complete ISBT sample barcode (=W13131200096800).

VISION® includes the complete ISBT barcode in results uploaded to the LIS.

Note: VISION® has a configurable option to strip the '=' and the trailing flag characters from an ISBT barcode when it is displayed on the user interface and reports. This configuration setting does not affect the ISBT barcodes in LIS messages; LIS order and result messages must always contain the complete ISBT barcode.

3.4 Vision ASTM

ASTM records are the components that make up ASTM Messages. In the tables that show the field sequences, **shaded rows** indicate fields that are not supported by the ORTHO VISION® System. No data is sent in these field positions, and they are marked *Unused*.

Yellow rows mean that there is a difference between Vision ASTM and Enhanced ASTM.

The column labeled “#” contains the field index within the record. An ASTM field can contain multiple ASTM components separated by the repeat delimiter. Components are indexed as X.Y, where X is the field index and Y is the index of the component within the field. “D” and “U” indicate if the fields/components are optional in the record when downloading to the ORTHO VISION® System (D) and uploading to the LIS (U). “R” indicates the repeatability.

Table 1 can be used as a key for the values in the D, U, and R columns.

Table 1: Abbreviations used in record tables

D	U	R
R = Required Field X = Required Field for cross match test Q= Only applies to QC Orders	A = Always Sent X = Always Sent for cross match test	Y=Field can repeat
O = Optional Field	S = Sometimes Sent P = When Provided	N= Field does not repeat
N = Never Used	N = Never Sent	-
- NA = not applicable	- NA = not applicable	-

In the record tables, the values for D, U, and R are included in three horizontal rows below the column headings.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				

The column “Field” contains the field name (*cf.* also [2]). The column “Notes” contains descriptions and restrictions for the field

The ORTHO VISION® System ignores incoming Comment, Result, Request Information, Scientific, and Manufacturing information record types.

Delimiters are variable on download and defined in the message header (see Section 6.4 of ASTM E 1394). In the following, the ORTHO VISION® System upload delimiters are used for the purpose of describing escape sequences.

3.4.1 Vision ASTM H Header Record

The ORTHO VISION® System supports the following fields (not shaded) of the ASTM Header (H) Record. Defined in Section 7.1 of ASTM E 1394.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; x: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
1	R	A	N	Record Type ID	= "H" or "h"
2	R	A	N	Field Delimiters	= " ^&", these are the Field, Repeat, Component and Escape Delimiters. All delimiters are fixed on upload. All delimiters can be variable on download.
3	N	N	N	Message Control ID	- <i>Unused</i>
4	N	N	N	Access Password	- <i>Unused</i>
5	O	A	N	Sender Name/ID	System Name field from system configuration
5.1	-	A	N	-	= "OCD", manufacturer name
5.2	-	A	N	-	= "VISION", product name
5.3	-	A	N		Software Version
5.4	-	A	N		Instrument ID
6	N	N	N	Sender Street Address	- <i>Unused</i>
7	N	N	N	Reserved Field	- <i>Unused</i>
8	N	N	N	Sender Telephone Number	- <i>Unused</i>
9	N	N	N	Characteristics of Sender	- <i>Unused</i>
10	N	N	N	Receiver ID	- <i>Unused</i>
11	N	N	N	Comment	- <i>Unused</i>
12	N	A	N	Processing ID	= "P" P-Production message
13	N	A	N	Version Number	ASTM protocol version "LIS2-A"
14	O	A	N	Date and Time of Message	Date and Time of transmission YYYYMMDDHHMMSS

3.4.2 Vision ASTM L Trailer Record

The ORTHO VISION® System supports the following fields (not shaded) of the ASTM Trailer (T) Record. Defined in Section 13.1 of ASTM E 1394.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; x: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
1	R	A	N	Record Type ID	= "L" or "I"
2	N	N	N	Sequence number	- <i>Unused</i>
3	N	N	N	Termination Code	- <i>Unused</i>

3.4.3 Vision ASTM P Patient Record

The ORTHO VISION® System supports the following fields (not shaded) of the ASTM Patient (P) Record. Defined in Section 8.1 of ASTM E 1394.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
1	R	A	N	Record Type ID	= "P" or "p"
2	O	A	N	Sequence number	Patient sequence number. Set to 1 for the first patient record; 2 for the 2 nd record; and so on.
3	O	P	N	Practice Assigned Patient ID	= Patient ID If present, the unique identifier for the patient. A Patient ID may contain one or more alphanumeric, symbols, and embedded blank characters up to a maximum length of 20 characters. Leading zeros in a Patient ID are retained and stored as part of the Patient ID. Patient IDs composed of only blank characters are treated as NULL. The System strips leading and trailing blanks. Embedded blanks are not removed and are considered as part of the Patient ID.
4	N	N	N	Lab Assigned Patient ID	- <i>Unused</i>

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
5.1	O	S	N	Patient ID No. 3	= National ID
5.2	O	S	N	-	= Medical Record
5.3	O	S	N	-	= Other ID
6.1	O	S	N	Patient Name	= Last Name
6.2	O	S	N	-	= First Name
6.3	O	S	N	-	= Middle Initial
6.4	N	N	N	-	= Suffix
6.5	N	N	N	-	= Title
7	O	P	N	Mother's Maiden Name	= Mother's Maiden Surname. May be required to distinguish between patients with the same birth date and last name.
8	O	P	N	Birth date	= Actual birth date (YYYYMMDD, YYYYMMDDHHMM, or YYYYMMDDHHMMSS). The upload format is: YYYYMMDDHHMMSS.
9	O	S	N	Patient Sex	= NULL, M, F, or U NULL=U. If Null, uploads U.
10	N	N	N	Patient Race/Ethnic Origin	- <i>Unused</i>
11	N	N	N	Patient Address	- <i>Unused</i>
12	N	N	N	Reserved Field	- <i>Unused</i>
13	N	N	N	Patient Telephone Number	- <i>Unused</i>
14	O	P	N	Attending Physician	The System allows only one physician.
14.1	O	P	N	-	Physician ID
14.2	O	P	N	-	Last Name
14.3	O	P	N	-	First Name
14.4	O	P	N	-	Middle Initial
15	O	P	N	Patient's Birth Name	= Patient's Birth Surname. May be required to distinguish between patients with the same birth date and last name.
16	N	N	N	Special Field 2	- <i>Unused</i>

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
17	N	N	N	Patient Height	- <i>Unused</i>
18	N	N	N	Patient Weight	- <i>Unused</i>
19	N	N	N	Patient's Diagnosis	- <i>Unused</i>
20	N	N	N	Patient Active Medications	- <i>Unused</i>
21	N	N	N	Patient's Diet	- <i>Unused</i>
22	N	N	N	Practice Field 1	- <i>Unused</i>
23	N	N	N	Practice Field 2	- <i>Unused</i>
24	N	N	N	Admission and Discharge Dates	- <i>Unused</i>
25	N	N	N	Admission Status	- <i>Unused</i>
26	N	N	N	Location	- <i>Unused</i>
27	N	N	N	Nature of Alternative Diagnosis Code	- <i>Unused</i>
28	N	N	N	Alternative Diagnosis Code	- <i>Unused</i>
29	N	N	N	Patient Religion	- <i>Unused</i>
30	N	N	N	Marital Status	- <i>Unused</i>
31	N	N	N	Isolation Status	- <i>Unused</i>
32	N	N	N	Language	- <i>Unused</i>
33	N	N	N	Hospital Service	- <i>Unused</i>
34	N	N	N	Hospital Institution	- <i>Unused</i>
35	N	N	N	Dosage Category	- <i>Unused</i>

3.4.4 Vision ASTM O Order Record

The ORTHO VISION® System supports the following fields (not shaded) of the ASTM Order (O) Record. Defined in Section 9.4 of ASTM E 1394.

Note: The information with asterisks (*) in the D column are explained in the **Notes** column in the same row.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
1	R	A	N	Record Type ID	= "O" or "o"
2	R	A	N	Sequence Number	Starts at 1. Sequence number increases by one for each result within an order. Reset with each new patient record.
3	R	A	Y	Specimen ID	= Sample ID. A maximum of two Sample IDs can be identified. A unique BRC Sample ID is generated by the ORTHO VISION® System for each execution of a BRC profile. The generated BRC Sample ID always starts with "BRC" and contains a local based date/timestamp and a sequence number. For example: BRC_19/03/20140809_01
4	N	N	N	Instrument Specimen ID	- <i>Unused</i>
5	R	A	N	Universal Test ID	-
5.1	R	A	Y	Profile Name	= Profile name Profiles are configured on the ORTHO VISION® System and there is no standardization. Profiles must be known by the System and profile names are case sensitive (ABO and abo are considered different profile names). On upload: Only one profile per Order record. On download: Only one donor list per order record. Cross Match: On download, if the profile contains a cross match test then the following components are required: Number of donor samples One or more pairs containing the donor Sample ID and Sample type.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
5.2	X	S	N	Number (N) of Donor Samples	= number of donor samples or NULL This component is required if donor samples are included in the order. A crossmatch test may contain zero or more donor samples. If no donor samples are included for a crossmatch order, then the order will be held until a donor sample is manually added.
5.a	X *	P	N	n th Donor Specimen ID	= SampleID of n th donor $a=2*n+1$, where $1<n\leq N$ is the n th donor ID *Required for each additional donor (see 5.1, cross match)
5.b	X *	P	N	Sample type of n th Donor ID	= Sample type of n th donor $b=2*n+2$, where $1<n\leq N$ is the n th donor ID *Required for each additional donor (see 5.1, cross match)
5.c	Q	N	N	Number (M) of Cassette Lots to use	= number of subsequent Cassette ID/lots. $c=2*n+3$, where $1<n\leq N$ is the n th donor ID. If 0 or NULL the System automatically ¹ determines which lot(s) to use. Cassette Lot information is ignored for non QC Orders (Action Code is not = Q)
5.d	Q *	N	N	m th Cassette ID	= Cassette ID $d=2*n+2*m+2$, where $1\leq m\leq M$ is the m th Cassette Lot to specify, n is the n th donor ID * Required for each Cassette.

¹ The system select lots (not expired) with the earliest expiry date first.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
5.e	Q *	N	N	m th Cassette Lot ID	= Cassette Lot number e=2*n+2*m+3, where 1≤m≤M is the m th Cassette Lot to specify, n is the n th donor ID * Required for each Cassette.
5.f	Q	N	N	Number (P) of Reagent Lots to use	= number of subsequent reagent ID/lots. f=2*n+2*m+4, where m is the m th Cassette Lot to specify, n is the n th donor ID. If 0 or NULL the System determines automatically ² which lot(s) to use. Reagent Lot information is ignored for non QC Orders (Action Code is not = Q)
5.g	Q *	N	N	p th Reagent ID	= Reagent ID g=2*N+2*M+2*p+3, where 1≤p≤P is the p th Reagent Lot ID to specify, N is the total number of donor samples, and M is the total number of Cassette IDs/Lots. * Required for each reagent.
5.h	Q *	N	N	p th Reagent Lot ID	= Reagent Lot number h=2*N+2*M+2*p+4, where 1≤p≤P is the p th Reagent Lot ID to specify, N is the total number of donor samples, and M is the total number of Cassette IDs/Lots. *Required for each reagent.
6	O	A	N	Priority	= NULL, S, A, R, C, P, or N. N is the same as R in LIS2-A, 9.4.6. NULL = R = N = C = P = routine priority. S = A = STAT priority.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
7	O	A	N	Requested/Order Date and Time	= date and time of Request/Order YYYYMMDDHHMMSSS The order creation time is provided from the ORTHO VISION® System.
8	O	N	Y	Specimen Collection Date and Time	= date and time specimen was collected (YYYYMMDDHHMMSS) On Upload: NULL
9	N	N	N	Collection End Time	- <i>Unused</i>
10	N	N	N	Collection Volume	- <i>Unused</i>
11	N	N	N	Collector ID	- <i>Unused</i>
12	R	N	N	Action Code	= C, N, A, Q N = A C: cancel the described order, N A: Add profiles on a known sample, new profiles on an unknown sample. Q: treat specimen as Q/C test specimen NULL on Result Upload
13	N	N	N	Danger Code	- <i>Unused</i>
14	O	S	Y	Relevant Clinical Info –	Expected Test Results Ignored for non QC Orders (Action Code is not = Q) Test names are case sensitive and must be known by the system. This information overwrites predefined expected test results. Must be set for each performed test of Non-Ortho QC Profile. An expected result has to be provided for each test executed by this order.
14.1	O	S	N	Test-Name	= Name of an Analysis type

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
14.2	O	S	N	Expected Result	= Expected Result
15	N	N	N	Date/Time Specimen Received	- <i>Unused</i>
16	R	A	Y	Specimen Descriptor	= Sample type Exactly one Specimen Descriptor required for each Specimen ID (see 3)
17	N	N	N	Ordering Physician	- <i>Unused</i>
18	N	N	N	Physician's Phone Number	- <i>Unused</i>
19	O	N	N	User Field 1	= NULL or S If S: save all the Cassettes for all the profiles for manual review.
20	N	S	N	User Field 2	= comment Error message if this order could not be processed by the ORTHO VISION® System.
21	N	N	N	Laboratory Field 1	- <i>Unused</i>
22	N	N	N	Laboratory Field 2	- <i>Unused</i>
23	N	A	N	Date/Time Results Reported or Last Modified	= YYYYMMDDHHMMSS
24	N	N	N	Instrument Charge	- <i>Unused</i>
25	N	N	N	Instrument Section ID	- <i>Unused</i>
26	N	A	N	Report Types	= P, F, R, or X. P: partial results F: final results R: repeat results X: order cancelled on the instrument
27	N	N	N	Reserved Field	- <i>Unused</i>
28	O	S	N	Location or Ward of Specimen Collection	=Collection Location Identifies the location the specimen was collected.
29	N	N	N	Nosocomial Infection Flag	- <i>Unused</i>
30	N	N	N	Specimen Service	- <i>Unused</i>
31	N	N	N	Specimen Institution	- <i>Unused</i>

3.4.5 Vision ASTM R Result Record

A result record is transmitted to the LIS for each executed test.

The ORTHO VISION® System supports the following fields (not shaded) of the ASTM Result (R) Record. Defined in Section 10.1 of ASTM E 1394.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
1	-	A	N	Record Type ID	= "R" or "r"
2	-	A	N	Sequence number	Initial value is 1, reset for each new order; maximum length is unlimited.
3	-	A	N	Test ID	-
3.1	-	A	N	Analysis	= Analysis type
3.2	-	S	N	Donor Specimen ID	Included only if cross match test. = Sample ID of donor The System writes one R record per reaction. Analysis results returned by the ORTHO VISION® System are configurable and may change.
4	-	A	N	Data or Measurement Value	= Analysis Result
5	-	N	N	Units of Measurement Value	- <i>Unused</i>
6	-	N	N	Reference Ranges	- <i>Unused</i>

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
7	-	S	Y	Result Abnormal Flags	= NULL, M, Q, S, T, X, E, I, F, C, P, NA, R M: The result has been entered manually or one of the well results has been modified manually. Q: Out of QC. The test included at least one reagent or Cassette whose periodic QC test is overdue. S: Out of maintenance service. There is an overdue maintenance task. T: Test mode. This result was simulated by the instrument. X: Errors from the imaging system E: Temperature/Humidity sensor reading out of the notification range I: Indeterminate results (no match from the AD) F: User defined protocol C: Discrepant result P: Above/below positive reaction threshold NA: Result expired R: Result expired for a result that has been entered manually N: Negative Reaction Threshold
8	-	N	N	Nature of Abnormality Testing	- <i>Unused</i>
9	-	A	N	Result Status	= F, R, X F: final result R: repeat result X: result rejected or cancelled ³

³ The status cancelled is returned in any of the following cases:

- The corresponding test was cancelled by a LIS order cancel message (Action code = C).
- The corresponding test was cancelled by the user.
- The corresponding test was cancelled due to an unexpected, non-recoverable error (e.g., analytic or redundant analytic validation failed).
- The result was rejected by the user.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
10	-	N	N	Date of Change in Instrument Normative Values or Units	- <i>Unused</i>
11	-	S	N	Operator Identification	*When "Include Operator" is set to No (Default). = Operator ID, Operator who accepts the test. Returns "Automatic" for tests accepted automatically.
11.1	-	S	N	Instrument Operator	*When "Include Operator" is set to Yes. The logged in operator who loaded the samples/reagents.
11.2	-	S	N	Verifier	*When "Include Operator" is set to Yes. The operator who accepts the test. Returns "Automatic" for tests accepted automatically.
12	-	N	N	Date/Time Test Started	- <i>Unused</i>
13	-	A	N	Date/Time Test Completed	= Date/Time of result YYYYMMDDHHMMSS
14	-	A	N	Instrument Identification	= Number
15	-	S	N	Test Name	When enabled, this will contain the name of the test that the result record is associated with (see section 4.1).

Note: In case of test restart (due to recovery) or unexpected errors, the system does not send cancel messages to the LIS.

3.4.6 Vision ASTM Q Request Record

The ORTHO VISION® System supports the following fields (not shaded) of the ASTM Request (Q) Record.

Defined in Section 12.1 of ASTM E 1394.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
1	-	A	N	Record Type ID	= "Q" or "q"
2	-	A	N	Sequence number	Initial value is 1, reset for each new message
3	-	-	N	Starting Range ID Number	-
3.1	-	N	N	Computer system patient ID	- <i>Unused</i>
3.2	-	A	N	Computer system specimen ID	= Computer System Sample ID. The Sample ID of interest, only one allowed per record.
4	-	N	N	Ending Range ID Number	- <i>Unused</i>
5	-	N	N	Universal Test ID	- <i>Unused</i>
6	-	N	N	Nature of Request Time Limits	- <i>Unused</i>
7	-	N	N	Beginning Request Results Date and Time	- <i>Unused</i>
8	-	N	N	Ending Request Results Date and Time	- <i>Unused</i>
9	-	N	N	Requesting Physician Name	- <i>Unused</i>
10	-	N	N	Requesting Physician Phone Number	- <i>Unused</i>
11	-	N	N	User Field 1	- <i>Unused</i>
12	-	N	N	User Field 2	- <i>Unused</i>
13	-	A	N	Request Information Status Codes	= O O: Requesting test orders and demographics only

3.4.7 Vision ASTM M Result Record

The ORTHO VISION® System supports the following fields (not shaded) of the ASTM Result (M) Record. Defined in Section 15.1 of ASTM E 1394.

Table 2. Results or Error

Grade	Meaning
0	"0" reaction
5	"(+)" reaction
10	"1+" reaction
20	"2+" reaction
30	"3+" reaction
40	"4+" reaction
-90	Well Not Found
-95	Wrong liquid level
-100	Light too low
-101	Light too high
-110	Contrast interference
-111	Empty column
-112	Too few cells
-113	Too many cells
-115	Mixed field
-116	Indeterminate
-117	Fibrin
-118	Bubble
-119	Cells detected
-201	Focus error
-203	Splash
-206	Tilt error
-207	Rotation error
-208	Skew error
-209	Well fluid error
-256	Cassette not detected
-260	Wrong position
-999	Not applicable

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
1	-	A	N	Record Type ID	= "M" or "m"
2	-	A	N	Sequence number	=1 for initial order, then reset for each new order; maximum length is unlimited
3	-	A	N	Result Well Name	= Name of the test well For cross match: Donor Sample ID
4.1	-	A	N	Type of Cassette	= Type of Cassette
4.2	-	A	N	Number of the well	= 1..6
4.3	-	A	N	Cassette ID Number	= serial # as given in the barcode.
4.4	-	A	N	Cassette Lot Number	-
4.5	-	A	N	Cassette Expiration Date	= YYYYMMDDHHMMSS
4.6	-	S	N	Mono Image File Name	= The file name of the Cassette image used in determining this result. The actual image data is stored in the Shared Images Folder even when this message not transferred through the shared folder interface.
4.7	-	S	N	Color Image File Name	= The file name of the Cassette image used in determining this result. The actual image data is stored in the Shared Images Folder even when this message not transferred through the shared folder interface.
5	-	A	Y	Reagent Information	-
5.1	-	A	N	Reagent Name	= Reagent Name
5.2	-	A	N	Reagent Lot Number	-
5.3	-	A	N	Reagent Expiration Date	=YYYYMMDDHHMMSS
6.1	-	A	N	Final Result or Error	= from Table 2 Results or Error
6.2	-	A	N	Manual Correction Flag	=M or A M: Manual correction A: Automatic correction
6.3	-	S	N	Read Result or Error	= from Table 2 Results or Error
6.4	-	S	N	Operator ID	= The Operator ID of the operator that made the correction.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
7	-	S	N	Test Name	When enabled, this will contain the name of the test that the result record is associated with (see section 4.1).

3.4.8 Vision ASTM C Comment Record

- The ORTHO VISION® System supports the following fields (not shaded) of the ASTM Comment (C) Record.
- Defined in Section 11 of ASTM E 1394.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; x: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
1	N	S	N	Record Type ID	= "C" or "c"
2	N	S	N	Sequence number	
3	N	S	N	Comment Source	I – Clinical Instrument System
4	N	S	N	Comment Text	<Operator-ID>: comment text
5	N	S	N	Comment Type	G- generic or result comment

3.4.9 Sample Vision ASTM Messages

Note: All results are simulated.

3.4.9.1 ASTM Keep-Alive Message

The ASTM Keep-Alive message is designed to prevent TCP/IP connections between the LIS and VISION® from being dropped during long periods where no messages are exchanged. Keep-Alive message is enabled by setting its interval value between 1-1440. A zero value disables sending the message. If the keep-alive message fails to be sent, VISION® will report an "Apsw26 Unable to Connect to the LIS".

```
VISION: <ENQ>
LIS : <ACK>
VISION: <STX>1H|\^&|||OCD^VISION^5.14.0.47342^JNumber|||P|LIS2-A|20220902174004<CR><ETX>21<CR><LF>
LIS : <ACK>
VISION: <STX>2L||<CR><ETX>86<CR><LF>
LIS : <ACK>
```

VISION: <EOT>

3.4.9.2 Host Query

A request for orders (Q Record) is sent from the instrument to the LIS, when Send Host Query is enabled, and samples or Quality controls are scanned for which the instrument does not have orders. The message contains only one Sample ID. Upon receipt of the query message the LIS sends any orders associated with the sample.

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber|||||P|LIS2-A|20210309140523
Q|1|^SID107|||||||O
L|
```

When VISION® scans an ISBT sample barcode, VISION® stores the complete ISBT 128 barcode, including the leading “=” sign, the Donor Identification Number (DIN), and Flag characters (=W13131200096100). The complete ISBT 128 barcode is uploaded in the Host Query message if not part of an order.

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber|||||P|LIS2-A|20210309140523
Q|1|^=W13131200096100|||||||O
L|
```

3.4.9.3 Minimal Order

A VISION® order requires a minimum of a Sample ID, sample type, and a profile. The minimal order message that VISION® will accept is as follows:

```
H|\^&
P|1
O|1|SID101||ABO-D|||||||CENTBLOOD
L
```

3.4.9.4 Example Order

```
H|\^&|||Mini LIS|||||||LIS2-A|20210309142633
P|1|PID123456||NID123456^MID123456^OID123456|Brown^Bobby^B|White|196501020304|U|||||PHY1234^Kildare^James^P|Blaine
O|1|SID305||ABO|N|20210309142633|||||N||||CENTBLOOD
L
```


3.4.9.5 Example Result

Result with “Strip trailing delimiters” set to “No”:

The Order record report type (O.26) is set to ‘F’ for final results and the Result records Result Status fields (R.6) are set to ‘F’ for final result.

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber| |||||P|LIS2-A|20210309142229
P|1|PID123456| |NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U| |||PHY1234^Kildare^James^P|Blaine| |||||
O|1|SID305| |ABO|N|20210309142136| |||||CENTBLOOD| |||||20210309142229| |F| |||
R|1|ABO|B| |||F| |Automatic| |20210309142229|JNumber
M|1|Anti-A|ABD Confirmation^4^000000^77777^20210828235959^20210309_142225Grey.jpg^20210309_142225Color.jpg| |0^A
M|2|Anti-B|ABD Confirmation^5^000000^77777^20210828235959^20210309_142225Grey.jpg^20210309_142225Color.jpg| |40^A
R|2|Rh|POS| |||F| |Automatic| |20210309142229|JNumber
M|1|Anti-D|ABD Confirmation^6^000000^77777^20210828235959^20210309_142225Grey.jpg^20210309_142225Color.jpg| |40^A
L| |
```

Result with “Strip trailing delimiters” set to “Yes”:

Notice the lack of trailing delimiters in the P, O, R, and L records.

This message was resent by the operator. The Order record report type (O.26) is set to ‘R’ and the Result records Result Status fields (R.6) are also set to ‘R’ for repeat results.

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber| |||||P|LIS2-A|20210309142437
P|1|PID123456| |NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U| |||PHY1234^Kildare^James^P|Blaine
O|1|SID305| |ABO|N|20210309142136| |||||CENTBLOOD| |||||20210309142229| |R
R|1|ABO|B| |||R| |Automatic| |20210309142229|JNumber
M|1|Anti-A|ABD Confirmation^4^000000^77777^20210828235959^20210309_142225Grey.jpg^20210309_142225Color.jpg| |0^A
M|2|Anti-B|ABD Confirmation^5^000000^77777^20210828235959^20210309_142225Grey.jpg^20210309_142225Color.jpg| |40^A
R|2|Rh|POS| |||R| |Automatic| |20210309142229|JNumber
M|1|Anti-D|ABD Confirmation^6^000000^77777^20210828235959^20210309_142225Grey.jpg^20210309_142225Color.jpg| |40^A
L
```

Result with “Transmit Column Grade Results” set to “No”:

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber| |||||P|LIS2-A|20210309142633
P|1|PID123456| |NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U| |||PHY1234^Kildare^James^P|Blaine
O|1|SID305| |ABO|N|20210309142136| |||||CENTBLOOD| |||||20210309142229| |R
R|1|ABO|B| |||R| |Automatic| |20210309142229|JNumber
R|2|Rh|POS| |||R| |Automatic| |20210309142229|JNumber
L
```


3.4.9.9 Result with 2 Samples

This result was manually edited and accepted by an operator:

- The grades for the ABO result were manually set; this is indicated by the Result record 'Result Abnormal Flags field (R.7) containing the Manual flag (M) along with the Manufactures record's 'Manual Correction Flag' set to 'M' in field (M.6.2). The ABO's Anti-A M-record contains "20^M^10^soladmin" in field (M.6). M.6.1 contains the value (40) entered by an operator. M.6.2 contains the Manual Flag (M) indicating that was manually set. M.6.3 contains the original grade result (0) read by the instrument. And M.6.4 contains the operator ID.
- The interpreted result for the Rh result was manually set; this is indicated by the Result record 'Result Abnormal Flags field (R.7) containing the Manual flag (M) along with the Rh Manufactures record's 'Manual Correction Flag' set to 'A' (automatic) in field (M.6.2).
- The ABScr result was not edited; this is indicated by the absence of the Manual flag (M) in the Result record 'Result Abnormal Flags field (R.7).
- The "Upload comments with result" option is enabled for this message.

The ABO results were manually accepted by 'soladmin'; this is indicated by the Result records 'Operator Identification' field (R.11) set to the operator ID of 'soladmin' and not set to 'Automatic'.

The ABScr results were automatically accepted; this is indicated by the Result record 'Operator Identification' field (R.11) set to 'Automatic'.

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber| |||||P|LIS2-A|20210309151053
P|1|PID123456| |NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U| |||PHY1234^Kildare^James^P|Blaine| |||||
O|1|SID303|SID304| |Type && Screen|N|20210309150806| |||||PACKEDCELLS\PLASMA| |||||20210309151049| ||F| |||
C|1|I|User1: Adjusted grade for Anti-A column|G
C|2|I|User1: Manually set Rh to POS based on image review|G
R|1|ABO|A| |M| |F| |soladmin| |20210309150832|JNumber
M|1|Anti-A|ABD Confirmation^1^000000^77777^20210828235959^20210309_150832Grey.jpg^20210309_150832Color.jpg| |20^M^10^soladmin
M|2|Anti-B|ABD Confirmation^2^000000^77777^20210828235959^20210309_150832Grey.jpg^20210309_150832Color.jpg| |0^A
R|2|Rh|POS| |M| |F| |soladmin| |20210309150832|JNumber
M|1|Anti-D|ABD Confirmation^3^000000^77777^20210828235959^20210309_150832Grey.jpg^20210309_150832Color.jpg| |0^A
R|3|ABScr|POS| |||F| |Automatic| |20210309150844|JNumber
M|1|0.8-Sel I|AHG Polyspecific^1^000037^77777^20210828235959^20210309_150838Grey.jpg^20210309_150838Color.jpg|0.8% Sel I^1234^20210828235959|20^A
M|2|0.8-Sel II|AHG Polyspecific^2^000037^77777^20210828235959^20210309_150838Grey.jpg^20210309_150838Color.jpg|0.8% Sel II^1234^20210828235959|40^A
L| |
```

3.4.9.10 Order with Multiple Profiles

The LIS can send orders containing one or more profiles. The VISION® System treats each profile as separate orders containing identical demographics and responds with individual result messages for each profile.

```
H|^&|||Mini LIS|LIS2-A|20210309155210
P|1|PID123456|NID123456^MID123456^OID123456|Brown^Bobby^B|White|196501020304|M
O|1|SID304|ABO\ABScr|N|20210309155210||||N|||CENTBLOOD
O|2|SID305|Pheno|N|20210309155210||||N|||CENTBLOOD
P|2|PID789012|NID789012^MID789012^OID789012|Forbin^Charles|Fisher|19410403|M
O|1|SID306|ABO\ABScr|N|20210309155210||||N|||CENTBLOOD
O|2|SID307|Pheno|N|20210309155210||||N|||CENTBLOOD
L
```

VISION® will create six orders from the message above: one for each profile.

Sample ID	Patient	Profile Name	Status	Priority	Results		Order Type	Patient ID
SID304	Bobby Brown	ABO	Accepted	Routine	ABO AB	Rh NEG	IH	PID123456
SID304	Bobby Brown	ABScr	Accepted	Routine	ABScr POS		IH	PID123456
SID305	Bobby Brown	Pheno	Accepted	Routine	Rh POS*	Pheno CCEe*	IH	PID123456
SID306	Charles Forbin	ABO	Accepted	Routine	ABO O	Rh POS	IH	PID789012
SID306	Charles Forbin	ABScr	Accepted	Routine	ABScr POS		IH	PID789012
SID307	Charles Forbin	Pheno	Accepted	Routine	Rh POS	Pheno CcEe	IH	PID789012

3.4.9.11 Profile with Multiple Tests

The Profile “T&S” below is defined with the test 10001 “ABO(FWD)/Rh-00” and 10066 “08 AbScr Sel Poly”. Each test has its own set of R-records and each R-record has its own set of M-records. In the result message below, “Upload test name” is enabled for both Result and Manufacture records.

The same Ctrl result (well 4 in M.4.2) for test “ABO(FWD)/Rh-00” is listed under both the ABO and Rh interpreted Result record. When the same Ctrl well is listed under more than one analysis result, the M-records will be identical, with the possible exception of the M-record sequence number. The M-record sequence number is specific to its position under each R-record.

When sending a message that uses delimiters in a text field, the sender must escape the delimiter. The profile “T&S” is written as “T&E&S” in the order to escape the ‘&’ in the profile name (ASTM standard escape sequence). The ASTM standard specifies “T&E&S” or “T&X26&S” for escape sequences.

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber|P|LIS2-A|20210309164727
P|1|PID123456|NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U|||PHY1234^Kildare^James^P|Blaine|
O|1|SID305|T&X26&S|N|20210309164646|CENTBLOOD|20210309164727|F|
R|1|ABScr|POS|F|Automatic|20210309164727|JNumber|08 AbScr Sel Poly
M|1|0.8-Sel I|AHG Polyspecific^1^000039^77777^20210828235959^20210309_164720Grey.jpg^20210309_164720Color.jpg|0.8% Sel I^1234^20210828235959|10^A|08
AbScr Sel Poly
M|2|0.8-Sel II|AHG Polyspecific^2^000039^77777^20210828235959^20210309_164720Grey.jpg^20210309_164720Color.jpg|0.8% Sel II^1234^20210828235959|0^A|08
AbScr Sel Poly
R|2|ABO|B|F|Automatic|20210309164716|JNumber|ABO(FWD)/Rh-00
M|1|Anti-A|ABO-Rh/Reverse^1^000012^77777^20210828235959^20210309_164715Grey.jpg^20210309_164715Color.jpg|0^A|ABO(FWD)/Rh-00
M|2|Anti-B|ABO-Rh/Reverse^2^000012^77777^20210828235959^20210309_164715Grey.jpg^20210309_164715Color.jpg|40^A|ABO(FWD)/Rh-00
M|3|Ctrl|ABO-Rh/Reverse^4^000012^77777^20210828235959^20210309_164715Grey.jpg^20210309_164715Color.jpg|0^A|ABO(FWD)/Rh-00
R|3|Rh|POS|F|Automatic|20210309164716|JNumber|ABO(FWD)/Rh-00
M|1|Anti-D|ABO-Rh/Reverse^3^000012^77777^20210828235959^20210309_164715Grey.jpg^20210309_164715Color.jpg|40^A|ABO(FWD)/Rh-00
M|2|Ctrl|ABO-Rh/Reverse^4^000012^77777^20210828235959^20210309_164715Grey.jpg^20210309_164715Color.jpg|0^A|ABO(FWD)/Rh-00
L|
```

3.4.9.12 Donor Confirmation Test

```
H|\^&|||Mini LIS|LIS2-A|20210309164445
P|1
O|1|=W13131200096100|Group+Rh|N|20210309164445|CENTBLOOD
O|1|=W13131200096200|Group+Rh|N|20210309164445|CENTBLOOD
L
```

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber|P|LIS2-A|20210309165812
P|1|U|
O|1|=W13131200096200|Group+Rh|N|20210309165647|CENTBLOOD|20210309165811|F|
R|1|ABO|O|F|Automatic|20210309165811|JNumber
M|1|Anti-A|ABD Confirmation^1^000002^77777^20210828235959^20210309_165807Grey.jpg^20210309_165807Color.jpg|0^A
M|2|Anti-B|ABD Confirmation^2^000002^77777^20210828235959^20210309_165807Grey.jpg^20210309_165807Color.jpg|0^A
R|2|Rh|POS|F|Automatic|20210309165811|JNumber
M|1|Anti-D|ABD Confirmation^3^000002^77777^20210828235959^20210309_165807Grey.jpg^20210309_165807Color.jpg|40^A
```

L|

The “Group+Rh” profile is defined to use the test 10007 “ABD Conf-10” which consumes only three wells for each test. VISION® can run two “ABD Conf-10” tests per Cassette. Therefore, the well number fields (M.4.2) will be either 1-2-3 (above) or 4-5-6 (below).

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber|P|LIS2-A|20210309165811
P|1|U|
O|1|=W13131200096100|Group+Rh|N|20210309165647|CENTBLOOD|20210309165810|F|
R|1|ABO|AB|F|Automatic|20210309165810|JNumber|ABD Conf-10
M|1|Anti-A|ABD Confirmation^4^000001^77777^20210828235959^20210309_165801Grey.jpg^20210309_165801Color.jpg|40^A|ABD Conf-10
M|2|Anti-B|ABD Confirmation^5^000001^77777^20210828235959^20210309_165801Grey.jpg^20210309_165801Color.jpg|40^A|ABD Conf-10
R|2|Rh|NEG|F|Automatic|20210309165810|JNumber|ABD Conf-10
M|1|Anti-D|ABD Confirmation^6^000001^77777^20210828235959^20210309_165801Grey.jpg^20210309_165801Color.jpg|0^A|ABD Conf-10
L|
```

3.4.9.13 Dilution Series Test

```
H|\^&|||Mini LIS|LIS2-A|20210309165812
P|1
O|1|SID304|DS|N|20210309165812|N|CENTBLOOD
L
```

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber|P|LIS2-A|20210309171203
P|1|U|
O|1|SID304|DS|N|20210309170748|CENTBLOOD|20210309171158|F|
R|1|DilSeries|Done|P|F|soladmin|20210309171127|JNumber
M|1|DS-Neat|AHG anti-IgG^1^000019^77777^20210828235959^20210309_171116Grey.jpg^20210309_171116Color.jpg|0.8% Sel I^1234^20210828235959|30^A
M|2|DS-2|AHG anti-IgG^2^000019^77777^20210828235959^20210309_171116Grey.jpg^20210309_171116Color.jpg|0.8% Sel I^1234^20210828235959|30^A
M|3|DS-4|AHG anti-IgG^3^000019^77777^20210828235959^20210309_171116Grey.jpg^20210309_171116Color.jpg|0.8% Sel I^1234^20210828235959|20^A
M|4|DS-8|AHG anti-IgG^4^000019^77777^20210828235959^20210309_171116Grey.jpg^20210309_171116Color.jpg|0.8% Sel I^1234^20210828235959|20^A
M|5|DS-16|AHG anti-IgG^5^000019^77777^20210828235959^20210309_171116Grey.jpg^20210309_171116Color.jpg|0.8% Sel I^1234^20210828235959|5^A
M|6|DS-32|AHG anti-IgG^6^000019^77777^20210828235959^20210309_171116Grey.jpg^20210309_171116Color.jpg|0.8% Sel I^1234^20210828235959|5^A
M|7|DS-64|AHG anti-IgG^1^000020^77777^20210828235959^20210309_171122Grey.jpg^20210309_171122Color.jpg|0.8% Sel I^1234^20210828235959|0^A
M|8|DS-128|AHG anti-IgG^2^000020^77777^20210828235959^20210309_171122Grey.jpg^20210309_171122Color.jpg|0.8% Sel I^1234^20210828235959|0^A
M|9|DS-256|AHG anti-IgG^3^000020^77777^20210828235959^20210309_171122Grey.jpg^20210309_171122Color.jpg|0.8% Sel I^1234^20210828235959|0^A
M|10|DS-512|AHG anti-IgG^4^000020^77777^20210828235959^20210309_171122Grey.jpg^20210309_171122Color.jpg|0.8% Sel I^1234^20210828235959|0^A
M|11|DS-1024|AHG anti-IgG^5^000020^77777^20210828235959^20210309_171122Grey.jpg^20210309_171122Color.jpg|0.8% Sel I^1234^20210828235959|0^A
M|12|DS-Ctrl|AHG anti-IgG^6^000020^77777^20210828235959^20210309_171122Grey.jpg^20210309_171122Color.jpg|0.8% Sel I^1234^20210828235959|0^A
L|
```

3.4.9.14 Pheno Test

```
H|\^&|||Mini LIS|LIS2-A|20210310120708
P|1|PID123456|NID123456^MID123456^OID123456|Brown^Bobby^B|White|196501020304|U|PHY1234^Kildare^James^P|Blaine
O|1|SID305|Pheno|N|20210310120708|N|CENTBLOOD
```

Default	R 2 Pheno CCEE P F Automatic 20180912152530 JNumber
Set 1	R 2 Pheno C+c-E+e- P R Automatic 20180912152530 JNumber
Set 2	R 2 Pheno C+E+c-e- P R Automatic 20180912152530 JNumberrr
Set 3	R 2 Pheno 2,3,-4,-5 P R Automatic 20180912152530 JNumber

H|\&|||OCD^VISION^5.13.1.46935^JNumber|||||P|LIS2-A|20210310122924
P|1|PID123456||NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U||||PHY1234^Kildare^James^P|Blaine||||||||||||||||
O|1|SID305||Pheno|N|20210310122908|||||CENTBLOOD|||||20210310122924||||F||||
R|1|Rh|POS|||P||F||Automatic||20210310122924|JNumber
M|1|Anti-D|Rh-hr^1^000034^77777^20210828235959^20210310_122923Grey.jpg^20210310_122923Color.jpg||20^A
M|2|Ctrl|Rh-hr^6^000034^77777^20210828235959^20210310_122923Grey.jpg^20210310_122923Color.jpg||0^A
R|2|Pheno|CcEe|||P||F||Automatic||20210310122924|JNumber
M|1|Anti-C|Rh-hr^2^000034^77777^20210828235959^20210310_122923Grey.jpg^20210310_122923Color.jpg||10^A
M|2|Anti-E|Rh-hr^3^000034^77777^20210828235959^20210310_122923Grey.jpg^20210310_122923Color.jpg||30^A
M|3|Anti-c|Rh-hr^4^000034^77777^20210828235959^20210310_122923Grey.jpg^20210310_122923Color.jpg||5^A
M|4|Anti-e|Rh-hr^5^000034^77777^20210828235959^20210310_122923Grey.jpg^20210310_122923Color.jpg||5^A
M|5|Ctrl|Rh-hr^6^000034^77777^20210828235959^20210310_122923Grey.jpg^20210310_122923Color.jpg||0^A
L||

H|\^&|||Mini LIS|||||LIS2-A|20210310113149
P|1|PID123456|NID123456^MID123456^OID123456|Brown^Bobby^B|White|196501020304|U||||PHY1234^Kildare^James^P|Blaine
O|1|SID305||Jka|N|20210310113149||||N||||CENTBLOOD
L

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber| ||||P|LIS2-A|20210310120708
P|1|PID123456||NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U| |||PHY1234^Kildare^James^P|Blaine| |||||
O|1|SID305||Jka|N|20210310120633| |||||CENTBLOOD| |||||20210310120708| ||F| |||
R|1|Jka|POS| |P|F|Automatic| |20210310120707|JNumber
M|1|Anti-Jka|Reverse diluent^1^000077^77777^20210828235959^20210310_120657Grey.jpg^20210310_120657Color.jpg|Anti-Jka^123^20210828235959|5^A
L| |
```

Orders can be cancelled by the LIS or by VISION®. When the order is cancelled, the ORTHO VISION® System sends a cancelation message to the LIS. The LIS cannot cancel an order once it has started processing.

J66623EN

H|\^&||Mini LIS|||||LIS2-A|20210310122924
P|1|PID123456||NID123456^MID123456^OID123456|Brown^Bobby^B|White|196501020304|U||||PHY1234^Kildare^James^P|Blaine
O|1|SID999||ABO|N|20210310122924||||C||||CENTBLOOD
L

LIS Order Cancellation Request:

The cancellation request should be identical to the original order but with the Order action code (O.12) set to Cancel (C).

H|\^&||Mini LIS|||||LIS2-A|20210310122924
P|1|PID123456||NID123456^MID123456^OID123456|Brown^Bobby^B|White|196501020304|U||||PHY1234^Kildare^James^P|Blaine
O|1|SID999||ABO|N|20210310122924||||C||||CENTBLOOD
L

Response sent to the LIS when LIS cancels an order:

H|\^&||OCD^VISION^5.13.1.46935^JNumber|||||P|LIS2-A|20210310131317
P|1|PID123456||NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U||||PHY1234^Kildare^James^P|Blaine|||||||
O|1|SID999||ABO|N|20210310131312||||||CENTBLOOD|||||20210310131317||X||||
L|

Response sent to the LIS when the ORTHO VISION® System cancels an order:

H|\^&||OCD^VISION^5.13.1.46935^JNumber|||||P|LIS2-A|20210310131152
P|1|PID123456||NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U||||PHY1234^Kildare^James^P|Blaine|||||||
O|1|SID999||ABO|N|20210310131142||||||CENTBLOOD|||||20210310131152||F||||
R|1|ABO||||X|soladmin||20210310131152|JNumber
R|2|Rh||||X|soladmin||20210310131152|JNumber
L|

3.4.9.17 Quality Control Examples

Quality control tests can be ordered through the LIS or manually entered on VISION®. QC tests are ordered against a Profile.

Quality Control Order:

3.4.9.17.1 When VISION® is managing quality controls:

- Set 'Barcode Interpretation – Ortho QC' to Enable
- Set the Profile's QC Mode to MBC

Quality Control Order (Managed by VISION®):

Quality Control orders have the Order Action Code (O.12) set to Q.

```
H|\^&||Mini LIS|||||LIS2-A|20210310132933
P|1|||||
O|1|QC811516811551||Type && Screen|S|20210310132933|||Q|||ORTHO CONFIDENCE WB|||||
L||
```

```
H|\^&||Mini LIS|||||LIS2-A|20210310132933
P|1|||||
O|1|QC1365568111232||Type && Screen|S|20210310132933|||Q|||ORTHO DAILY QC|||||
L||
```

Quality Control Order with Cassette and reagent lots specified:

When VISION® is managing Quality Controls, an LIS can send Quality Control orders to VISION® and specify required Cassette and reagent lots. VISION® will hold the order until the specified Cassettes and reagents are loaded on VISION®. See Cassettes and Reagents tables in section 4.

Profile	Number of donor samples	Number of Cassette Types	Cassette ID	Cassette Lot	Cassette ID	Cassette Lot	Number of Reagent Types	Reagent ID	Reagent Lot	Reagent ID	Reagent Lot
Type && Screen	^0	^2	^00	^00016	^22	^00061	^2	^29	^0363	^30	^0363

```
H|\^&||Mini LIS|||||LIS2-A|20210310133809
P|1|||||
O|1|QC536582022651||Type && Screen^0^2^00^00016^22^00061^2^29^0363^30^0363|S|20210310133809|||Q|||ORTHO CONFIDENCE WB|||||
L||
```

```
H|\^&|||Mini LIS|||||LIS2-A|20210310133809
P|1|||||
O|1|QC1365568111232||Type && Screen^0^2^00^00061^22^00061^2^29^0363^30^0363|S|20210310133809|||Q|||ORTHO DAILY QC|
L|
```

3.4.9.17.2 User Defined Quality Control Orders

User Defined Quality Control Order with expected results (Managed by VISION®):

The LIS must include expected results in field O.14 when ordering non-Ortho QC Sample IDs (result not known from barcode). Expected results must be included even if expected results are configured in Vision.

When User Defined Quality Controls are managed by the LIS, their orders are the same as non-QC orders.

```
H|\^&|||Mini LIS|||||LIS2-A|20210310140816
P|1|||||
O|1|UDQC6789013||Type && Screen|S|20210310140816|||Q||ABO^A\Rh^NEG\ABScr^POS||CENTBLOOD|
L|
```

User Defined Quality Control Result with expected results:

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber|P|LIS2-A|20210310140927
P|1|||||U|||||
O|1|UDQC6789013||Type && Screen|A|20210310140842|||CENTBLOOD|||20210310140927||F|||
R|1|ABScr|POS|||F|Automatic|20210310140927|JNumber
M|1|0.8-Sel I|AHG Polyspecific^5^000038^77777^20210828235959^20210310_140923Grey.jpg^20210310_140923Color.jpg|0.8% Sel I^1234^20210828235959|10^A
M|2|0.8-Sel II|AHG Polyspecific^6^000038^77777^20210828235959^20210310_140923Grey.jpg^20210310_140923Color.jpg|0.8% Sel II^1234^20210828235959|30^A
R|2|ABO|AB|||F|Automatic|20210310140927|JNumber
M|1|Anti-A|ABD Confirmation^1^000004^77777^20210828235959^20210310_140916Grey.jpg^20210310_140916Color.jpg||40^A
M|2|Anti-B|ABD Confirmation^2^000004^77777^20210828235959^20210310_140916Grey.jpg^20210310_140916Color.jpg||40^A
R|3|Rh|NEG|||F|Automatic|20210310140927|JNumber
M|1|Anti-D|ABD Confirmation^3^000004^77777^20210828235959^20210310_140916Grey.jpg^20210310_140916Color.jpg||0^A
L|
```

3.4.9.17.3 When the LIS is managing quality controls:

- Set 'Barcode Interpretation – Ortho QC' to Disabled
- Set the Profile's QC Mode to no QC

A quality control order from the LIS is the same as any other order, the Order Action Code (O.12) is not set to Q. Since barcode interpretation is turned off, the LIS order needs to contain a sample fluid type that aligns with the QC fluid; use 'CENTBLOOD' instead of 'ORTHO DAILY QC'.

Because the LIS is managing QCs, the action code in the order should not be set to 'Q' (treat specimen as Q/C test).

Quality Control Result (Managed by LIS):

```
H|\^&|||Mini LIS|||||LIS2-A|20210310140927
P|1|||||U|||||
O|1|QC1365568111232||Type && Screen|S|20210310140927||||N||||CENTBLOOD|||||
L||
```

3.4.9.17.4 BRC Results

BRC tests cannot be ordered by the LIS, however their results are uploaded to the LIS.

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber|||||P|LIS2-A|20210310145050
P|1|||||U|||||
O|1|BRC_3/10/20211445_01||00:BRC 00 Neg|A|20210310144524|||||20210310145049|||F||||
R|1|BRC|Pass||||F||Automatic||20210310145049|JNumber
M|1|BRC-|ABO-Rh/Reverse^1^000009^77777^20210828235959^20210310_145043Grey.jpg^20210310_145043Color.jpg|B Cells^1234^20210828235959|0^A
M|2|BRC-|ABO-Rh/Reverse^2^000009^77777^20210828235959^20210310_145043Grey.jpg^20210310_145043Color.jpg|A1 Cells^1234^20210828235959|0^A
M|3|BRC-|ABO-Rh/Reverse^3^000009^77777^20210828235959^20210310_145043Grey.jpg^20210310_145043Color.jpg|BRC-E3^1234^20210828235959|0^A
M|4|BRC-|ABO-Rh/Reverse^4^000009^77777^20210828235959^20210310_145043Grey.jpg^20210310_145043Color.jpg|BRC-E3^1234^20210828235959|0^A
M|5|BRC-|ABO-Rh/Reverse^5^000009^77777^20210828235959^20210310_145043Grey.jpg^20210310_145043Color.jpg|BRC-S1^1234^20210828235959|A1
Cells^1234^20210828235959|0^A
M|6|BRC-|ABO-Rh/Reverse^6^000009^77777^20210828235959^20210310_145043Grey.jpg^20210310_145043Color.jpg|BRC-S1^1234^20210828235959|B
Cells^1234^20210828235959|0^A
L||
```

3.4.9.18 Results with multiple reagents

Vision does not guarantee that the list of reagents in an M-record will be reported in the same sequence every time. The reagent name and not its position in the uploaded result message should always be used to access reagent lots and expiration dates. Below, BLISS is listed first in the first two M-records but last in the third M-record.

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber|||||P|LIS2-A|20210310150718
P|1|||||U|||||
O|1|SID305||ABScr-2|N|20210310150516|||||CENTBLOOD|||||20210310150718|||F||||
R|1|ABScr|POS||||F||Automatic||20210310150718|JNumber
M|1|Fic-Unt 1|AHG Polyspecific^1^000040^77777^20210828235959^20210310_150717Grey.jpg^20210310_150717Color.jpg|BLISS^1234^20210828235959\Fic Unt 1^1234^20210828235959|30^A
M|2|Fic-Unt 2|AHG Polyspecific^2^000040^77777^20210828235959^20210310_150717Grey.jpg^20210310_150717Color.jpg|BLISS^1234^20210828235959\Fic Unt 2^1234^20210828235959|0^A
M|3|Fic-Unt 3|AHG Polyspecific^3^000040^77777^20210828235959^20210310_150717Grey.jpg^20210310_150717Color.jpg|Fic Unt 3^1234^20210828235959|20^A|BLISS^1234^20210828235959
L||
```

3.4.9.19 Include Operator is set to 'Yes'

When "Include Operator" is set to 'Yes' the logged in operator that loaded the samples/reagents and the operator that accepted the results are both included in result messages. The name "Automatic" is used when Vision automatically accepts results.

Below, user1 loaded the samples and reagents. The ABScr results were automatically accepted, and user2 manually accepted the ABO results.

```
H|\&|||OCD^VISION^5.13.1.46935^JNumber|||||P|LIS2-A|20210311120645
P|1|||||U|||||
O|1|SID305||Type && Screen|N|20210311120600|||||CENTBLOOD|||||20210311120644|||F||||
R|1|ABO|A|||P||F||user1^user2||20210311120621|JNumber
M|1|Anti-A|ABD Confirmation^4^000008^77777^20210828235959^20210311_120619Grey.jpg^20210311_120619Color.jpg||10^A
M|2|Anti-B|ABD Confirmation^5^000008^77777^20210828235959^20210311_120619Grey.jpg^20210311_120619Color.jpg||0^A
R|2|Rh|NEG|||||F||user1^user2||20210311120621|JNumber
M|1|Anti-D|ABD Confirmation^6^000008^77777^20210828235959^20210311_120619Grey.jpg^20210311_120619Color.jpg||0^A
R|3|ABScr|POS|||||F||user1^Automatic||20210311120631|JNumber
M|1|0.8-Sel I|AHG Polyspecific^1^000044^77777^20210828235959^20210311_120624Grey.jpg^20210311_120624Color.jpg|0.8% Sel I^1234^20210828235959|0^A
M|2|0.8-Sel II|AHG Polyspecific^2^000044^77777^20210828235959^20210311_120624Grey.jpg^20210311_120624Color.jpg|0.8% Sel II^1234^20210828235959|10^A
L|
```

In some cases, SYSTEM will be returned as the operator.

3.4.9.20 Partial Results

Partial results may be uploaded for **profiles** containing two or more tests. A partial result is a result message containing tests results for some but not all the tests in a **profile**. Partial results may be uploaded when the upload mode is configured for Automatic Partial or when an operator manually sends a partial result to the LIS.

A result message will never contain partial **test** results. Test 1007 (10: ABD Conf-10) has two analyses; ABO and Rh. Result messages containing a test result for test 1007 will always contain the ABO and Rh results.

A partial result is indicated in the message below by the Order record Report Types field (O.26) set to P for partial results. The Result records Result Status field (R.8) are set to F for finale result. The partial result message contains all the results for the ABO/Rh test.

```
H|\&|||OCD^VISION^5.13.1.46935^JNumber|||||P|LIS2-A|20210311133300
P|1|||||U|||||
O|1|SID305||Type && Screen|N|20210311133233|||||CENTBLOOD|||||20210311133300|||P||||
R|1|ABO|A|||F||Automatic||20210311133259|JNumber
M|1|Anti-A|ABD Confirmation^1^000000^77777^20210828235959^20210311_133255Grey.jpg^20210311_133255Color.jpg||40^A
M|2|Anti-B|ABD Confirmation^2^000000^77777^20210828235959^20210311_133255Grey.jpg^20210311_133255Color.jpg||0^A
R|2|Rh|NEG|||||F||Automatic||20210311133259|JNumber
M|1|Anti-D|ABD Confirmation^3^000000^77777^20210828235959^20210311_133255Grey.jpg^20210311_133255Color.jpg||0^A
L|
```

For profiles that have more than 2 tests, there can be additional partial results uploaded. The Order record Report Types field (O.26) will be set to P for partial results. The newly completed tests will have their Result records Result Status fields (R.8) is set to F for final result. The results that were reported previously will be included and will have their Result records Result Status fields (R.8) is set to R for repeat result.

When the last test in the profile is reported, the Order record Report Types field (O.26) will be set to F for final result.

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber| |||||P|LIS2-A|20210311133429
P|1|||||U|||||
O|1|SID305||Type && Screen|N|20210311133233| |||||CENTBLOOD| |||||20210311133429||F| |||
R|1|ABScr|POS| ||P|F|soladmin| |20210311133411|JNumber
M|1|0.8-Sel I|AHG Polyspecific^1^000037^77777^20210828235959^20210311_133407Grey.jpg^20210311_133407Color.jpg|0.8% Sel I^1234^20210828235959|5^A
M|2|0.8-Sel II|AHG Polyspecific^2^000037^77777^20210828235959^20210311_133407Grey.jpg^20210311_133407Color.jpg|0.8% Sel II^1234^20210828235959|20^A
R|2|ABO|A| |||R|Automatic| |20210311133259|JNumber
M|1|Anti-A|ABD Confirmation^1^000000^77777^20210828235959^20210311_133255Grey.jpg^20210311_133255Color.jpg|40^A
M|2|Anti-B|ABD Confirmation^2^000000^77777^20210828235959^20210311_133255Grey.jpg^20210311_133255Color.jpg|0^A
R|3|Rh|NEG| |||R|Automatic| |20210311133259|JNumber
M|1|Anti-D|ABD Confirmation^3^000000^77777^20210828235959^20210311_133255Grey.jpg^20210311_133255Color.jpg|0^A
L| |
```

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3.5 ASTM and Enhanced ASTM

This section describes the predecessor ASTM format. This format can be used by enabling the ASTM or Enhanced ASTM compatibility mode.

3.5.1 ASTM H Header Record

No compatibility issues. See [Vision ASTM H Header Record](#).

3.5.2 ASTM L Trailer Record

No compatibility issues. See [Vision ASTM L Trailer Record](#).

3.5.3 ASTM P Patient Record

The ORTHO VISION® System supports the following fields (not shaded) of the ASTM Patient (P) Record. Defined in Section 8.1 of ASTM E 1394.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
1	R	A	N	Record Type ID	= "P" or "p"
2	O	A	N	Sequence number	Patient sequence number. Set to 1 for the first patient record; 2 for 2 nd record; and so on.
3	O	P	N	Practice Assigned Patient ID	= Patient ID If present, the unique identifier for the patient. If empty all following fields of this record are ignored.
4	N	N	N	Lab Assigned Patient ID	- <i>Unused</i>
5.1	O	P	N	Patient ID No. 3	= National ID
5.2	O	P	N	-	= Medical Record
5.3	O	P	N	-	= Other ID
6.1	O	P	N	Patient Name	= Last Name
6.2	O	P	N	-	= First Name
6.3	O	P	N	-	= Middle Initial
6.4	N	N	N	-	= Suffix
6.5	N	N	N	-	= Title
7	O	P	N	Mother's Maiden Name	= Mother's Maiden Surname. May be required to distinguish between patients with the same birth date and last name.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
8	O	S	N	Birth Date	= actual birth date (YYYYMMDD, YYYYMMDDHHMM, or YYYYMMDDHHMMSS). Birthdate upload format is always YYYYMMDDHHMMSS.
9	O	A	N	Patient Sex	= NULL, M, F, or U NULL=U If Null, uploads U.
10	N	N	N	Patient Race/Ethnic Origin	-Unused
11	N	N	N	Patient Address	-Unused
12	N	N	N	Reserved Field	-Unused
13	N	N	N	Patient Telephone Number	-Unused
14.1	N	N	N	Attending Physician ID	- Unused
14.2	N	N	N	-	-Unused
14.3	N	N	N	-	-Unused
14.4	N	N	N	-	-Unused
15	O	P	N	Patient's Birth Name	= Patient's Birth Surname. May be required to distinguish between patients with the same birth date and last name.
16	N	N	N	Special Field 2	-Unused
17	N	N	N	Patient Height	-Unused
18	N	N	N	Patient Weight	-Unused
19	N	N	N	Patient's Diagnosis	-Unused
20	N	N	N	Patient Active Medications	-Unused
21	N	N	N	Patient's Diet	-Unused
22	N	N	N	Practice Field 1	-Unused
23	N	N	N	Practice Field 2	-Unused
24	N	N	N	Admission and Discharge Dates	-Unused
25	N	N	N	Admission Status	-Unused
26	N	N	N	Location	-Unused
27	N	N	N	Nature of Alternative Diagnosis Code	-Unused

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
28	N	N	N	Alternative Diagnosis Code	-Unused
29	N	N	N	Patient Religion	-Unused
30	N	N	N	Marital Status	-Unused
31	N	N	N	Isolation Status	-Unused
32	N	N	N	Language	-Unused
33	N	N	N	Hospital Service	-Unused
34	N	N	N	Hospital Institution	-Unused
35	N	N	N	Dosage Category	-Unused

3.5.4 ASTM O Order Record

The ORTHO VISION® System supports the following fields (not shaded) of the ASTM Order (O) Record. Defined in Section 9.4 of ASTM E 1394.

Note: The information with asterisks (*) in the D column are explained in the **Notes** column in the same row.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
1	R	A	N	Record Type ID	= "O" or "o"
2	N	A	N	Sequence Number	Starts at 1. Sequence number increases by one for each result within an order. Reset for new patient.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
3	R	A	N	Specimen ID	= Sample ID A Sample ID may contain one or more alphanumeric, symbols, and embedded blanks characters up to a maximum length of 20 characters. Leading zeros in a Sample ID are retained and stored as part of the Sample ID. Sample IDs composed of only blank characters are treated as NULL. The System strips leading and trailing blanks. Embedded blanks are not removed and are considered as part of the Sample ID. Sample IDs are case sensitive and leading zeros are not ignored. A unique BRC Sample ID is generated by the ORTHO VISION® System for each execution of a BRC profile. The generated BRC Sample ID always starts with "BRC" and contains a locale based date/timestamp and a sequence number. For example: BRC_19/03/20140809_01
4	N	N	N	Instrument Specimen ID	<i>-Unused</i>
5	R	A	Y	Universal Test ID	-
5.1	R	A	Y	Profile Name	= Profile name Profiles must be known by the System. Profile names are case sensitive (ABO and abo are treated as different profile names). On upload: Only one profile per Order record. On download: Only one donor list per order record. Cross Match: On download, if the profile contains a cross match test, the following components are required: Number of donor samples, and one or more pairs containing the donor Sample ID and Sample type.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
5.2	X	N	N	Number (N) of Donor Samples	= number of donor samples or NULL This component is required if donor samples are included in the order. A crossmatch test may contain zero or more donor samples. If no donor samples are included for a crossmatch order, then the order will be held until a donor sample is manually added.
5.3	X	N	N	First Donor Specimen ID	= Sample ID of 1 st donor
5.4	X	N	N	Sample type of First Donor ID	= Sample type of 1 st donor
5.a	X *	N	N	n th Donor Specimen ID	= Sample ID of n th donor a=2*n+1, where 1<n≤N is the n th donor ID *Required for each additional donor (see 5.1, cross match)
5.b	X *	N	N	Sample type of n th Donor ID	= Sample type of n th donor b=2*n+2, where 1<n≤N is the n th donor ID *Required for each additional donor (see 5.1, cross match)
6	O	A	N	Priority	= NULL, S, A, R, C, P, or N. N is the same as R in LIS2-A, 9.4.6. NULL = R = N = C = P = routine priority. S = A = STAT priority.
7	O	A	N	Requested/Order Date and Time	= Date and time the Request/Order was recorded YYYYMMDDHHMMSS If Null, the ORTHO VISION® System uses the current time and always uploads a value.
8	N	N	N	Specimen Collection Date and Time	NULL in ASTM Enhanced compatibility mode YYYYMMDDHHMMSS
9	N	N	N	Collection End Time	-Unused
10	N	N	N	Collection Volume	-Unused
11	N	N	N	Collector ID	-Unused

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
12	R	N	N	Action Code	C, N, A. N = A C: cancel the described order. N A: Add profiles on a known sample, new profiles on an unknown sample. Q: Option not supported in ASTM Enhanced compatibility mode On Upload: NULL
13	N	N	N	Danger Code	- <i>Unused</i>
14	N	N	N	Relevant Clinical Info	- <i>Unused</i>
15	N	N	N	Date/Time Specimen Received	- <i>Unused</i>
16	R	A	N	Specimen Descriptor	=Sample type
17	N	N	N	Ordering Physician	- <i>Unused</i>
18	N	N	N	Physician's Phone Number	- <i>Unused</i>
19	O	N	N	User Field 1	= NULL or S S: save all the Cassettes for all the profiles
20	N	S	Y	User Field 2	= Comment Error message if this order could not be processed by the ORTHO VISION® System.
21	N	N	N	Laboratory Field 1	- <i>Unused</i>
22	N	N	N	Laboratory Field 2	- <i>Unused</i>
23	N	A	N	Date/Time Results Reported or Last Modified	= YYYYMMDDHHMMSS
24	N	N	N	Instrument Charge	- <i>Unused</i>
25	N	N	N	Instrument Section ID	- <i>Unused</i>
26	N	A	N	Report Types	= P, F, R, or X P: partial results F: final results R: repeat results X: order cancelled on the instrument
27	N	N	N	Reserved Field	- <i>Unused</i>

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
28	N	N	N	Location or Ward of Specimen Collection	NULL in compatibility mode
29	N	N	N	Nosocomial Infection Flag	- <i>Unused</i>
30	N	N	N	Specimen Service	- <i>Unused</i>
31	N	N	N	Specimen Institution	- <i>Unused</i>

3.5.5 ASTM R Result Record

The ORTHO VISION® System supports the following fields (not shaded) of the ASTM Result (R) Record. Defined in Section 10.1 of ASTM E 1394.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
1	-	A	N	Record Type ID	= "R" or "r"
2	-	A	N	Sequence number	Initial value is 1, reset for each new order; maximum length is unlimited.
3.1	-	A	N	Name of Analysis	= Analysis type
3.2	-	S	N	Donor Specimen ID	Included only if cross match test. = Sample ID of donor The System writes one R record per reaction.
4	-	A	N	Data or Measurement Value	= Analysis Result
5	-	N	N	Units of Measurement Value	- <i>Unused</i>
6	-	N	N	Reference Ranges	- <i>Unused</i>
7	-	S	N	Result Abnormal Flags	= NULL or M M: The result has been entered manually or one of the well results has been modified manually.
8	-	N	N	Nature of Abnormality Testing	- <i>Unused</i>
9	-	A	N	Result Status	= F, R, or X F: final result R: repeat result X: test cancelled or result rejected ⁴
10	-	N	N	Date of Change in Instrument Normative Values or Units	- <i>Unused</i>

⁴ The status cancelled is returned in any of the following cases:

- The corresponding test was cancelled by a LIS order cancel message (Action code = C).
- The corresponding test was cancelled by the user.
- The corresponding test was cancelled due to an unexpected, non-recoverable error (e.g., analytic or redundant analytic validation failed).
- The result was rejected by the user.

Note: In case of test restart (due to recovery) or unexpected errors, the does not send cancel messages to the LIS

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
11	-	S	N	Operator Identification	When "Include Operator" is set to No (Default). = Operator ID, Operator who accepts the test. Returns "Automatic" for tests accepted automatically.
11.1	-	S	N	Instrument Operator	When "Include Operator" is set to Yes. The logged in operator that loaded the samples/reagents.
11.2	-	S	N	Verifier	When "Include Operator" is set to Yes. The operator that accepted the test. Returns "Automatic" for tests accepted automatically.
12	-	N	N	Date/Time Test Started	- <i>Unused</i>
13	-	A	N	Date/Time Test Completed	= Date/Time of result YYYYMMDDHHMMSS
14	-	A	N	Instrument Identification	= Instrument ID

3.5.6 Enhanced ASTM Q Request Record

No compatibility issues. See [Vision ASTM Q Request Record](#)

3.5.7 Enhanced ASTM M Manufacturer Record

This Record is not supported by the ASTM format. Whenever a result record is transmitted, there is one record M (manufacturer) for each well result utilized for the result.

The ORTHO VISION® System supports the following fields of the Extended Result (M) Record. Defined in Section 15.1 of ASTM E 1394.

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
1	-	A	N	Record Type ID	= "M" or "m"
2	-	A	N	Sequence number	=1 for initial order, then reset for each new order; maximum length is unlimited
3	-	A	N	Result Well Name	= Name of the test well For cross match: donor Sample ID
4.1	-	A	N	Type of Cassette	= Type of Cassette
4.2	-	A	N	Number of the well	= 1..6

#	D	U	R	Field	Notes
D	R: Required; X: Required for Xmatch; O: Optional; N: Never used; - NA				
U	A: Always sent; X: Always sent for Xmatch; S: Sometimes sent; N: Never sent; -NA				
R	Y: Field can repeat; N: Field does not repeat				
4.3	-	A	N	Cassette ID Number	= serial # as given in the barcode.
4.4	-	A	N	Cassette Lot Number	-
4.5	-	A	N	Cassette Expiration Date	YYYYMMDDHHMMSS
4.6	-	S	N	Image File Name	= The file name of the Cassette image used in determining this result.
4.7	-	N	N	Color Image File Name	Not available in Enhanced ASTM Mode
5	-	A	Y	Reagent Information	
5.1	-	A	N	Reagent Name	= Reagent Name
5.2	-	A	N	Reagent Lot Number	-
5.3	-	A	N	Reagent Expiration Date	YYYYMMDDHHMMSS
6.1	-	A	N	Final Result or Error	= from Table 2 Results or Error
6.2	-	A	N	Manual Correction Flag	= M or A M: Manual correction A: Automatic correction
6.3	-	S	N	Read Result or Error	-
6.4	-	S	N	Operator ID	= The Operator ID of the operator that made the correction.

3.5.8 Sample ASTM and Enhanced ASTM Messages

The difference between Enhanced ASTM and ASTM formatted messages is that the ASTM formatted messages do not include the Manufacture's (M) record that contains the reaction grades.

The main differences between Vision ASTM and Enhanced ASTM formatted messages are:

- Enhanced ASTM does not support the attending physician field (P.14) in the Patient record.
- Enhanced ASTM does not export the color image file name in field M.4.7 of the Manufacture's (M) record.
- The Message Terminator Record in Enhanced ASTM and ASTM upload messages is always "L".

Other differences are described below.

Enhanced ASTM and ASMT example messages are not included when the only difference are those listed above.

See section [3.4.9 Sample Vision ASTM Messages](#) on page 47 for any example message not included below.

3.5.8.1 ASTM Keep-Alive Message

The ASTM Keep-Alive message is designed to prevent TCP/IP connections between the LIS and VISION® from being dropped during long periods where no messages are exchanged. Keep-Alive message is enabled by setting its interval value between 1-1440. A zero value disables sending the message. If the keep-alive message fails to be sent, VISION® will report an "Apsw26 Unable to Connect to the LIS".

```
VISION: <ENQ>
LIS    : <ACK>
VISION: <STX>1H|\^&|||OCD^VISION^5.14.0.47342^JNumber|||P|LIS2-A|20220902174004<CR><ETX>21<CR><LF>
LIS    : <ACK>
VISION: <STX>2L||<CR><ETX>86<CR><LF>
LIS    : <ACK>
VISION: <EOT>
```

3.5.8.2 Host Query

Same as Vision ASTM. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

3.5.8.3 Minimal Order

Same as Vision ASTM. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

3.5.8.4 Example Order

```
H|\^&||Mini LIS|||||LIS2-A|20210311133429
P|1|PID123456||NID123456^MID123456^OID123456|Brown^Bobby^B|White|196501020304|U|||PHY1234^Kildare^James^P|Blaine
O|1|SID305||ABO|N|20210311133429||||N|||CENTBLOOD
L
```

3.5.8.5 Example ASTM Result Message

```
H|\^&||OCD^VISION^5.13.1.46935^JNumber|||||P|LIS2-A|20210311135953
P|1|PID123456||NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U|||||Blaine|||||||
O|1|SID305||ABO|N|20210311135932|||||CENTBLOOD|||||20210311135952||F||||
R|1|ABO|AB||||F|Automatic||20210311135952|JNumber
R|2|Rh|NEG||||F|Automatic||20210311135952|JNumber
L
```

3.5.8.6 Example Result

Enhanced ASTM Result with “Strip trailing delimiters” set to “No”:

The Order record report type (O.26) is set to ‘F’ for final results and all Result record Result Status field (R.6) is set to ‘F’ for final results.

```
H|\^&||OCD^VISION^5.13.1.46935^JNumber|||||P|LIS2-A|20210311140214
P|1|PID123456||NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U|||||Blaine|||||||
O|1|SID305||ABO|N|20210311140148|||||CENTBLOOD|||||20210311140214||F||||
R|1|ABO|O||||F|Automatic||20210311140214|JNumber
M|1|Anti-A|ABD Confirmation^4^000001^77777^20210828235959^20210311_140208Grey.jpg||0^A
M|2|Anti-B|ABD Confirmation^5^000001^77777^20210828235959^20210311_140208Grey.jpg||0^A
R|2|Rh|POS||||F|Automatic||20210311140214|JNumber
M|1|Anti-D|ABD Confirmation^6^000001^77777^20210828235959^20210311_140208Grey.jpg||40^A
L
```

Enhanced ASTM Result with “Strip trailing delimiters” set to “Yes”:

Notice the lack of trailing delimiters in the P, O, and R records. This message was resent by the operator. The Order record report type (O.26) is set to ‘F’ for final results and the Result record Result Status field (R.6) is set to ‘R’ for repeat results. This is different than what Vision ASTM does. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber| |||||P|LIS2-A|20210311140340
P|1|PID123456| |NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U| |||||Blaine
O|1|SID305| |ABO|N|20210311140148| |||||CENTBLOOD| |||||20210311140214| ||F
R|1|ABO|O| |||||R| |Automatic| |20210311140214|JNumber
M|1|Anti-A|ABD Confirmation^4^000001^77777^20210828235959^20210311_140208Grey.jpg| |0^A
M|2|Anti-B|ABD Confirmation^5^000001^77777^20210828235959^20210311_140208Grey.jpg| |0^A
R|2|Rh|POS| |||||R| |Automatic| |20210311140214|JNumber
M|1|Anti-D|ABD Confirmation^6^000001^77777^20210828235959^20210311_140208Grey.jpg| |40^A
L
```

3.5.8.7 Crossmatch Order

Enhanced ASTM and ASTM crossmatch orders are the same as Vision ASTM.

```
H|\^&|||Mini LIS| |||||LIS2-A|20210311140340
P|1|PID123456| |NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U| |||||PHY1234^Kildare^James^P|Blaine| |||||
O|1|SID101| |XM^2^=W13131200096100^PACKEDCELLS^=W13131200096200^PACKEDCELLS|N|20210311140340| |||||N| |||||PLASMA| |||||
L| |
```

3.5.8.8 Crossmatch Result

Vision ASTM echoes back the donor list in the Order record whereas Enhanced ASTM and ASTM formats do not.

```
H|\^&|||OCD^VISION^5.13.1.46935^JNumber| |||||P|LIS2-A|20210311141250
P|1|PID123456| |NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U| |||||Blaine
O|1|SID101| |XM|N|20210311141214| |||||PLASMA| |||||20210311141249| ||F
R|1|XM^=W13131200096100|CMP| |||||F| |Automatic| |20210311141249|JNumber
M|1|=W13131200096100|AHG Polyspecific^5^000037^77777^20210828235959^20210311_141242Grey.jpg|BLISS^1234^20210828235959|0^A
R|2|XM^=W13131200096200|CMP| |||||F| |Automatic| |20210311141249|JNumber
M|1|=W13131200096200|AHG Polyspecific^6^000037^77777^20210828235959^20210311_141242Grey.jpg|BLISS^1234^20210828235959|0^A
L
```

3.5.8.9 Order with 2 Samples

Same as Vision ASTM. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

3.5.8.10 Result with 2 Samples

Same as Vision ASTM. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

3.5.8.11 Order with Multiple Profiles

Same as Vision ASTM. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

3.5.8.12 Profile with Multiple Tests

Same as Vision ASTM. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

3.5.8.13 Donor Confirmation Test

Same as Vision ASTM. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

3.5.8.14 Dilution Series Test

Same as Vision ASTM. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

3.5.8.15 Ortho Rare Antisera Result

Same as Vision ASTM. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

3.5.8.16 Cancelled Orders

Same as Vision ASTM. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

3.5.8.17 Quality Control Examples

Quality control tests can be ordered through the LIS or manually entered on VISION®. QC tests are ordered against a Profile.

Quality Control Order (Managed by VISION®):

When Quality Controls are managed by Vision and is configured for Enhanced ASTM or ASTM; all Quality Control orders must be manually entered on VISION®.

Quality Control Result (Managed by LIS):

Same as Vision ASTM. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

3.5.8.18 BRC Results

Same as Vision ASTM. See section [3.4.9 Sample Vision ASTM Messages](#) on page 47.

3.6 Compatibility with ORTHO AutoVue® and ORTHO AutoVue® Ultra Systems

This section provides a quick reference for the comparison of the ORTHO VISION® System LIS to the ORTHO AutoVue® and ORTHO AutoVue® Ultra Systems LIS. This section can be used as an aide by LIS experts to transition an existing LIS from the ORTHO AutoVue® Innova and ORTHO AutoVue® Ultra Systems to the ORTHO VISION® System.

Note: The order creation time is provided from the ORTHO VISION® System.

3.6.1 ORTHO VISION® System LIS Interface compatibility with ORTHO AutoVue®

The ORTHO VISION® System (VISION® System) Laboratory Information System (LIS) interface was designed to be compatible with Laboratory Information Systems that currently support the ORTHO AutoVue® Innova/Ultra System (AutoVue). VISION® is a new instrument; it is not a new version of AutoVue. VISION® was designed to be compatible, not interchangeable. Compatible so that it would be relatively easy to connect to an LIS that has an existing AutoVue driver. An LIS that already has a working AutoVue driver can leverage that driver when creating a VISION® driver. This document describes some of the differences between the two interfaces.

3.6.2 A Quick Look

The following example messages (Example 1 through 3) highlight some of the differences with the ASTM message content. These differences are discussed in the rest of the section.

Example 1: Order

```
H|^&||Mini LIS|||||LIS2-A|20140530144204
P|1|PID123456||NID123456^MID123456^OID123456|Brown^Bobby^B|White|196501020304|U|||||
O|1|SID05||ABO-D|N|20140530144925||||N|||CENTBLOOD|||||
L|
```

Example 2: AutoVue Enhanced ASTM Result

```
H|\^&||OCD^AV2G^1.0^J123456|||||P|1|20140202232445
P|1|PID123456|NID123456^MID123456^OID123456|Brown^Bobby||19650102030400|U|||||||
O|1|SID05|ABO-D|N|20140202232352|||||CENTBLOOD|||||20140202232435||F||||
R|1|ABO|A||M|F|theron|20140202232435|J123456
M|1|Anti-A|ABO-Rh/Reverse^1^000001^12345^20340202235959^22046516_001.bmp||0^A
M|2|Anti-B|ABO-Rh/Reverse^2^000001^12345^20340202235959^22046516_001.bmp||40^A
M|3|Ctrl|ABO-Rh/Reverse^4^000001^12345^20340202235959^22046516_001.bmp||40^A
R|2|Rh|POS||M|F|theron|20140202232435|J123456
M|1|Anti-D|ABO-Rh/Reverse^3^000001^12345^20340202235959^22046516_001.bmp||40^A
M|2|Ctrl|ABO-Rh/Reverse^4^000001^12345^20340202235959^22046516_001.bmp||40^A
L
```

Example 3: VISION® System Enhanced ASTM Result

```
H|\&|||OCD^VISION^0.84.0.39963^J123456| |||||P|LIS2-A|20140530145835
P|1PID123456|NID123456^MID123456^OID123456|Brown^Bobby^B|White|19650102030400|U| |||||
O|1SID005|ABO-D|N|20140530145736| |||||CENTBLOOD| |||||20140530145835| |F| |||
R|1ABO|AB| |||F|Automatic| |20140530145834|J123456
M|1Anti-A|ABO-Rh/Reverse^1^200006^00001^20150101235959^20140530_145826Grey.jpg| |40^A
M|2Anti-B|ABO-Rh/Reverse^2^200006^00001^20150101235959^20140530_145826Grey.jpg| |40^A
M|3Ctrl|ABO-Rh/Reverse^4^200006^00001^20150101235959^20140530_145826Grey.jpg| |0^A
R|2Rh|POS| |||F|Automatic| |20140530145834|J123456
M|1Anti-D|ABO-Rh/Reverse^3^200006^00001^20150101235959^20140530_145826Grey.jpg| |40^A
M|2Ctrl|ABO-Rh/Reverse^4^200006^00001^20150101235959^20140530_145826Grey.jpg| |0^A
L
```

3.6.3 Header Record

Example 4 highlights the differences in header records:

- The product name (H.5.2) reflects the name of the instrument.
- The software version (H.5.3) reflects each instruments software version.
- The Version number (H.13) reflects the ASTM protocol supported by the instrument. In AutoVue, a value of 1 indicates E1394-97.

Example 4: Header Records

AutoVue	H \^& OCD^AV2G^1.0^J123456 P 1 20140202232445
VISION® System	H \^& OCD^VISION^0.62.0.36863^J123456 P LIS2-A 20140131113948

3.6.4 Patient Record

The VISION® System supports the patients middle initial (P.6.3) and mother's maiden name (P.7), AutoVue does not. See Example 5.

Example 5: Order Patient Record

AutoVue	P 1 PID123456 NID123456^MID123456^OID123456 Brown^Bobby 19650102030400 U
VISION® System	P 1 PID123456 NID123456^MID123456^OID123456 Brown^Bobby^B White 19650102030400 U

The VISION® System and AutoVue allow the LIS to send orders without patient demographics. Result messages from these orders differ between the two systems (see examples 6 and 7):

- AutoVue ignores other patient demographics fields when an order does not contain a patient ID (P.3 is null).
- The VISION® System does not ignore other patient demographics fields when an order does not contain a patient ID (P.3 is null).
- When the patient ID (P.3) is null, AutoVue returns a zero for the patient id (P.3) and nulls for all other patient demographics fields.
- The VISION® System returns Unknown (U) for patient sex (P.9) when it receives an order that does not contain the patient sex (P.9 is null).

Example 6: Patient Record without Patient Demographics

LIS Order	P 1
AutoVue	P 1 0
VISION® System	P 1 U

Example 7: Patient Record without Patient ID

LIS Order	P 1 NID5^MID5^OID5 Brown^Bobby^B White 19650102030405 M
AutoVue	P 1 0
VISION® System	P 1 NID5^MID5^OID5 Brown^Bobby^B White 19650102030400 M

3.6.5 Manufacturer Information Record

Example 8 highlights the differences in Manufacturer Information Records.

- The Cassette image file name (M.4.6) and image type differ between AutoVue and the VISION® System. The AutoVue image type is BMP and the VISION® System uses JPEG (JPG). The file base names differ as well (see M.4.6 in example 8 below).

Example 8: Enhanced ASTM Manufacturer Information Record

AutoVue	M 1 Anti-A ABO-Rh/Reverse^1^000001^12345^20340202235959^22046516_001.bmp 0^A
VISION® System	M 1 Anti-A ABO-Rh/Reverse^1^100001^00001^20150101235959^20140131_113947Grey.jpg 40^A

3.6.6 Download File Pattern is Case Sensitive

Windows file systems are case-preserving and case-insensitive. The VISION® System compares the download file pattern to the file name returned by Windows in a case-sensitive manner. The download file pattern LIS.DNL is considered by the VISION® System as different from a file named lis.dnl.

AutoVue is case-insensitive.

3.6.7 LIS Order Edits

Patient demographic information may not be correct when it is first entered into an LIS. When the information is corrected, an LIS may update orders on AutoVue instruments with the correct information.

AutoVue allows the LIS to update patient demographics as follows:

- LIS downloads an order.
- Demographic information is corrected on the LIS.
- LIS downloads the order again with the corrected demographic information.
- AutoVue prompts the operator to accept or reject the changes. When the operator accepts the changes the associated patient demographics are updated in AutoVue.

AutoVue enforces that a sample can be associated with only one patient and one or more orders. A patient can be associated with one or more samples and orders.

The VISION® System does not support LIS order edits. If an LIS sends an order edit (update), the edit will be treated as a new order and will be scheduled.

3.6.8 Operator Rejected Results

- The AutoVue result rejected notification message, sets the rejected result records for each analysis with the Result Status set to test canceled (R.9 set to X).
- The VISION® System only uploads accepted results. The VISION® System does not upload a message to the LIS if all results are rejected.

[illegible]

	M 1 Anti-K ADK^5^000002^02345^20340212235959^123.bmp 40^A M 2 Ctrl ADK^6^000002^02345^20340212235959^123.bmp 0^A R 4 ABScr POS F theron 20140212155313 J123456 M 1 Surg 1 AHG anti-IgG^1^000005^22310^20330507235959^123.bmp BLISS^0028^20160128235959\Surg 1^0126^20160128235959 0^A M 2 Surg 2 AHG anti-IgG^2^000005^22310^20330507235959^123.bmp BLISS^0028^20160128235959\Surg 2^0126^20160128235959 40^A M 3 Surg 3 AHG anti-IgG^3^000005^22310^20330507235959^123.bmp BLISS^0028^20160128235959\Surg 3^0126^20160128235959 40^A R 5 Auto POS F theron 20140212155313 J123456 M 1 Auto AHG anti-IgG^4^000005^22310^20330507235959^123.bmp BLISS^0028^20160128235959 40^A
VISION® System Result Message	P U O 1 Donor01 BG+AutoControl N 20140213151036 CENTBLOOD 20140213151319 F R 1 ABScr POS F soladmin 20140213151201 J123456 M 1 Surg 1 AHG anti-IgG^1^100002^00001^20150101235959^20140213_151158Grey.jpg BLISS^0021^20160121235959\Surg 1^0021^20160121235959 0^A M 2 Surg 2 AHG anti-IgG^2^100002^00001^20150101235959^20140213_151158Grey.jpg BLISS^0021^20160121235959\Surg 2^0021^20160121235959 10^A M 3 Surg 3 AHG anti-IgG^3^100002^00001^20150101235959^20140213_151158Grey.jpg BLISS^0021^20160121235959\Surg 3^0021^20160121235959 30^A R 2 Auto NEG F soladmin 20140213151201 J123456 M 1 Auto AHG anti-IgG^4^100002^00001^20150101235959^20140213_151158Grey.jpg BLISS^0021^20160121235959 0^A

3.6.9 Error Response

The VISION® System cancels an order and optionally includes an error description when it encounters an error reading the order.

AutoVue does not have this feature.

Example 10: Error Response

H \^& OCD^VISION^0.84.0.39963^J123456 P LIS2-A 20140527110118 P 1 PID123456 NID123456^MID123456^OID123456 Brown^Bobby^B White 19650102030400 U O 1 SID005 ABO-F N 20140527110118 CENTBLOOD Profile with name [ABO-F] not found! 20140527110118 X
--

3.6.10 Trace File Names

AutoVue trace file name format is Prefix + Timestamp + Extension. Prefixes are DNL for download, UP for upload, and QU for query files. A timestamp follows the ASTM date/time format of YYYYMMDDhhmmss (i.e. 20140605103721). The file extension is the same for all types; ".TRA".

Example AutoVue trace file names:

- DNL20140605103721.TRA
- UP20140605103838.TRA
- QU20140605103058.TRA

VISION® System

The download and upload trace file name format is Prefix + Timestamp + SequenceNumber + Extension. Prefixes are DNL for download, and UP for upload (same as AutoVue). A timestamp follows the ASTM date/time format of YYYYMMDDhhmmss (i.e. 20140605103721). The SequenceNumber is only written when more than one file is written within the same second. The file extension is the same for all types; ".tra".

The query trace file name format is Prefix + SequenceNumber + Timestamp + Extension. Prefix is QU for query trace files (same as AutoVue). The SequenceNumber is three digits followed by a dash and is always written. The timestamp follows the ASTM date/time format of YYYYMMDDhhmmss (i.e. 20140605103721). The file extension is the same for all types; ".tra".

Example VISION® System trace file names:

- DNL20140530132501.tra
- DNL20140530132941-0.tra
- UP20140530132501.tra
- QU001-20140530135928.tra

3.6.11 Physical Layer

The VISION® System can be configured to communicate to an LIS by either TCP/IP via Ethernet, Serial RS232 or Network File sharing.

AutoVue can be configured to communicate to an LIS by either FTP, Serial RS232 or Network File sharing.

Physical Layer				
	Serial RS232	TCP/IP via Ethernet	File sharing	FTP
AutoVue	Kermit	Not Supported	via network shared folders	Connects to LIS FTP server
VISION® System	ASTM E1381-02	ASTM E1381-02	via network shared folders	Not Supported

4. Test Specific Content

The following sections contain test specific names included in all LIS Result messages.

4.1 BioVue Tests

The following are the BioVue tests, organized in small tables by Unit Results.

Test names are prefixed with their Cassette IDs. For example, in the test name “00: ABO(FWD)/Rh-00” below, the “00:” prefix is the BioVue Cassette ID for the “ABO-Rh/Reverse” Cassette. See section 4.2 for a list of Cassettes.

Note: Tests are not available in all regions.

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
10001	00: ABO(FWD)/Rh-00	ABO Rh	Anti-A Anti-B Ctrl		Anti-D Ctrl
10002	48: ABO(FWD)-ABODD-48	ABO Rh	Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl
10003	40: ABO(FWD)-ADDK-40	ABO Rh Kell	Anti-A Anti-B Ctrl	Anti-K Ctrl	Anti-D Anti-D Ctrl
10004	44: ABO(FWD)-D/CDE-44	ABO Rh	Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl
10005	00: ABO(FWD)-00	ABO	Anti-A Anti-B Ctrl		

				Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh	
10006	44: ABO(FWD)-44	ABO	Anti-A Anti-B Anti-A+B Ctrl			
10007	10: ABD Conf-10	ABO Rh	Anti-A Anti-B		Anti-D	
10008	10: ABD Conf New Susp-10	ABO Rh	Anti-A Anti-B		Anti-D	
10013	66: 4 ABO(RVS)-A1,A2,B	ABO	A1-Cells A2-Cells B-Cells			
10014	66: 4 ABO(RVS)-A1,A2,B,O	ABO	A1-Cells A2-Cells B-Cells O-Cells			
10015	66: 4 ABO(RVS)-A1,B	ABO	A1-Cells B-Cells			
10016	66: 4 ABO(RVS)-A1,B,O	ABO	A1-Cells B-Cells O-Cells			
10017	66: 08 ABO(RVS)-A1,A2,B	ABO	A1-Cells A2-Cells B-Cells			
10018	66: 08 ABO(RVS)-A1,A2,B,O	ABO	A1-Cells A2-Cells B-Cells O-Cells			
10019	66: 08 ABO(RVS)-A1,B	ABO	A1-Cells B-Cells			

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
10020	66: 08 ABO(RVS)-A1,B,O	ABO	A1-Cells B-Cells O-Cells		
10021	00: 4 ABO(FWD/RVS)/Rh-00	ABO Rh	Anti-A Anti-B Ctrl A1-Cells B-Cells		Anti-D Ctrl
10022	00: 08 ABO(FWD/RVS)/Rh-00	ABO Rh	Anti-A Anti-B Ctrl A1-Cells B-Cells		Anti-D Ctrl
10023	44,66: 4 ABO(FWD)-44 + (RVS)-A1,A2,B	ABO	A1-Cells A2-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl		
10024	44,66: 4 ABO(FWD)-D/CDE-44 + (RVS)-A1,A2,B	ABO Rh	A1-Cells A2-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl
10025	48,66: 4 ABO(FWD)-ABODD-48 + (RVS)-A1,A2,B	ABO Rh	A1-Cells A2-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
10026	40,66: 4 ABO(FWD)-ADDK-40 + (RVS)-A1,A2,B	ABO Rh Kell	A1-Cells A2-Cells B-Cells Anti-A Anti-B Ctrl	Anti-K Ctrl	Anti-D Anti-D Ctrl
10027	44,66: 4 ABO(FWD)-44 + (RVS)-A1,A2,B,O	ABO	A1-Cells A2-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		
10028	44,66: 4 ABO(FWD)-D/CDE-44 + (RVS)-A1,A2,B,O	ABO Rh	A1-Cells A2-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl
10029	48,66: 4 ABO(FWD)-ABODD-48 + (RVS)-A1,A2,B,O	ABO Rh	A1-Cells A2-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
10030	40,66: 4 ABO(FWD)-ADDK-40 + (RVS)-A1,A2,B,O	ABO Rh Kell	A1-Cells A2-Cells B-Cells O-Cells Anti-A Anti-B Ctrl	Anti-K Ctrl	Anti-D Anti-D Ctrl
10031	44,66: 4 ABO(FWD)-44 + (RVS)-A1,B	ABO	A1-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl		
10032	44,66: 4 ABO(FWD)-D/CDE-44 + (RVS)-A1,B	ABO Rh	A1-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl
10033	48,66: 4 ABO(FWD)-ABODD-48 + (RVS)-A1,B	ABO Rh	A1-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl
10034	40,66: 4 ABO(FWD)-ADDK-40 + (RVS)-A1,B	ABO Rh Kell	A1-Cells B-Cells Anti-A Anti-B Ctrl	Anti-K Ctrl	Anti-D Anti-D Ctrl

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
10035	44,66: 4 ABO(FWD)-44 + (RVS)-A1,B,O	ABO	A1-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		
10036	44,66: 4 ABO(FWD)-D/CDE-44 + (RVS)-A1,B,O	ABO Rh	A1-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl
10037	48,66: 4 ABO(FWD)-ABODD-48 + (RVS)-A1,B,O	ABO Rh	A1-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl
10038	40,66: 4 ABO(FWD)-ADDK-40 + (RVS)-A1,B,O	ABO Rh Kell	A1-Cells B-Cells O-Cells Anti-A Anti-B Ctrl	Anti-K Ctrl	Anti-D Anti-D Ctrl
10039	44,66: 08 ABO(FWD)-44 + (RVS)-A1,A2,B	ABO	A1-Cells A2-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl		

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
10040	44,66: 08 ABO(FWD)-D/CDE-44 + (RVS)-A1,A2,B	ABO Rh	A1-Cells A2-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl
10041	48,66: 08 ABO(FWD)-ABODD-48 + (RVS)-A1,A2,B	ABO Rh	A1-Cells A2-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl
10042	40,66: 08 ABO(FWD)-ADDK-40 + (RVS)-A1,A2,B	ABO Rh Kell	A1-Cells A2-Cells B-Cells Anti-A Anti-B Ctrl	Anti-K Ctrl	Anti-D Anti-D Ctrl
10043	44,66: 08 ABO(FWD)-44 + (RVS)-A1,A2,B,O	ABO	A1-Cells A2-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
10044	44,66: 08 ABO(FWD)-D/CDE-44 + (RVS)-A1,A2,B,O	ABO Rh	A1-Cells A2-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl
10045	48,66: 08 ABO(FWD)-ABODD-48 + (RVS)-A1,A2,B,O	ABO Rh	A1-Cells A2-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl
10046	40,66: 08 ABO(FWD)-ADDK-40 + (RVS)-A1,A2,B,O	ABO Rh Kell	A1-Cells A2-Cells B-Cells O-Cells Anti-A Anti-B Ctrl	Anti-K Ctrl	Anti-D Anti-D Ctrl
10047	44,66: 08 ABO(FWD)-44 + (RVS)-A1,B	ABO	A1-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl		

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
10048	44,66: 08 ABO(FWD)-D/CDE-44 + (RVS)-A1,B	ABO Rh	A1-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl
10049	48,66: 08 ABO(FWD)-ABODD-48 + (RVS)-A1,B	ABO Rh	A1-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl
10050	40,66: 08 ABO(FWD)-ADDK-40 + (RVS)-A1,B	ABO Rh Kell	A1-Cells B-Cells Anti-A Anti-B Ctrl	Anti-K Ctrl	Anti-D Anti-D Ctrl
10051	44,66: 08 ABO(FWD)-44 + (RVS)-A1,B,O	ABO	A1-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		
10052	44,66: 08 ABO(FWD)-D/CDE-44 + (RVS)-A1,B,O	ABO Rh	A1-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
10053	48,66: 08 ABO(FWD)-ABODD-48 + (RVS)-A1,B,O	ABO Rh	A1-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl
10054	40,66: 08 ABO(FWD)-ADDK-40 + (RVS)-A1,B,O	ABO Rh Kell	A1-Cells B-Cells O-Cells Anti-A Anti-B Ctrl	Anti-K Ctrl	Anti-D Anti-D Ctrl
10055	66: 4 ABO(RVS)-6 cell	ABO	A1-Cells A2-Cells B-Cells O-Cells O-Cells O-Cells		
10056	66: 08 ABO(RVS)-6 cell	ABO	A1-Cells A2-Cells B-Cells O-Cells O-Cells O-Cells		

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
10057	44,66: 4 ABO(FWD)-44 + (RVS)-6 cell	ABO	A1-Cells A2-Cells B-Cells O-Cells O-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		
10058	44,66: 4 ABO(FWD)-D/CDE-44 + (RVS)-6 cell	ABO Rh	A1-Cells A2-Cells B-Cells O-Cells O-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl
10059	48,66: 4 ABO(FWD)-ABODD-48 + (RVS)-6 cell	ABO Rh	A1-Cells A2-Cells B-Cells O-Cells O-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
10060	40,66: 4 ABO(FWD)-ADDK-40 + (RVS)-6 cell	ABO Rh Kell	A1-Cells A2-Cells B-Cells O-Cells O-Cells O-Cells Anti-A Anti-B Ctrl	Anti-K Ctrl	Anti-D Anti-D Ctrl
10061	44,66: 08 ABO(FWD)-44 + (RVS)-6 cell	ABO	A1-Cells A2-Cells B-Cells O-Cells O-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl		
30017	66: UPR ABO-Sim 2 (3)	ABO	Hem-A1 Hem-A2 Hem-B		
30018	66: UPR ABO-Sim 3 (4)	ABO	Hem-A1 Hem-A2 Hem-B Hem-O		
30019	66: UPR ABO-Sim 4 (2)	ABO	Hem-A1 Hem-B		
30020	66: UPR ABO-Sim 5 (3-O)	ABO	Hem-A1 Hem-B Hem-O		

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
30044	44,66: UPR ABO/D-44 + Sim 4	ABO Rh	Hem-A1 Hem-A2 Hem-B Hem-O Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl
30045	48,66: UPR ABO/DD-48 + Sim 4	ABO Rh	Hem-A1 Hem-A2 Hem-B Hem-O Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl
30048	44,66: UPR ABO/D-44 + Sim 2	ABO Rh	Hem-A1 Hem-B Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl
30049	48,66: UPR ABO/DD-48 + Sim 2	ABO Rh	Hem-A1 Hem-B Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl

			Unit Results		
Test ID	Test Name	Analyses	ABO	Kell	Rh
30052	44,66: UPR ABO/D-44 + Sim 3-O	ABO Rh	Hem-A1 Hem-B Hem-O Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-CDE Ctrl
30053	48,66: UPR ABO/DD-48 + Sim 3-O	ABO Rh	Hem-A1 Hem-B Hem-O Anti-A Anti-B Anti-A+B Ctrl		Anti-D Anti-D Ctrl

Test ID	Test Name	Analyses	Unit Results				
			ABScr	Auto	Kell	Pheno	Rh
10009	11: Rh-hr-11	Rh Pheno				Anti-C Anti-E Anti-c Anti-e Ctrl	Anti-D Ctrl
10010	77: Rh/K-77	Pheno Kell			Anti-K Ctrl	Anti-C Anti-E Anti-c Anti-e Ctrl	
10066	22: 08 AbScr Sel Poly	ABScr	0.8-Sel I 0.8-Sel II				
10067	33: 08 AbScr Sel IgG	ABScr	0.8-Sel I 0.8-Sel II				
10068	22: 08 AbScr Surg Poly	ABScr	0.8-Surg1 0.8-Surg2 0.8-Surg3				
10069	33: 08 AbScr Surg IgG	ABScr	0.8-Surg1 0.8-Surg2 0.8-Surg3				
10070	22: 08 AbScr BVSF Unt Poly	ABScr	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3				
10071	33: 08 AbScr BVSF Unt IgG	ABScr	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3				
10072	22: 08 ABSCR4 Poly	ABScr	0.8-ABSCR4 Cell 1 0.8-ABSCR4 Cell 2 0.8-ABSCR4 Cell 3 0.8-ABSCR4 Cell 4				

Test ID	Test Name	Analyses	Unit Results				
			ABScr	Auto	Kell	Pheno	Rh
10073	33: 08 ABSCR4 IgG	ABScr	0.8-ABSCR4 Cell 1 0.8-ABSCR4 Cell 2 0.8-ABSCR4 Cell 3 0.8-ABSCR4 Cell 4				
10074	22: 08 AbScr Sel+Auto Poly	ABScr Auto	0.8-Sel I 0.8-Sel II	Auto			
10075	33: 08 AbScr Sel+Auto IgG	ABScr Auto	0.8-Sel I 0.8-Sel II	Auto			
10076	22: 08 AbScr Surg+Auto Poly	ABScr Auto	0.8-Surg1 0.8-Surg2 0.8-Surg3	Auto			
10077	33: 08 AbScr Surg+Auto IgG	ABScr Auto	0.8-Surg1 0.8-Surg2 0.8-Surg3	Auto			
10078	22: 08 AbScr BVSF Unt+Auto Poly	ABScr Auto	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3	Auto			
10079	33: 08 AbScr BVSF Unt+Auto IgG	ABScr Auto	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3	Auto			
10080	22: 4 AbScr Sel Poly	ABScr	Sel I Sel II				
10081	22: 4 AbScr Surg Poly	ABScr	Surg 1 Surg 2 Surg 3				
10082	33: 4 AbScr Sel IgG	ABScr	Sel I Sel II				
10083	33: 4 AbScr Surg IgG	ABScr	Surg 1 Surg 2 Surg 3				

Test ID	Test Name	Analyses	Unit Results				
			ABScr	Auto	Kell	Pheno	Rh
10084	22: 4 AbScr Surg+Dia Poly	ABScr	Surg 1 Surg 2 Surg 3 Diego				
10085	33: 4 AbScr Surg+Dia IgG	ABScr	Surg 1 Surg 2 Surg 3 Diego				
10086	22: 4 AbScr Sel+Dia Poly	ABScr	Sel I Sel II Diego				
10087	33: 4 AbScr Sel+Dia IgG	ABScr	Sel I Sel II Diego				
10088	22: 4 AbScr Dia Poly	ABScr	Diego				
10089	33: 4 AbScr Dia IgG	ABScr	Diego				
10090	22: 4 AbScr BVSF Unt Poly	ABScr	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3				
10091	33: 4 AbScr BVSF Unt IgG	ABScr	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3				
10092	22: 4 AbScr BVSF Unt+Dia Poly	ABScr	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3 Diego				
10093	33: 4 AbScr BVSF Unt+Dia IgG	ABScr	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3 Diego				

Test ID	Test Name	Analyses	Unit Results				
			ABScr	Auto	Kell	Pheno	Rh
10094	22: 4 ABSCR4 Poly	ABScr	ABSCR4 Cell 1 ABSCR4 Cell 2 ABSCR4 Cell 3 ABSCR4 Cell 4				
10095	33: 4 ABSCR4 IgG	ABScr	ABSCR4 Cell 1 ABSCR4 Cell 2 ABSCR4 Cell 3 ABSCR4 Cell 4				
10096	22: 4 AbScr Sel+Auto Poly	ABScr Auto	Sel I Sel II	Auto			
10097	22: 4 AbScr Surg+Auto Poly	ABScr Auto	Surg 1 Surg 2 Surg 3	Auto			
10098	33: 4 AbScr Sel+Auto IgG	ABScr Auto	Sel I Sel II	Auto			
10099	33: 4 AbScr Surg+Auto IgG	ABScr Auto	Surg 1 Surg 2 Surg 3	Auto			
10100	22: 4 AbScr Surg+Dia+Auto Poly	ABScr Auto	Surg 1 Surg 2 Surg 3 Diego	Auto			
10101	33: 4 AbScr Surg+Dia+Auto IgG	ABScr Auto	Surg 1 Surg 2 Surg 3 Diego	Auto			
10102	22: 4 AbScr Sel+Dia+Auto Poly	ABScr Auto	Sel I Sel II Diego	Auto			
10103	33: 4 AbScr Sel+Dia+Auto IgG	ABScr Auto	Sel I Sel II Diego	Auto			

Test ID	Test Name	Analyses	Unit Results				
			ABScr	Auto	Kell	Pheno	Rh
10104	22: 4 AbScr Dia+Auto Poly	ABScr Auto	Diego	Auto			
10105	33: 4 AbScr Dia+Auto IgG	ABScr Auto	Diego	Auto			
10106	22: 4 AbScr BVSF Unt+Dia+Auto Poly	ABScr Auto	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3 Diego	Auto			
10107	33: 4 AbScr BVSF Unt+Dia+Auto IgG	ABScr Auto	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3 Diego	Auto			
10108	22: 4 AbScr BVSF Unt+Auto Poly	ABScr Auto	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3	Auto			
10109	33: 4 AbScr BVSF Unt+Auto IgG	ABScr Auto	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3	Auto			
10110	88: 4 BVSF Trt Neut	ABScr	Fic-Trt 1 Fic-Trt 2 Fic-Trt 3				
10111	88: 08 BVSF Trt Neut	ABScr	Fic-Trt 1 Fic-Trt 2 Fic-Trt 3				
10112	55: 08 BVSF Poly/Neut	ABScr	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3 Fic-Trt 1 Fic-Trt 2 Fic-Trt 3				

			Unit Results				
Test ID	Test Name	Analyses	ABScr	Auto	Kell	Pheno	Rh
10113	55: 4 BVSF Poly/Neut	ABScr	Fic-Unt 1 Fic-Unt 2 Fic-Unt 3 Fic-Trt 1 Fic-Trt 2 Fic-Trt 3				
10114	88: Bro 2セルスクリーン	ABScr	Br-Sel I Br-Sel II				
10115	88: Bro 3セルスクリーン	ABScr	Br-Surg 1 Br-Surg 2 Br-Surg 3				
10116	88: Bro Dia Neut	ABScr	Br-Diego				
10117	88: Bro 2セルスクリーン+Dia	ABScr	Br-Sel I Br-Sel II Br-Diego				
10118	88: Bro 3セルスクリーン+Dia	ABScr	Br-Surg 1 Br-Surg 2 Br-Surg 3 Br-Diego				
10119	55: Bro 2セルスクリーン PLN	ABScr	Sel I Sel II Br-Sel I Br-Sel II				
10120	55: Bro 3セルスクリーン PLN	ABScr	Surg 1 Surg 2 Surg 3 Br-Surg 1 Br-Surg 2 Br-Surg 3				

Test ID	Test Name	Analyses	Unit Results				
			ABScr	Auto	Kell	Pheno	Rh
10121	55: Bro 2セル+Dia PLN	ABScr	Sel I Sel II Diego Br-Sel I Br-Sel II Br-Diego				
10122	55: Bro 2セル+自己 PLN	ABScr Auto	Sel I Sel II Br-Sel I Br-Sel II	Auto Auto			
10123	88: Bro Dia+自己	ABScr Auto	Br-Diego	Auto			
10124	88: Bro 2セルスクリーン+自己	ABScr Auto	Br-Sel I Br-Sel II	Auto			
10125	88: Bro 3セルスクリーン+自己	ABScr Auto	Br-Surg 1 Br-Surg 2 Br-Surg 3	Auto			
10126	88: Bro 2セル+Dia+自己	ABScr Auto	Br-Sel I Br-Sel II Br-Diego	Auto			
10127	88: Bro 3セル+Dia+自己	ABScr Auto	Br-Surg 1 Br-Surg 2 Br-Surg 3 Br-Diego	Auto			
30068	22: UPR 8 DEP_Nat 3 Poly	ABScr	DE_Nat-1 DE_Nat-2 DE_Nat-3				
30076	22: UPR 8 DEP_Nat 3 + Auto Poly	ABScr Auto	DE_Nat-1 DE_Nat-2 DE_Nat-3	08_Auto			

			Unit Results				
Test ID	Test Name	Analyses	ABScr	Auto	Kell	Pheno	Rh
30112	55: UPR 8 DEP_Mixte	ABScr	DE_Nat-1 DE_Nat-2 DE_Nat-3 DE_Enz-1 DE_Enz-2 DE_Enz-3				
31001	66: 4 RAS Rh/Kell Rvs	Pheno Kell			Anti-K 3rd Rh Ctrl	Anti-C 2nd Anti-E 2nd Anti-c 2nd Anti-e 2nd Rh Ctrl	

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
10128	22: 08 Panel A Poly	Ident				Cell-1
						Cell-2
						Cell-3
						Cell-4
						Cell-5
						Cell-6
						Cell-7
						Cell-8
						Cell-9
						Cell-10
						Cell-11
10129	33: 08 Panel A IgG	Ident				Cell-1
						Cell-2
						Cell-3
						Cell-4
						Cell-5
						Cell-6
						Cell-7
						Cell-8
						Cell-9
						Cell-10
						Cell-11
10130	22: 08 Panel B Poly	Ident				Cell-12
						Cell-13
						Cell-14
						Cell-15
						Cell-16
						Cell-17
						Cell-18
						Cell-19
						Cell-20
						Cell-21
						Cell-22

			Unit Results			
Test ID	Test Name	Analyses	Auto	BRC	DilSeries	Ident
10131	33: 08 Panel B IgG	Ident				Cell-12 Cell-13 Cell-14 Cell-15 Cell-16 Cell-17 Cell-18 Cell-19 Cell-20 Cell-21 Cell-22
10132	22: 08 Panel C Unt Poly	Ident				Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10133	33: 08 Panel C Unt IgG	Ident				Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
10134	22: 08 Panel A+Auto Poly	Ident Auto	Auto			Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10135	33: 08 Panel A+Auto IgG	Ident Auto	Auto			Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10136	22: 08 Panel B+Auto Poly	Ident Auto	Auto			Cell-12 Cell-13 Cell-14 Cell-15 Cell-16 Cell-17 Cell-18 Cell-19 Cell-20 Cell-21 Cell-22

			Unit Results			
Test ID	Test Name	Analyses	Auto	BRC	DilSeries	Ident
10137	33: 08 Panel B+Auto IgG	Ident Auto	Auto			Cell-12 Cell-13 Cell-14 Cell-15 Cell-16 Cell-17 Cell-18 Cell-19 Cell-20 Cell-21 Cell-22
10138	22: 08 Panel C Unt+Auto Poly	Ident Auto	Auto			Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10139	33: 08 Panel C Unt+Auto IgG	Ident Auto	Auto			Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
10140	22: 4 Panel A Poly	Ident				Cell-1
						Cell-2
						Cell-3
						Cell-4
						Cell-5
						Cell-6
						Cell-7
						Cell-8
						Cell-9
						Cell-10
						Cell-11
10141	33: 4 Panel A IgG	Ident				Cell-1
						Cell-2
						Cell-3
						Cell-4
						Cell-5
						Cell-6
						Cell-7
						Cell-8
						Cell-9
						Cell-10
						Cell-11
10142	22: 4 Panel B Poly	Ident				Cell-12
						Cell-13
						Cell-14
						Cell-15
						Cell-16
						Cell-17
						Cell-18
						Cell-19
						Cell-20
						Cell-21
						Cell-22

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
10143	33: 4 Panel B IgG	Ident				Cell-12 Cell-13 Cell-14 Cell-15 Cell-16 Cell-17 Cell-18 Cell-19 Cell-20 Cell-21 Cell-22
10144	22: 4 Panel C Unt Poly	Ident				Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10145	33: 4 Panel C Unt IgG	Ident				Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
10146	22: 4 Panel A+Auto Poly	Ident Auto	Auto			Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10147	33: 4 Panel A+Auto IgG	Ident Auto	Auto			Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10148	22: 4 Panel B+Auto Poly	Ident Auto	Auto			Cell-12 Cell-13 Cell-14 Cell-15 Cell-16 Cell-17 Cell-18 Cell-19 Cell-20 Cell-21 Cell-22

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
10149	33: 4 Panel B+Auto IgG	Ident Auto	Auto			Cell-12
						Cell-13
						Cell-14
						Cell-15
						Cell-16
						Cell-17
						Cell-18
						Cell-19
						Cell-20
						Cell-21
						Cell-22
10150	22: 4 Panel C Unt+Auto Poly	Ident Auto	Auto			Cell-1
						Cell-2
						Cell-3
						Cell-4
						Cell-5
						Cell-6
						Cell-7
						Cell-8
						Cell-9
						Cell-10
						Cell-11
10151	33: 4 Panel C Unt+Auto IgG	Ident Auto	Auto			Cell-1
						Cell-2
						Cell-3
						Cell-4
						Cell-5
						Cell-6
						Cell-7
						Cell-8
						Cell-9
						Cell-10
						Cell-11

			Unit Results			
Test ID	Test Name	Analyses	Auto	BRC	DilSeries	Ident
10152	88: 4 Panel C Trt Neut	Ident				Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10153	88: 08 Panel C Trt Neut	Ident				Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10154	88: 08 Panel A IS Neut	Ident				Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11

			Unit Results			
Test ID	Test Name	Analyses	Auto	BRC	DilSeries	Ident
10155	88: 08 Panel B IS Neut	Ident				Cell-12 Cell-13 Cell-14 Cell-15 Cell-16 Cell-17 Cell-18 Cell-19 Cell-20 Cell-21 Cell-22
10156	88: 08 Panel C Unt IS Neut	Ident				Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10157	88: 08 Panel A+Auto IS Neut	Ident Auto	Auto			Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
10158	88: 08 Panel B+Auto IS Neut	Ident Auto	Auto			Cell-12 Cell-13 Cell-14 Cell-15 Cell-16 Cell-17 Cell-18 Cell-19 Cell-20 Cell-21 Cell-22
10159	88: 08 Panel C Unt+Auto IS Neut	Ident Auto	Auto			Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10160	88: 4 Panel A IS Neut	Ident				Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11

			Unit Results			
Test ID	Test Name	Analyses	Auto	BRC	DilSeries	Ident
10161	88: 4 Panel B IS Neut	Ident				Cell-12 Cell-13 Cell-14 Cell-15 Cell-16 Cell-17 Cell-18 Cell-19 Cell-20 Cell-21 Cell-22
10162	88: 4 Panel C Unt IS Neut	Ident				Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10163	88: 4 Panel A+Auto IS Neut	Ident Auto	Auto			Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
10164	88: 4 Panel B+Auto IS Neut	Ident Auto	Auto			Cell-12
						Cell-13
						Cell-14
						Cell-15
						Cell-16
						Cell-17
						Cell-18
						Cell-19
						Cell-20
						Cell-21
						Cell-22
10165	88: 4 Panel C Unt+Auto IS Neut	Ident Auto	Auto			Cell-1
						Cell-2
						Cell-3
						Cell-4
						Cell-5
						Cell-6
						Cell-7
						Cell-8
						Cell-9
						Cell-10
						Cell-11
10166	88: 4 Panel A+Brom Neut	Ident				Cell-1
						Cell-2
						Cell-3
						Cell-4
						Cell-5
						Cell-6
						Cell-7
						Cell-8
						Cell-9
						Cell-10
						Cell-11

			Unit Results			
Test ID	Test Name	Analyses	Auto	BRC	DilSeries	Ident
10167	88: 4 Panel B+Brom Neut	Ident				Cell-12 Cell-13 Cell-14 Cell-15 Cell-16 Cell-17 Cell-18 Cell-19 Cell-20 Cell-21 Cell-22
10168	88: 4 Panel C Unt+Brom Neut	Ident				Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10169	88: 4 Panel A+Brom+Auto Neut	Ident Auto	Auto			Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
10170	88: 4 Panel B+Brom+Auto Neut	Ident Auto	Auto			Cell-12 Cell-13 Cell-14 Cell-15 Cell-16 Cell-17 Cell-18 Cell-19 Cell-20 Cell-21 Cell-22
10171	88: 4 Panel C Unt+Brom+Auto Neut	Ident Auto	Auto			Cell-1 Cell-2 Cell-3 Cell-4 Cell-5 Cell-6 Cell-7 Cell-8 Cell-9 Cell-10 Cell-11
10172	22: 08 Auto Poly	Auto	Auto			
10173	33: 08 Auto IgG	Auto	Auto			
10174	22: 4 Auto Poly	Auto	Auto			
10175	33: 4 Auto IgG	Auto	Auto			
10177	88: Bro 自己对照	Auto	Auto			
20001	10: BRC ABD Surg	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC+		

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
20002	20: BRC Newborn Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		
20003	10: BRC ABD Sel	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC+		
20004	20: BRC Newborn Surg	BRC		BRC+ BRC+ BRC+ BRC+ BRC- BRC-		
20005	00: BRC 00 Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		
20006	20: BRC Newborn Sel	BRC		BRC+ BRC+ BRC+ BRC+ BRC- BRC-		
20007	00: BRC 00 Surg Pos	BRC		BRC+ BRC+ BRC+ BRC- BRC+ BRC+		

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
20008	10: BRC ABD Fic	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC+		
20009	00: BRC 00 Sel Pos	BRC		BRC+ BRC+ BRC+ BRC- BRC+ BRC+		
20010	10: BRC 10 WkD	BRC		BRC+		
20011	20: BRC Newborn Fic	BRC		BRC+ BRC+ BRC+ BRC+ BRC- BRC-		
20012	20: BRC 20 WkD	BRC		BRC+ BRC-		
20013	40: BRC ADK Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		
20014	44: BRC ABO-Rh Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
20015	00: BRC 00 Fic Pos	BRC		BRC+ BRC+ BRC+ BRC- BRC+ BRC+		
20016	20: BRC 20 B of A+B	BRC		BRC+ BRC-		
20017	40: BRC ADK Surg	BRC		BRC+ BRC+ BRC+ BRC+ BRC+ BRC-		
20018	44: BRC ABO-Rh Surg	BRC		BRC+ BRC+ BRC+ BRC+ BRC-		
20019	00: BRC 00 WkD	BRC		BRC+ BRC-		
20020	40: BRC ADK Sel	BRC		BRC+ BRC+ BRC+ BRC+ BRC+ BRC-		
20021	44: BRC ABO-Rh Sel	BRC		BRC+ BRC+ BRC+ BRC+ BRC-		
20022	66: BRC Rvs 2 cell	BRC		BRC- BRC- BRC+ BRC+		

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
20023	40: BRC ADK Fic	BRC		BRC+ BRC+ BRC+ BRC+ BRC+ BRC-		
20024	44: BRC ABO-Rh Fic	BRC		BRC+ BRC+ BRC+ BRC+ BRC-		
20025	66: BRC Rvs 3 cell	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC+		
20026	44: BRC ABO-Rh E	BRC		BRC+ BRC+ BRC-		
20027	66: BRC Rvs 4 Neg	BRC		BRC- BRC- BRC- BRC-		
20028	44: BRC ABO-Rh C	BRC		BRC+ BRC+ BRC-		
20029	66: BRC Rvs 4 Pos	BRC		BRC+ BRC+ BRC+ BRC-		
20030	40: BRC 40 WkD	BRC		BRC+ BRC+ BRC-		
20031	44: BRC 44 WkD	BRC		BRC+ BRC-		

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
20032	66: BRC Rvs 6 Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		
20033	66: BRC Rvs 6 Pos	BRC		BRC+ BRC+ BRC+ BRC- BRC+ BRC+		
20034	66: BRC Rvs A1,B,O	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC-		
20035	22: BRC C3d Poly	BRC		BRC+		
20036	30: BRC C3d DAT	BRC		BRC- BRC+ BRC-		
20037	55: BRC C3d Poly/Neut	BRC		BRC+ BRC-		
20038	11: BRC Rh-hr Surg Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
20039	77: BRC Rh/K Pos	BRC		BRC+ BRC+ BRC+ BRC+ BRC+ BRC-		
20040	11: BRC Rh-hr Surg Pos	BRC		BRC+ BRC+ BRC+ BRC+ BRC+ BRC-		
20041	77: BRC Rh/K Surg Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		
20042	11: BRC Rh-hr Sel Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		
20043	77: BRC Rh/K Sel Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		
20044	11: BRC Rh-hr Sel Pos	BRC		BRC+ BRC+ BRC+ BRC+ BRC+ BRC-		

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
20045	77: BRC Rh/K Fic Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		
20046	11: BRC Rh-hr Fic Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		
20047	11: BRC Rh-hr Fic Pos	BRC		BRC+ BRC+ BRC+ BRC+ BRC+ BRC-		
20048	11: BRC 11 WkD	BRC		BRC+ BRC-		
20049	30: BRC IAT Surg	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC-		
20050	30: BRC IAT Sel	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC-		

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
20051	22: BRC 3 Poly Ltd	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC-		
20052	30: BRC IAT Fic	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC-		
20053	22: BRC 2 Poly	BRC		BRC- BRC- BRC+ BRC+		
20054	33: BRC 3 IgG Ltd	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC-		
20055	88: BRC 88 Fic	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC-		
20056	22: BRC BVSF Poly Ltd	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC-		

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
20057	33: BRC 2 IgG	BRC		BRC- BRC- BRC+ BRC+		
20058	33: BRC BVSF IgG Ltd	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC-		
20059	55: BRC 55 Fic Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		
20060	55: BRC 55 Fic Pos	BRC		BRC+ BRC+ BRC- BRC+ BRC+ BRC-		
20061	22: BRC 0.8% 3 Poly Ltd	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC-		
20062	33: BRC 0.8% 3 IgG Ltd	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC-		

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
20063	22: BRC 0.8% 2 Poly	BRC		BRC- BRC- BRC+ BRC+		
20064	33: BRC 0.8% 2 IgG	BRC		BRC- BRC- BRC+ BRC+		
20065	55: BRC 55 Bro 2cell Pos	BRC		BRC+ BRC+ BRC+ BRC+		
20066	88: BRC 88 Bro 2 cell	BRC		BRC- BRC- BRC+ BRC+		
20067	55: BRC 55 Bro 2cell Neg	BRC		BRC- BRC- BRC- BRC-		
20068	88: BRC 88 Bro 3 cell	BRC		BRC- BRC- BRC- BRC+ BRC+ BRC-		
20069	55: BRC 55 Bro 3cell Pos	BRC		BRC+ BRC+ BRC- BRC+ BRC+ BRC-		

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
20070	55: BRC 55 Bro 3cell Neg	BRC		BRC- BRC- BRC- BRC- BRC- BRC-		
30132	22: UPR 8 ID_Nat 11 Poly	Ident				ID_NT-1 ID_NT-2 ID_NT-3 ID_NT-4 ID_NT-5 ID_NT-6 ID_NT-7 ID_NT-8 ID_NT-9 ID_NT-10 ID_NT-11
30138	22: UPR 8 ID_Nat 11 + Auto Poly	Ident Auto	08_Auto			ID_NT-1 ID_NT-2 ID_NT-3 ID_NT-4 ID_NT-5 ID_NT-6 ID_NT-7 ID_NT-8 ID_NT-9 ID_NT-10 ID_NT-11

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
30153	88: UPR 8 ID_Enz 11 Neut	Ident				ID_TR-1 ID_TR-2 ID_TR-3 ID_TR-4 ID_TR-5 ID_TR-6 ID_TR-7 ID_TR-8 ID_TR-9 ID_TR-10 ID_TR-11
80001	66: DS BV Rvs 08 RT RRBC	DilSeries			DS-Neat DS-2 DS-4 DS-8 DS-16 DS-32 DS-64 DS-128 DS-256 DS-512 DS-1024 DS-Ctrl	
80003	66: DS BV Rvs 4 RT RRBC	DilSeries			DS-Neat DS-2 DS-4 DS-8 DS-16 DS-32 DS-64 DS-128 DS-256 DS-512 DS-1024 DS-Ctrl	

				Unit Results		
Test ID	Test Name	Analyses	Auto	BRC	DilSeries	Ident
80005	33: DS BV IAT IgG 08 37 RRBC	DilSeries			DS-Neat DS-2 DS-4 DS-8 DS-16 DS-32 DS-64 DS-128 DS-256 DS-512 DS-1024 DS-Ctrl	
80007	33: DS BV IAT IgG 4 37 RRBC	DilSeries			DS-Neat DS-2 DS-4 DS-8 DS-16 DS-32 DS-64 DS-128 DS-256 DS-512 DS-1024 DS-Ctrl	
80009	22: DS BV IAT Poly 08 37 RRBC	DilSeries			DS-Neat DS-2 DS-4 DS-8 DS-16 DS-32 DS-64 DS-128 DS-256 DS-512 DS-1024 DS-Ctrl	

				Unit Results		
Test ID	Test Name	Analyses	Auto	BRC	DilSeries	Ident
80011	22: DS BV IAT Poly 4 37 RRBC	DilSeries			DS-Neat DS-2 DS-4 DS-8 DS-16 DS-32 DS-64 DS-128 DS-256 DS-512 DS-1024 DS-Ctrl	
80021	88: DS BV Neut 08 37 RRBC	DilSeries			DS-Neat DS-2 DS-4 DS-8 DS-16 DS-32 DS-64 DS-128 DS-256 DS-512 DS-1024 DS-Ctrl	
80022	88: DS BV Neut 4 37 RRBC	DilSeries			DS-Neat DS-2 DS-4 DS-8 DS-16 DS-32 DS-64 DS-128 DS-256 DS-512 DS-1024 DS-Ctrl	

				Unit Results		
Test ID	Test Name	Analyses	Auto	BRC	DilSeries	Ident
80101	66: DS BV EFS Rvs 08 RT RRBC	DilSeries			DS-Neat DS-2 DS-4 DS-8 DS-16 DS-32 DS-64 DS-128 DS-256 DS-512 DS-1024 DS-Ctrl	
80105	33: DS BV EFS IAT IgG 08 37 RRBC	DilSeries			DS-Neat DS-2 DS-4 DS-8 DS-16 DS-32 DS-64 DS-128 DS-256 DS-512 DS-1024 DS-Ctrl	
80109	22: DS BV EFS IAT Poly 08 37 RRBC	DilSeries			DS-Neat DS-2 DS-4 DS-8 DS-16 DS-32 DS-64 DS-128 DS-256 DS-512 DS-1024 DS-Ctrl	

Test ID	Test Name	Analyses	Unit Results			
			Auto	BRC	DilSeries	Ident
80121	88: DS BV EFS Neut 08 37 RRBC	DilSeries			DS-Neat DS-2 DS-4 DS-8 DS-16 DS-32 DS-64 DS-128 DS-256 DS-512 DS-1024 DS-Ctrl	

Test ID	Test Name	Analyses	Unit Results					
			Fya	Fyb	Jka	Jkb	Lea	Leb
10189	33: 08 RAS Fya IgG	Fya	Anti-Fya					
10190	33: 08 RAS Fyb IgG	Fyb		Anti-Fyb				
10191	66: 08 RAS Jka Rvs	Jka			Anti-Jka			
10192	66: 08 RAS Jkb Rvs	Jkb				Anti-Jkb		
10199	66: 08 RAS Lea Rvs	Lea					Anti-Lea	
10200	66: 08 RAS Leb Rvs	Leb						Anti-Leb
10203	33: 4 RAS Fya IgG	Fya	Anti-Fya					
10204	33: 4 RAS Fyb IgG	Fyb		Anti-Fyb				
10205	66: 4 RAS Jka Rvs	Jka			Anti-Jka			
10206	66: 4 RAS Jkb Rvs	Jkb				Anti-Jkb		
10213	66: 4 RAS Lea Rvs	Lea					Anti-Lea	
10214	66: 4 RAS Leb Rvs	Leb						Anti-Leb
10217	33: 08 RAS Fya IgG + Control	Fya	Anti-Fya					

			Unit Results					
Test ID	Test Name	Analyses	Fya Ctrl	Fyb	Jka	Jkb	Lea	Leb
10218	33: 08 RAS Fyb IgG + Control	Fyb		Anti-Fyb Ctrl				
10219	66: 08 RAS Jka Rvs + Control	Jka			Anti-Jka Ctrl			
10220	66: 08 RAS Jkb Rvs + Control	Jkb				Anti-Jkb Ctrl		
10226	66: 08 RAS Lea Rvs + Control	Lea					Anti-Lea Ctrl	
10227	66: 08 RAS Leb Rvs + Control	Leb						Anti-Leb Ctrl
10230	33: 4 RAS Fya IgG + Control	Fya	Anti-Fya Ctrl					
10231	33: 4 RAS Fyb IgG + Control	Fyb		Anti-Fyb Ctrl				
10232	66: 4 RAS Jka Rvs + Control	Jka			Anti-Jka Ctrl			
10233	66: 4 RAS Jkb Rvs + Control	Jkb				Anti-Jkb Ctrl		
10239	66: 4 RAS Lea Rvs + Control	Lea					Anti-Lea Ctrl	
10240	66: 4 RAS Leb Rvs + Control	Leb						Anti-Leb Ctrl
10243	33: 08 RAS Fya/Fyb IgG + Control	Fya Fyb	Anti-Fya Ctrl	Anti-Fyb Ctrl				
10244	66: 08 RAS Jka/Jkb Rvs + Control	Jka Jkb			Anti-Jka Ctrl	Anti-Jkb Ctrl		
10246	66: 08 RAS Lea/Leb Rvs +	Lea					Anti-Lea	Anti-Leb

			Unit Results					
Test ID	Test Name	Analyses	Fya	Fyb	Jka	Jkb	Lea	Leb
	Control	Leb					Ctrl	Ctrl
10248	33: 4 RAS Fya/Fyb IgG + Control	Fya Fyb	Anti-Fya Ctrl	Anti-Fyb Ctrl				
10249	66: 4 RAS Jka/Jkb Rvs + Control	Jka Jkb			Anti-Jka Ctrl	Anti-Jkb Ctrl		
10252	66: 4 RAS Lea/Leb Rvs + Control	Lea Leb					Anti-Lea Ctrl	Anti-Leb Ctrl

			Unit Results						
Test ID	Test Name	Analyses	DVI	K 2nd	Kell	M	MNS3	MNS4	P1
10011	90: Kell-90	Kell			Anti-K				
10012	95: Kell+Control-95	Kell			Ctrl Anti-K				
10193	33: 08 RAS MNS3 IgG	MNS3					Anti-MNS3		
10194	33: 08 RAS MNS4 IgG	MNS4						Anti-MNS4	
10195	66: 08 RAS K 2nd Rvs	K 2nd		Anti-K 2nd					
10197	88: 08 RAS M Neut	M				Anti-M			
10201	66: 08 RAS P1 Rvs	P1							Anti-P1
10207	33: 4 RAS MNS3 IgG	MNS3					Anti-MNS3		
10208	33: 4 RAS MNS4 IgG	MNS4						Anti-MNS4	
10209	66: 4 RAS K 2nd Rvs	K 2nd		Anti-K 2nd					
10211	88: 4 RAS M Neut	M				Anti-M			
10215	66: 4 RAS P1 Rvs	P1							Anti-P1
10221	33: 08 RAS MNS3 IgG + Control	MNS3					Anti-MNS3 Ctrl		
10222	33: 08 RAS MNS4 IgG + Control	MNS4						Anti-MNS4 Ctrl	
10223	66: 08 RAS K 2nd Rvs + Control	K 2nd		Anti-K 2nd Ctrl					
10225	88: 08 RAS M Neut + Control	M				Anti-M Ctrl			
10228	66: 08 RAS P1 Rvs + Control	P1							Anti-P1 Ctrl
10229	66: 08 RAS DVI Rvs + Control	DVI	Anti-DVI Ctrl						

			Unit Results						
Test ID	Test Name	Analyses	DVI	K 2nd	Kell	M	MNS3	MNS4	P1
10234	33: 4 RAS MNS3 IgG + Control	MNS3					Anti-MNS3 Ctrl		
10235	33: 4 RAS MNS4 IgG + Control	MNS4						Anti-MNS4 Ctrl	
10236	66: 4 RAS K 2nd Rvs + Control	K 2nd		Anti-K 2nd Ctrl					
10238	88: 4 RAS M Neut + Control	M				Anti-M Ctrl			
10241	66: 4 RAS P1 Rvs + Control	P1							Anti-P1 Ctrl
10242	66: 4 RAS DVI Rvs + Control	DVI	Anti-DVI Ctrl						
10245	33: 08 RAS MNS3/MNS4 IgG + Control	MNS3 MNS4					Anti-MNS3 Ctrl	Anti-MNS4 Ctrl	
10247	66: 08 RAS K 2nd/DVI Rvs + Control	K 2nd DVI	Anti-DVI Ctrl	Anti-K 2nd Ctrl					
10250	33: 4 RAS MNS3/MNS4 IgG + Control	MNS3 MNS4					Anti-MNS3 Ctrl	Anti-MNS4 Ctrl	
10251	66: 4 RAS K 2nd/DVI Rvs + Control	K 2nd DVI	Anti-DVI Ctrl	Anti-K 2nd Ctrl					

Test ID	Test Name	Analyses	Unit Results			
			Anti-k (cellano)	DVI	Weak D	XM
10178	22: 4 Min XM Poly	XM				Pat-Cells
10179	33: 4 Min XM IgG	XM				Pat-Cells
10180	22: 4 Maj XM Poly	XM				Dnr-Cells
10181	33: 4 Maj XM IgG	XM				Dnr-Cells
10182	88: Bro ㄅㄚ(主)	XM				Dnr-Cells
10183	22: 08 Maj XM Poly	XM				Dnr-Cells
10184	33: 08 Maj XM IgG	XM				Dnr-Cells
10185	22: 08 Min XM Poly	XM				Pat-Cells
10186	33: 08 Min XM IgG	XM				Pat-Cells
10187	66: 4 IS XM Rvs	XM				Dnr-Cells
10188	66: 08 IS XM Rvs	XM				Dnr-Cells
10196	33: 08 RAS Weak D IgG	Weak D			Anti-Dwk	
10202	66: 08 RAS DVI Rvs	DVI		Anti-DVI		
10210	33: 4 RAS Weak D IgG	Weak D			Anti-Dwk	
10216	66: 4 RAS DVI Rvs	DVI		Anti-DVI		
10224	33: 08 RAS Weak D IgG + Control	Weak D			Anti-Dwk Ctrl	
10237	33: 4 RAS Weak D IgG + Control	Weak D			Anti-Dwk Ctrl	
10253	88: 08 RAS Anti-k (cellano) Neut	Anti-k (cellano)	Anti-k (cellano)			
10254	88: 4 RAS Anti-k (cellano) Neut	Anti-k (cellano)	Anti-k (cellano)			

Test ID	Test Name	Analyses	Unit Results					
			ABO	Auto	C3	IgG	Poly	Rh
10062	22: DAT Poly	Poly					Poly	
10063	30: DAT IgG, C3b,C3d	IgG C3			C3b,C3d Ctrl	IgG Ctrl		
10064	33: DAT IgG	IgG				IgG		
10065	20: Newborn	ABO Rh IgG	Anti-A Anti-B Anti-A+B Ctrl			Ctrl IgG		Anti-D Ctrl
10176	30: 4 Auto IgG, C3b,C3d	Auto		IgG C3b,C3d Ctrl				
10262	22: DAT Poly EFS	Poly					Poly	
10263	30: DAT IgG, C3b,C3d EFS	IgG C3			C3b,C3d Ctrl	IgG Ctrl		
10264	33: DAT IgG EFS	IgG				IgG		

			Unit Results			
Test ID	Test Name	Analyses	ABO	Kell	Pheno	Rh
10255	14: Rh/K II-14	Pheno Kell		Anti-K 3rd Rh Ctrl	Anti-C 2nd Anti-E 2nd Anti-c 2nd Anti-e 2nd Rh Ctrl	
10256	00,66: 4 ABO(FWD/RVS)/Rh-00 + (RVS)-O	ABO Rh	Anti-A Anti-B Ctrl A1-Cells B-Cells O-Cells			Anti-D Ctrl
10257	00,66: 08 ABO(FWD/RVS)/Rh-00 + (RVS)-O	ABO Rh	Anti-A Anti-B Ctrl A1-Cells B-Cells O-Cells			Anti-D Ctrl

Test ID	Test Name	Analyses	Unit Results			
			ABO	Kell	Pheno	Rh
10258	46,77: ABD/Rev-46 + Rh/K-77	ABO Rh Pheno Kell	Anti-A Anti-B Anti-A+B A1-Cells B-Cells Ctrl	Anti-K Ctrl	Anti-C Anti-E Anti-c Anti-e Ctrl	Anti-D Ctrl
30258	46,77: UPR ABD/Rev-46 + Sim 2	ABO Rh Pheno Kell	Anti-A Anti-B Anti-A+B Hem-A1 Hem-B Ctrl	Anti-K Ctrl	Anti-C Anti-E Anti-c Anti-e Ctrl	Anti-D Ctrl

Test ID	Test Name	Analyses	Unit Results			
			C3	IgG	Poly	Undefined
10289	22: 08 RAS Poly + Control	Undefined				Undefined Ctrl
10290	22: 4 RAS Poly + Control	Undefined				Undefined Ctrl
10291	22: 4 RAS Poly Decon + Control	Undefined				Undefined Ctrl
10292	22: 08 RAS Poly Decon + Control	Undefined				Undefined Ctrl
10462	22: DAT Poly MF OFF	Poly			Poly	
10463	30: DAT IgG, C3b,C3d MF OFF	IgG C3	C3b,C3d Ctrl	IgG Ctrl		
10464	33: DAT IgG MF OFF	IgG		IgG		

Test ID	Test Name	Analyses	Unit Results	
			ABO	Rh
60002	52: ABODDVI [Donor]	ABO Rh	Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
60007	12: ABDVI [Donor]	ABO Rh	Anti-A Anti-B	Anti-D(DVI)
60008	12: ABDVI [Donor] New Suspension	ABO Rh	Anti-A Anti-B	Anti-D(DVI)
60025	52,66: 4 ABODDVI [Donor] + (RVS) A1A2B	ABO Rh	A1-Cells A2-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
60029	52,66: 4 ABODDVI [Donor] + (RVS) A1A2BO	ABO Rh	A1-Cells A2-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl

Test ID	Test Name	Analyses	Unit Results	
			ABO	Rh
60033	52,66: 4 ABODDVI [Donor] + (RVS) A1B	ABO Rh	A1-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
60037	52,66: 4 ABODDVI [Donor] + (RVS) A1BO	ABO Rh	A1-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
60041	52,66: 08 ABODDVI [Donor] + (RVS) A1A2B	ABO Rh	A1-Cells A2-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
60045	52,66: 08 ABODDVI [Donor] + (RVS) A1A2BO	ABO Rh	A1-Cells A2-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl

Test ID	Test Name	Analyses	Unit Results	
			ABO	Rh
60049	52,66: 08 ABODDVI [Donor] + (RVS) A1B	ABO Rh	A1-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
60053	52,66: 08 ABODDVI [Donor] + (RVS) A1BO	ABO Rh	A1-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
60059	52,66: 4 ABODDVI [Donor] + (RVS) 6 cell	ABO Rh	A1-Cells A2-Cells B-Cells O-Cells O-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl

Test ID	Test Name	Analyses	Unit Results		
			ABO	IgG	Rh
60065	24: DVI Newborn	ABO Rh IgG	Anti-A Anti-B Anti-A+B Ctrl	Ctrl IgG	Anti-D(DVI) Ctrl

Test ID	Test Name	Unit Results		
		Analyses	ABO	Rh
70003	52: DVI Grouping	ABO Rh	Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
70025	52,66: 4 DVI Grouping + (RVS) A1A2B	ABO Rh	A1-Cells A2-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
70029	52,66: 4 DVI Grouping + (RVS) A1A2BO	ABO Rh	A1-Cells A2-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
70033	52,66: 4 DVI Grouping + (RVS) A1B	ABO Rh	A1-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl

Test ID	Test Name	Unit Results		
		Analyses	ABO	Rh
70037	52,66: 4 DVI Grouping + (RVS) A1BO	ABO Rh	A1-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
70041	52,66: 08 DVI Grouping + (RVS) A1A2B	ABO Rh	A1-Cells A2-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
70045	52,66: 08 DVI Grouping + (RVS) A1A2BO	ABO Rh	A1-Cells A2-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
70049	52,66: 08 DVI Grouping + (RVS) A1B	ABO Rh	A1-Cells B-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl

Test ID	Test Name	Unit Results		
		Analyses	ABO	Rh
70053	52,66: 08 DVI Grouping + (RVS) A1BO	ABO Rh	A1-Cells B-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl
70059	52,66: 4 DVI Grouping + (RVS) 6 cell	ABO Rh	A1-Cells A2-Cells B-Cells O-Cells O-Cells O-Cells Anti-A Anti-B Anti-A+B Ctrl	Anti-D Anti-D(DVI) Ctrl

4.2 Cassettes

The following is a list of available Cassettes. Cassette names appear in the manufacturer record in field 4.1.

ID	Name	Short Name
00	ABO-Rh/Reverse	ABO-Rh/Rev
10	ABD Confirmation	ABD
11	Rh-hr	Rh-hr
12	ABD(DVI) Donor Conf	ABD(DVI) Donor
14	Rh/K II	Rh/K II
20	ABO-Rh/DAT newborn	Newborn
22	AHG Polyspecific	Poly
24	DVI Newborn	DVI Newborn
30	DAT	DAT/IAT
33	AHG anti-IgG	IgG
40	ADK	ADDK
44	ABO-Rh	ABOD/CDE
46	ABD/Reverse Cassette	ABF
48	ABODD	ABO-DD
52	DVI Grouping	DVI Grouping
55	AHG Polyspecific/Neutral	Poly/Neut
66	Reverse diluent	Reverse
77	Rh/K	Rh/K
88	Neutral	Neutral
90	Kell	Kell
95	Kell/Control	Kell/Ctrl

4.3 Reagents

The following is a list of available reagents. The reagent name appears in the manufacturing record in field 5.1.

Vision does not guarantee that the list of reagents in an M-record will be reported in the same sequence everytime. The reagent name and not its position in the uploaded result message should always be used to access reagent lots and expiration dates. See [3.4.8.17 Results with multiple reagents](#) for an example message.

Liquid Name	Reagent ID	Liquid Family
0.8 RCD	00	0.8% Red Cell Diluent
Panel A Cell 1	01	4% Resolve Panel A
Panel A Cell 2	02	4% Resolve Panel A
Panel A Cell 3	03	4% Resolve Panel A
Panel A Cell 4	04	4% Resolve Panel A
Panel A Cell 5	05	4% Resolve Panel A
Panel A Cell 6	06	4% Resolve Panel A
Panel A Cell 7	07	4% Resolve Panel A
Panel A Cell 8	08	4% Resolve Panel A
Panel A Cell 9	09	4% Resolve Panel A
Panel A Cell 10	10	4% Resolve Panel A
Panel A Cell 11	11	4% Resolve Panel A
A1 Cells	12	4% Affirmagen
A2 Cells	13	4% Affirmagen
B Cells	14	4% Affirmagen
O Cells	15	4% Affirmagen
Fic Unt 1	16	4% BioVue Screen

Liquid Name	Reagent ID	Liquid Family
Fic Unt 2	17	4% BioVue Screen
Fic Unt 3	18	4% BioVue Screen
Fic Trt 1	19	4% BioVue Screen
Fic Trt 2	20	4% BioVue Screen
Fic Trt 3	21	4% BioVue Screen
Sel I	23	4% Selectogen
Sel II	24	4% Selectogen
BLISS	28	BLISS
0.8% Sel I	29	0.8% Selectogen
0.8% Sel II	30	0.8% Selectogen
0.8% Surg1	31	0.8% Surgiscreen
0.8% Surg2	32	0.8% Surgiscreen
0.8% Surg3	33	0.8% Surgiscreen
Surg 1	34	4% Surgiscreen
Surg 2	35	4% Surgiscreen
Surg 3	36	4% Surgiscreen
Panel B Cell 12	40	4% Resolve Panel B
Panel B Cell 13	41	4% Resolve Panel B
Panel B Cell 14	42	4% Resolve Panel B
Panel B Cell 15	43	4% Resolve Panel B
Panel B Cell 16	44	4% Resolve Panel B
Panel B Cell 17	45	4% Resolve Panel B
Panel B Cell 18	46	4% Resolve Panel B

Liquid Name	Reagent ID	Liquid Family
Panel B Cell 19	47	4% Resolve Panel B
Panel B Cell 20	48	4% Resolve Panel B
Panel B Cell 21	49	4% Resolve Panel B
Panel B Cell 22	50	4% Resolve Panel B
0.8% Panel A Cell 1	51	0.8% Resolve Panel A
0.8% Panel A Cell 2	52	0.8% Resolve Panel A
0.8% Panel A Cell 3	53	0.8% Resolve Panel A
0.8% Panel A Cell 4	54	0.8% Resolve Panel A
0.8% Panel A Cell 5	55	0.8% Resolve Panel A
0.8% Panel A Cell 6	56	0.8% Resolve Panel A
0.8% Panel A Cell 7	57	0.8% Resolve Panel A
0.8% Panel A Cell 8	58	0.8% Resolve Panel A
0.8% Panel A Cell 9	59	0.8% Resolve Panel A
0.8% Panel A Cell 10	60	0.8% Resolve Panel A
0.8% Panel A Cell 11	61	0.8% Resolve Panel A
0.8% Panel B Cell 12	62	0.8% Resolve Panel B
0.8% Panel B Cell 13	63	0.8% Resolve Panel B
0.8% Panel B Cell 14	64	0.8% Resolve Panel B
0.8% Panel B Cell 15	65	0.8% Resolve Panel B
0.8% Panel B Cell 16	66	0.8% Resolve Panel B
0.8% Panel B Cell 17	67	0.8% Resolve Panel B
0.8% Panel B Cell 18	68	0.8% Resolve Panel B
0.8% Panel B Cell 19	69	0.8% Resolve Panel B

Liquid Name	Reagent ID	Liquid Family
0.8% Panel B Cell 20	70	0.8% Resolve Panel B
0.8% Panel B Cell 21	71	0.8% Resolve Panel B
0.8% Panel B Cell 22	72	0.8% Resolve Panel B
Diego	73	4% Diego
Bromelin	76	Bromelin
BRC-S1	77	BRC - Control Serum
BRC-S2	78	BRC - Control Serum
BRC-S3	79	BRC - Control Serum
BRC-E1	81	BRC - Control Red Blood Cells
BRC-E2	82	BRC - Control Red Blood Cells
BRC-E3	83	BRC - Control Red Blood Cells
BRC-E4 and/or Du Cells	84	BRC - Control Red Blood Cells
BRC-E5	85	BRC - Control Red Blood Cells
0.8 Fic Unt 1	87	0.8% BioVue Screen
0.8 Fic Unt 2	88	0.8% BioVue Screen
0.8 Fic Unt 3	89	0.8% BioVue Screen
IgG BRC-S3	90	Dilute BRC-S3
Poly BRC-S3	91	Dilute BRC-S3
0.8% A1 Cells	92	0.8% Affirmagen
0.8% A2 Cells	93	0.8% Affirmagen
0.8% B Cells	94	0.8% Affirmagen
0.8% O Cells	95	0.8% Affirmagen
0.8 Fic Trt 1	96	0.8% BioVue Screen

Liquid Name	Reagent ID	Liquid Family
0.8 Fic Trt 2	97	0.8% BioVue Screen
0.8 Fic Trt 3	98	0.8% BioVue Screen
0.8% Panel C Cell 1	121	0.8% Resolve Panel C
0.8% Panel C Cell 2	122	0.8% Resolve Panel C
0.8% Panel C Cell 3	123	0.8% Resolve Panel C
0.8% Panel C Cell 4	124	0.8% Resolve Panel C
0.8% Panel C Cell 5	125	0.8% Resolve Panel C
0.8% Panel C Cell 6	126	0.8% Resolve Panel C
0.8% Panel C Cell 7	127	0.8% Resolve Panel C
0.8% Panel C Cell 8	128	0.8% Resolve Panel C
0.8% Panel C Cell 9	129	0.8% Resolve Panel C
0.8% Panel C Cell 10	130	0.8% Resolve Panel C
0.8% Panel C Cell 11	131	0.8% Resolve Panel C
0.8% Panel C Enz Cell 1	132	0.8% Resolve Panel C
0.8% Panel C Enz Cell 2	133	0.8% Resolve Panel C
0.8% Panel C Enz Cell 3	134	0.8% Resolve Panel C
0.8% Panel C Enz Cell 4	135	0.8% Resolve Panel C
0.8% Panel C Enz Cell 5	136	0.8% Resolve Panel C
0.8% Panel C Enz Cell 6	137	0.8% Resolve Panel C
0.8% Panel C Enz Cell 7	138	0.8% Resolve Panel C
0.8% Panel C Enz Cell 8	139	0.8% Resolve Panel C
0.8% Panel C Enz Cell 9	140	0.8% Resolve Panel C
0.8% Panel C Enz Cell 10	141	0.8% Resolve Panel C

Liquid Name	Reagent ID	Liquid Family
0.8% Panel C Enz Cell 11	142	0.8% Resolve Panel C
0.8% ABSCR4 Cell 1	143	0.8% ABSCR4
0.8% ABSCR4 Cell 2	144	0.8% ABSCR4
0.8% ABSCR4 Cell 3	145	0.8% ABSCR4
0.8% ABSCR4 Cell 4	146	0.8% ABSCR4
Anti-Fya	147	Fya
Anti-Fyb	148	Fyb
Anti-Jka	149	Jka
Anti-Jkb	150	Jkb
Anti-MNS3	151	MNS3
Anti-MNS4	152	MNS4
Anti-M	153	M
Anti-Lea	155	Lea
Anti-Leb	156	Leb
Anti-Dwk	157	Weak D
Anti-K 2nd	158	K 2nd
Anti-P1	159	P1
Anti-DVI	160	DVI
Anti-k (cellano)	162	Anti-k (cellano)
Anti-C 2nd	165	Anti-C 2nd
Anti-E 2nd	166	Anti-E 2nd
Anti-c 2nd	167	Anti-c 2nd
Anti-e 2nd	168	Anti-e 2nd

Liquid Name	Reagent ID	Liquid Family
Anti-Kell 3rd	169	Anti-Kell 3rd
Rh Control	170	Rh Control
Panel C Cell 1	171	4% Resolve Panel C
Panel C Cell 2	172	4% Resolve Panel C
Panel C Cell 3	173	4% Resolve Panel C
Panel C Cell 4	174	4% Resolve Panel C
Panel C Cell 5	175	4% Resolve Panel C
Panel C Cell 6	176	4% Resolve Panel C
Panel C Cell 7	177	4% Resolve Panel C
Panel C Cell 8	178	4% Resolve Panel C
Panel C Cell 9	179	4% Resolve Panel C
Panel C Cell 10	180	4% Resolve Panel C
Panel C Cell 11	181	4% Resolve Panel C
Panel C Enz Cell 1	182	4% Resolve Panel C
Panel C Enz Cell 2	183	4% Resolve Panel C
Panel C Enz Cell 3	184	4% Resolve Panel C
Panel C Enz Cell 4	185	4% Resolve Panel C
Panel C Enz Cell 5	186	4% Resolve Panel C
Panel C Enz Cell 6	187	4% Resolve Panel C
Panel C Enz Cell 7	188	4% Resolve Panel C
Panel C Enz Cell 8	189	4% Resolve Panel C
Panel C Enz Cell 9	190	4% Resolve Panel C
Panel C Enz Cell 10	191	4% Resolve Panel C

Liquid Name	Reagent ID	Liquid Family
Panel C Enz Cell 11	192	4% Resolve Panel C
ABSCR4 Cell 1	193	4% ABSCR4
ABSCR4 Cell 2	194	4% ABSCR4
ABSCR4 Cell 3	195	4% ABSCR4
ABSCR4 Cell 4	196	4% ABSCR4
Ortho Sera Reagent Control	199	Ortho Sera Reagent Control
UPR ABO_Sim A1	0601	0.8% UPR ABO Simonin
UPR ABO_Sim A2	0602	0.8% UPR ABO Simonin
UPR ABO_Sim B	0603	0.8% UPR ABO Simonin
UPR ABO_Sim O	0604	0.8% UPR ABO Simonin
UPR 0.8% DEP_Nat 1	3031	UPR 0.8% DEP
UPR 0.8% DEP_Nat 2	3032	UPR 0.8% DEP
UPR 0.8% DEP_Nat 3	3033	UPR 0.8% DEP
UPR 0.8% ID_Nat 1	3131	UPR 0.8% ID
UPR 0.8% ID_Nat 2	3132	UPR 0.8% ID
UPR 0.8% ID_Nat 3	3133	UPR 0.8% ID
UPR 0.8% ID_Nat 4	3134	UPR 0.8% ID
UPR 0.8% ID_Nat 5	3135	UPR 0.8% ID
UPR 0.8% ID_Nat 6	3136	UPR 0.8% ID
UPR 0.8% ID_Nat 7	3137	UPR 0.8% ID
UPR 0.8% ID_Nat 8	3138	UPR 0.8% ID
UPR 0.8% ID_Nat 9	3139	UPR 0.8% ID
UPR 0.8% ID_Nat 10	3140	UPR 0.8% ID

Liquid Name	Reagent ID	Liquid Family
UPR 0.8% ID_Nat 11	3141	UPR 0.8% ID
UPR 0.8% DEP_Enz 1	4051	UPR 0.8% DEP
UPR 0.8% DEP_Enz 2	4052	UPR 0.8% DEP
UPR 0.8% DEP_Enz 3	4053	UPR 0.8% DEP
UPR 0.8% ID_Enz 1	4151	UPR 0.8% ID
UPR 0.8% ID_Enz 2	4152	UPR 0.8% ID
UPR 0.8% ID_Enz 3	4153	UPR 0.8% ID
UPR 0.8% ID_Enz 4	4154	UPR 0.8% ID
UPR 0.8% ID_Enz 5	4155	UPR 0.8% ID
UPR 0.8% ID_Enz 6	4156	UPR 0.8% ID
UPR 0.8% ID_Enz 7	4157	UPR 0.8% ID
UPR 0.8% ID_Enz 8	4158	UPR 0.8% ID
UPR 0.8% ID_Enz 9	4159	UPR 0.8% ID
UPR 0.8% ID_Enz 10	4160	UPR 0.8% ID
UPR 0.8% ID_Enz 11	4161	UPR 0.8% ID
Undefined	161	Undefined
Undefined Decontamination	200	Undefined Decontamination
Anti-H Mono (ATox)	5000	AntiToxin
Anti-A1 (ATox)	5001	AntiToxin
Anti-Cw Mono (ATox)	5002	AntiToxin
Anti-M Mono (ATox)	5003	AntiToxin
Anti-N Mono (ATox)	5004	AntiToxin
Anti-Lea Mono (ATox)	5005	AntiToxin

Liquid Name	Reagent ID	Liquid Family
Anti-MNS3 Mono (ATox)	5007	AntiToxin
Anti-MNS4 Mono (ATox)	5008	AntiToxin
Anti-Fya Poly (ATox)	5009	AntiToxin
Anti-Fyb Poly (ATox)	5100	AntiToxin
Anti-Jka Poly (ATox)	5101	AntiToxin
Anti-Jkb Poly (ATox)	5102	AntiToxin
Anti-Jka Mono (ATox)	5103	AntiToxin
Anti-Jkb Mono (ATox)	5104	AntiToxin
Anti-MNS3 Poly (ATox)	5105	AntiToxin
Anti-MNS4 Poly (ATox)	5106	AntiToxin
Anti-Kpa Poly (ATox)	5107	AntiToxin
Anti-Kpb Poly (ATox)	5108	AntiToxin
Anti-Lua Poly (ATox)	5109	AntiToxin
Anti-Lub Poly (ATox)	5200	AntiToxin
Anti-Cellano Poly (ATox)	5201	AntiToxin
Anti-Cellano Mono (ATox)	5202	AntiToxin
Anti-Coa Poly (ATox)	5203	AntiToxin
Anti-Cob Poly (ATox)	5204	AntiToxin
Anti-Wra Poly (ATox)	5205	AntiToxin
Anti-Dia Poly (ATox)	5206	AntiToxin
Anti-Cw Poly (ATox)	5207	AntiToxin
Anti-Leb Mono (ATox)	5208	AntiToxin
Anti-P1 Mono (ATox)	5209	AntiToxin

4.4 Unit Result Names

The following is a list of unit result names. Unit result names appear in the manufacturer record in field 3.

Unit Results
0.8-ABSCR4 Cell 1
0.8-ABSCR4 Cell 2
0.8-ABSCR4 Cell 3
0.8-ABSCR4 Cell 4
0.8-Sel I
0.8-Sel II
0.8-Surg1
0.8-Surg2
0.8-Surg3
08_Auto
A1-Cells
A2-Cells
ABSCR4 Cell 1
ABSCR4 Cell 2
ABSCR4 Cell 3
ABSCR4 Cell 4
Anti-A
Anti-A+B
Anti-B
Anti-C
Anti-c
Anti-C 2nd
Anti-c 2nd
Anti-CDE
Anti-D
Anti-D(DVI)
Anti-DVI

Unit Results
Anti-Dwk
Anti-E
Anti-e
Anti-E 2nd
Anti-e 2nd
Anti-Fya
Anti-Fyb
Anti-Jka
Anti-Jkb
Anti-K
Anti-K (cellano)
Anti-K 2nd
Anti-K 3rd
Anti-Lea
Anti-Leb
Anti-M
Anti-MNS3
Anti-MNS4
Anti-P1
Auto
B-Cells
BRC-
BRC+
Br-Diego
Br-Sel I
Br-Sel II
Br-Surg 1
Br-Surg 2
Br-Surg 3
C3b,C3d
Cell-1

Unit Results
Cell-10
Cell-11
Cell-12
Cell-13
Cell-14
Cell-15
Cell-16
Cell-17
Cell-18
Cell-19
Cell-2
Cell-20
Cell-21
Cell-22
Cell-3
Cell-4
Cell-5
Cell-6
Cell-7
Cell-8
Cell-9
Ctrl
DE_Enz-1
DE_Enz-2
DE_Enz-3
DE_Nat-1
DE_Nat-2
DE_Nat-3
Diego
Dnr-Cells
DS-1024

Unit Results
DS-128
DS-2
DS-16
DS-256
DS-32
DS-4
DS-512
DS-64
DS-8
DS-Ctrl
DS-Neat
Fic-Trt 1
Fic-Trt 2
Fic-Trt 3
Fic-Unt 1
Fic-Unt 2
Fic-Unt 3
Hem-A1
Hem-A2
Hem-B
Hem-O
ID_NT-1
ID_NT-10
ID_NT-11
ID_NT-2
ID_NT-3
ID_NT-4
ID_NT-5
ID_NT-6
ID_NT-7
ID_NT-8

Unit Results
ID_NT-9
ID_TR-1
ID_TR-10
ID_TR-11
ID_TR-2
ID_TR-3
ID_TR-4
ID_TR-5
ID_TR-6
ID_TR-7
ID_TR-8
ID_TR-9
IgG
O-Cells
Pat-Cells
Poly
Rh Ctrl
Sel I
Sel II
Surg 1
Surg 2
Surg 3
Undefined

4.5 Sample Types

The following is a list of sample types. The sample type appears in the order record in field 16. Additionally, the sample type also appears in field 5 for each donor sample.

Sample Type	Description
0.8CELLS_R	Sample
3CELLS_R	Sample
CENTBLOOD	Sample
EFS/UPR CQI	QCFluid
ORTHO CONFIDENCE WB	QCFluid
ORTHO DAILY QC	QCFluid
ORTHO DAILY QC JAPAN	QCFluid
ORTHO DAILY QC SERUM JAPAN	QCFluid
PACKEDCELLS	Sample
PLASMA	Sample

4.6 Analyses

The following is a list of analysis names and their values. The analysis name values will appear in the ASTM Result record fields 3 and 4.

Note: A value of '?' indicates that the result could not be interpreted. This can be caused by errors or an invalid combination of unit result values.

Analysis	Values
ABO	AB, A, B, O
ABScr	NEG, POS
Anti-k (cellano)	POS, NEG
Auto	POS, NEG
BRC	Pass
C3	POS, NEG
DilSeries	Done
DVI	POS, NEG
Fya	POS, NEG
Fyb	POS, NEG
Ident	Done
IgG	POS, NEG
Jka	POS, NEG
Jkb	POS, NEG
K 2nd	POS, NEG
Kell	POS, NEG
Lea	POS, NEG
Leb	POS, NEG
M	POS, NEG
MNS3	POS, NEG
MNS4	POS, NEG
P1	POS, NEG
Pheno	CCEE, CCEe, CCee, CcEE, CcEe, Ccee, ccEE, ccEe, ccee
Poly	POS, NEG
Rh	POS, NEG

Analysis	Values
Weak D	POS, NEG
XM	CMP, INCMP

4.7 Alternate Names for Analysis Values

The system can be configured to use the following alternate analysis values.

Result	Alternates
POS	+
NEG	-

Result	Set 1	Set 2	Set 3
CcEe	C+c+E+e+	C+E+c+e+	2,3,4,5
CCEE	C+c-E+e-	C+E+c-e-	2,3,-4,-5
CCEe	C+c-E+e+	C+E+c-e+	2,3,-4,5
CCee	C+c-E-e+	C+E-c-e+	2,-3,-4,5
CcEE	C+c+E+e-	C+E+c+e-	2,3,4,-5
Ccee	C+c+E-e+	C+E-c+e+	2,-3,4,5
ccEE	C-c+E+e-	C-E+c+e-	-2,3,4,-5
ccEe	C-c+E+e+	C-E+c+e+	-2,3,4,5
ccee	C-c+E-e+	C-E-c+e+	-2,-3,4,5

Ortho Clinical Diagnostics



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